

Neural-initialised static condensation with equilibrium correction for the analysis and thickness design of cut thin-walled TPMS lattices

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Abstract

Reduced models of structural components are increasingly learned from data, but a learned model is seldom a mechanical model: it rarely admits assembly or carries a bound, cannot be improved once trained, and its error in the quantities that analysis and design require is unknown. We ask how a learned component model can be given the structure of static condensation, symmetry, positive semidefiniteness, the rigid-body kernel and a lower bound by the exact Schur complement, together with controlled and improvable accuracy, and what then determines its error in the assembled compliance and in local design sensitivities. The answer lies in where the approximation is placed and how its condensed stiffness is formed. Approximating only the interior extension, on the complete retained space, leaves the displacement patterns exchanged between cells unchanged. Forming the condensed stiffness from the energy of that extension, $\hat{S} = F^T K F$ with F the extension and K the cell stiffness matrix, makes it a Ritz approximation bounded below by the exact Schur complement, with an error quadratic in the interior error. A fixed, geometry-specific two-grid correction applied inside that energy reduces the interior residual without losing any of these properties and cannot increase the error when its smoothing interval contains the spectrum. Learning supplies the trial field, the variational form the structure, and the correction the improvability. Following the interior error into the structural response shows that it enters the compliance weighted by the cell's energy share but the thickness sensitivity through the stiffness derivative, so that

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accurate compliance does not imply accurate local sensitivity and design use requires both. We realise the construction, neural-initialised condensation with equilibrium correction (NICE), for cut thin-walled Schwarz-P cells in a stabilised cut finite element model, with tens of thousands of retained degrees of freedom and a discrete space that changes with the cut. Under consistent tractions the mean directional energy error on 80 validation geometries is 0.074%, and in two-cell assemblies the compliance and thickness sensitivity errors stay below 0.28% and 1.5%. Applied without retraining to a network trained without it, the correction alone lowers that network's mean energy error from 6.89% to 0.0965%. An eight-cell design iteration takes 81–110 s on one GPU against 868–1,042 s for the whole-lattice Cholesky solution, thickness optimisation of an eight-cell block reproduces the design of exact condensation to 0.0051 per corner parameter, and a design iteration of a 110-cell plate with 32.7 million degrees of freedom takes 43 min.

Keywords: learned substructures, static condensation, neural-initialised condensation, equilibrium correction, cut finite elements, TPMS lattices, thickness sensitivity, thickness optimisation

1. Introduction

Structures assembled from repeated thin-walled cells, such as graded triply periodic minimal surface (TPMS) lattices Yu et al. (2019) with supports, external loads and cells cut by the specimen boundary, call for a reusable component model of one cell. The model must return the forces conjugate to the displacement imposed by its surroundings, so that the structure can be assembled, and the interior field, from which local design quantities such as thickness sensitivities are evaluated. Homogenisation, on which the design of graded TPMS lattices commonly rests Li et al. (2018), provides such a model only under a separation of scales, which a single layer of cut cells lacks; in the example of Section 5.10 it underestimates the compliance by 27–37%. Static condensation provides an exact one Guyan (1965), Irons (1965), at the cost of an interior factorisation for every geometry, which a design iteration repeats for every cell. Machine learning promises an inexpensive one, for example by mapping an element's nodal displacements directly to its forces Capuano & Rimoli (2019), but, as usually constructed, delivers a map without the structure of a mechanical model. A component model that is to replace static condensation in analysis and design must be sym-

metric and positive semidefinite with the rigid-body kernel, so that it can be assembled; it must be applicable to every new geometry without an interior factorisation; its error must be traceable to the compliance and to the sensitivities that analysis and design require; and it should be improvable at deployment without retraining. This paper shows that a learned component model acquires the structure of a mechanical model from where its approximation is placed and how its condensed stiffness is formed, and its improvability from a fixed correction of its interior field. It also quantifies what then determines the error in the assembled compliance and in the local thickness sensitivity.

For cut thin-walled cells the retained space is large: the box-face degrees of freedom carry exterior loads and intercell coupling, and those of a band of cut elements carry virtual work on the cut surface, so each cell here retains tens of thousands of degrees of freedom. Forming the Schur complement explicitly needs one interior solve per retained degree of freedom, and applying it needs an interior factorisation for every geometry. Two approximations reduce this cost. Reducing the retained degrees of freedom, as in port reduction Eftang & Patera (2013), Smetana & Patera (2016), Ballani et al. (2018) and in learned substructures described by a few boundary degrees of freedom Huang et al. (2023), Guo et al. (2026a), changes the displacement patterns that substructures can exchange and leaves a boundary model error independent of the interior. Approximating the interior extension on the complete retained space, the route of the static-condensation reduced basis element method Huynh et al. (2013), alters only the interior response to each pattern; inexact substructuring methods Dohrmann (2007), Li & Widlund (2007), Klawonn & Rheinbach (2007) also approximate subdomain and coarse solves, but only inside a preconditioner, so that their iteration converges to the exact discrete solution. We follow the second route with a learned, geometry-conditioned interior extension. The two kinds of error enter the structural response differently. An interior error is weighted by the energy the cell carries, so a cell with a small energy share can leave the compliance accurate while its thickness sensitivity, weighted by the stiffness derivative, remains inaccurate. This is the distinction on which goal-oriented error estimation rests Becker & Rannacher (2001), Oden & Prudhomme (2001); it has been observed for approximate reanalysis and inexact solves in topology optimisation Amir et al. (2009), Amir et al. (2010), Gogu (2015). We quantify it for learned substructures and adopt joint accuracy of compliance and local sensitivity as the criterion for design use.

The construction rests on three elements. Any admissible extension that reproduces rigid-body motion, evaluated in the energy form, yields a symmetric positive semidefinite condensed stiffness with the rigid-body kernel, bounded below by the exact Schur complement with an error quadratic in the interior error Toselli & Widlund (2005); this holds for static condensation itself, for multiscale finite elements Hou & Wu (1997), for component reduced-basis methods Huynh et al. (2013) and for learned shape functions evaluated as $N^T K N$ Huang et al. (2023). Learning supplies the trial extension: a geometry-conditioned network, exactly linear in the retained displacement, produces an initial interior field for each cell. A fixed, geometry-specific two-grid correction, Chebyshev smoothing around a coarse-grid Galerkin correction with a prescribed number of steps Xu (1992), then reduces the interior equilibrium residual at fixed retained displacement. Because the correction is a fixed linear map inside the condensed energy, the corrected condensed stiffness keeps every variational property, its error cannot exceed that of the uncorrected one when the smoothing interval contains the spectrum Golub & Varga (1961), Xu & Zikatanov (2002), and the transpose of the complete corrected extension returns the force conjugate to the stated energy. Unlike hybrid neural solvers, which combine learned predictions with relaxation, learned inverse actions or coarse spaces to accelerate one global solve Zhang, E. et al. (2024), Kopaničáková & Karniadakis (2025), Xing et al. (2026), the output is a condensed operator per cell that can be assembled, differentiated and improved. We call the construction neural-initialised condensation with equilibrium correction (NICE).

Learned substructures have been developed furthest as problem-independent machine learning (PIML), which predicts substructure shape functions for a few boundary degrees of freedom, imposes rigid-body motion constraints and, in its self-consistent form, evaluates the stiffness as $N^T K N$ Huang et al. (2022), Huang et al. (2023); its variants train without labelled shape functions by minimum potential energy Huang et al. (2024), enrich the boundary with Bézier functions Guo et al. (2026a), join oversampled bases by an overlapping partition of unity and correct the reconstructed fine-scale field by a few Jacobi-preconditioned conjugate-gradient iterations on the global equilibrium equations Guo et al. (2026b), impose symmetry equivariance Jiang, C. et al. (2026), treat isoparametric substructures Zhang, L. et al. (2024), transfer them to complex three-dimensional design domains Zhang et al. (2026) and optimise lattices Xu et al. (2025). Describing each substructure by a few boundary degrees of freedom gives coarse

models small enough for very large structures, at the price of a boundary model error that this line of work controls by refining the partition, enriching the interpolation or oversampling. Guo et al. (2026a, Table 1) also tried a learned interior with every boundary node retained: for $5 \times 5 \times 5$ -element substructures it was the most accurate option (displacement error 0.84% against 1.95% with the Bézier boundary), for $10 \times 10 \times 10$ -element ones less accurate (3.46% against 2.22%), which they tentatively attribute to the larger network output, and in both cases seven to fifteen times slower, so they adopted the Bézier boundary as the better balance of accuracy and cost. The present construction retains the complete trace of cut thin-walled cells, $1.7\text{--}3.4 \times 10^4$ degrees of freedom per cell, and restores the accuracy of the learned interior with the correction, at the price of an assembled model that carries all of these retained degrees of freedom. Unlike the global residual correction that Guo et al. (2026b) apply after the condensed solve, the correction here acts at fixed retained displacement inside the extension that defines each cell's condensed stiffness, so it improves the assembled stiffness and not only the recovered field.

Related structure-preserving models include learned elements that condense an embedded inner domain into a symmetric positive semidefinite interface stiffness Parish et al. (2024) and convex neural energy elements, which learn boundary energies from Schur-complement targets and recover interior fields separately, by exact lifting with one interior solve per element Jiang, H. et al. (2026); here one corrected extension defines both the assembled operator and the recovered field. Component reduced-basis methods certify an offline interior basis a posteriori Huynh et al. (2013), but presuppose a common discrete space across the parameter family, which a cut removes: the active elements and the retained set change from one geometry to the next, so NICE conditions one network on the discrete geometry of each cell and provides a computable residual (Eq. (5)) rather than a certified bound. Learned multigrid prolongations Greenfeld et al. (2019), Luz et al. (2020), machine-learning-assisted adaptive FETI-DP coarse spaces Heinlein et al. (2021), networks trained through the solver Um et al. (2020), geometry-aware neural operators and graph networks Li et al. (2023a), Li et al. (2023b), Pfaff et al. (2021), multigrid-structured operators He et al. (2024), learned Green's functions Boullé & Townsend (2023) and Ritz-type training E & Yu (2018) accelerate a global solve or learn solution maps of the whole problem; here the learned map is a per-cell extension, exactly linear in the retained displacement, so that its energy defines a condensed stiffness that can

be assembled. Spectral multiscale coarse bases Efendiev et al. (2013) have served in optimisation without scale separation as the coarse space of a two-level preconditioner for the fully resolved problem Alexandersen & Lazarov (2015); in analysis, numerical shape functions on unfitted coarse elements have been conditioned against small cuts Chen & Li (2026). Reduced-order cell operators have replaced the cell-wise local solves of inexact FETI-DP for lattices built by spline composition Hirschler et al. (2024), and for unfitted two-dimensional lattices BDDC has been combined with full cell stiffness matrices whose trimmed-cell integrals are approximated by a reduced-order model based on matrix discrete empirical interpolation Bonilla Moreno et al. (2027). Lattice cells have also been condensed into super-elements whose stiffness, interpolated in a density parameter by a reduced-order surrogate, supplies explicit stiffness derivatives for optimisation without scale separation Wu et al. (2019).

The contributions of this work are threefold. (i) A construction principle for learned component models, valid for any discrete model with a symmetric positive semidefinite stiffness and a positive definite interior block: on the complete retained space of a cell, a learned linear interior extension, a fixed geometry-specific two-grid correction and the transpose of the complete extension are combined in one condensed energy, and the resulting condensed stiffness is symmetric and positive semidefinite with the rigid-body kernel, bounded below by the exact Schur complement, assemblable and differentiable, and improvable without loss of these properties by a correction that cannot increase the error when its smoothing interval contains the spectrum. (ii) An error analysis from the component to the structure and to the design: the interior error enters the compliance through a load-specific share-weighted bound and the thickness sensitivity through the stiffness derivative, with a linear term that the energy error does not control, which makes joint accuracy of compliance and local sensitivity the design criterion; and the factorisation $\varepsilon = \delta^2 \kappa$ of the energy error into a relative displacement error and a stiffness ratio shows how interior displacement errors of about one percent become energy errors of 2–35% in the cut cells examined (Section 5.4). (iii) A realisation for cut thin-walled Schwarz-P cells in a stabilised cut finite element model, with a geometry-conditioned network that handles the cut-dependent discrete space and the cut band, verified on 80 validation geometries, on two-cell assemblies of selection and held-out cells and on eight-cell lattices, with an ablation of the retained representation, a cost comparison with the whole-lattice direct solution and conventional exact con-

densation, and thickness optimisations driven by the field-based sensitivities, checked against exact condensation and compared with a homogenised model (Section 5.10).

Section 2 states the discrete cut-cell model, exact condensation and the requirements on an approximate condensed stiffness; Section 3 derives the error relations that hold for any admissible extension; Section 4 constructs NICE; Section 5 reports the numerical examples; Section 6 discusses them and states the limitations; Section 7 concludes.

2. Discrete substructures and static condensation

The error relations of Section 3 and the construction of Section 4 apply to any discrete model with a symmetric positive semidefinite stiffness whose interior block is positive definite once the retained degrees of freedom are prescribed. The cut thin-walled TPMS cell of this section is the instance on which they are realised and tested: its retained space is large, its active elements and retained set change with the cut, and in its cut configurations interior displacement errors of about one percent can become energy errors of several percent (Section 5.4).

2.1. Geometry and discrete elastic problem

We consider three-dimensional thin-walled TPMS cells in the reference box $\mathcal{B} = [0, 1]^3$. Their material domain is defined by a Schwarz-P-type trigonometric level-set field, a spatially varying band parameter, and an optional planar cut:

$$\begin{aligned}\phi(x) &= \sum_{a=1}^3 \cos(2\pi x_a), & \tau(x) &= \sum_{c \in \{0,1\}^3} N_c^{Q_1}(x) \tau_c, \\ \Omega(\eta) &= \{x \in \mathcal{B} : |\phi(x)| \leq \tau(x), \mathbf{n} \cdot x \leq b_{\text{cut}}\}.\end{aligned}\tag{1}$$

The eight parameters τ_c , interpolated by the trilinear functions $N_c^{Q_1}$, control the implicit band width and hence the material distribution; we call them corner thickness parameters (band parameters, not pointwise wall thicknesses) and derivatives with respect to them thickness sensitivities. When the eight corner values are collected in vectors, such as $\boldsymbol{\tau}$ or the sensitivity vector $\mathbf{s}$, the corners are numbered $c = 1, \dots, 8$. The cut plane $\mathbf{n} \cdot x = b_{\text{cut}}$ has unit normal $\mathbf{n}$, pointing away from the retained material, and offset b_{cut} ;

the half-space constraint is omitted for an uncut cell, and η collects geometry, material and discretisation parameters.

The elastic problem uses continuous tensor-product Q_2 displacement functions on a Cartesian background mesh. Integration over the material domain supplies the bulk stiffness, and ghost-penalty stabilisation Burman (2010) couples neighbouring active elements to control small-cut effects Burman et al. (2015). The symmetric matrix K defines the stabilised discrete energy $\frac{1}{2}u^TKu$, and the reference in this work is the equilibrium of this discrete system: all errors are measured against it, not against the continuum problem. Figure 1 shows representative geometries and Table 1 the principal settings; Appendix A gives the integration and assembly details.

2.2. Retained degrees of freedom and equilibrium extension

For each cell, the retained degrees of freedom (DOFs), collected in the set P , are the active box-face displacement DOFs and all displacement DOFs of active elements carrying a positive-area cut-surface patch. These elements, the layer of active elements intersected by the cut plane, form the cut band: although some of its nodes lie away from the cut plane, their basis functions contribute to displacement and virtual work on that plane. This full retained space, with both intercell and exterior-boundary DOFs, is used throughout prediction and assembly.

Let I denote the remaining interior DOFs, $p = |P|$, $i = |I|$, and $n_a = p + i$. The selectors J_P and J_I extract the two sets from the active displacement vector, and $q = J_P u$ is the retained displacement. Boundary loads act through q ; interior body loads are zero. In the displayed (P, I) ordering, static condensation gives

$$K = \begin{bmatrix} K_{PP} & K_{PI} \\ K_{IP} & A \end{bmatrix}, \quad A = K_{II}, \quad E = \begin{bmatrix} I_p \\ -A^{-1}K_{IP} \end{bmatrix}, \quad (2)$$

$$S = K_{PP} - K_{PI}A^{-1}K_{IP} = E^TK E.$$

We assume $K \succeq 0$ and $A \succ 0$, so prescribing q removes every interior zero-energy motion. The equilibrium displacement extension $u = Eq$ uniquely minimises the discrete energy over $J_P u = q$, and Sq is its work-conjugate retained force. For a connected free substructure, we additionally assume that K has exactly six rigid-body modes, the columns of $R \in \mathbb{R}^{n_a \times 6}$, whose restriction $R_P = J_P R$ has rank six. Geometry generation enforces this: a

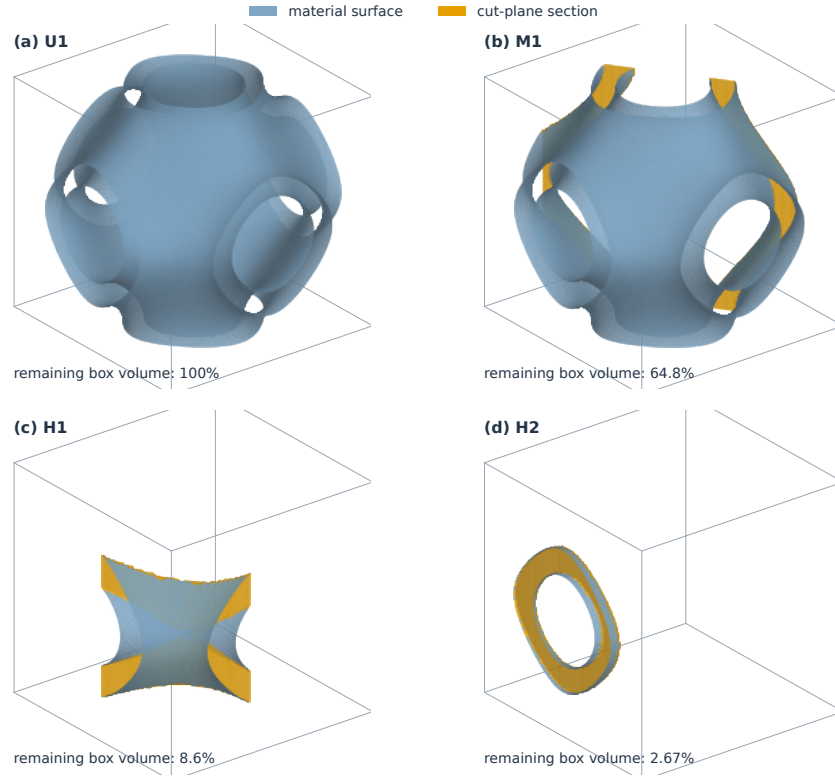


Figure 1: **Representative validation geometries.** The same unit-box scale and viewing direction are used for (a) uncut U1, (b) moderately cut M1, and (c,d) heavily cut H1 and H2. Blue denotes the material surface and orange the section on the cut plane. Percentages indicate the box volume remaining after the cut, relative to the unit box, before intersection with the thin-wall material. Surfaces are reconstructed from Eq. (1) (Supplementary Note S1).

cell is accepted only if its active background elements form a single face-connected set carrying retained box-face DOFs (Appendix A.1). This check establishes connectivity of the stabilised discrete model, in which material connected only through partially filled elements is coupled through shared basis functions and the ghost penalty.

2.3. Assembly and response measures

For substructure m , the Boolean assembly map B_m extracts $q_m = B_m U$ from the free global retained displacement U . Coincident box-face DOFs, identified by their background-grid position, are shared; retained cut-band DOFs outside the box faces remain local to their substructure. Because the thickness field on a shared face depends only on its four corner values and ϕ is periodic, the material patches of two neighbours coincide on every face not intersected by a cut; where a cut removes material from one side only, the unmatched face DOFs remain DOFs of the cell that carries them. Fixed homogeneous supports are incorporated into B_m . The reference and approximate assembled systems, which use the same maps B_m , are

$$\mathbb{K} = \sum_m B_m^T S_m B_m, \quad \hat{\mathbb{K}} = \sum_m B_m^T \hat{S}_m B_m, \quad \mathbb{K}U = f_g, \quad \hat{\mathbb{K}}\hat{U} = f_g. \quad (3)$$

Here $\hat{S}_m$ is the approximate condensed stiffness of Section 4 and f_g the prescribed force on the free retained DOFs. Local condensation commutes with assembly because the interior sets are disjoint, and the supported reference matrix $\mathbb{K}$ is assumed positive definite.

Compliance is $C = f_g^T U$, with $\hat{C} = f_g^T \hat{U}$ and relative error $e_C = |\hat{C}/C - 1|$. The local accuracy of the condensed stiffness is measured by the relative directional energy error of Section 3.1. Thickness sensitivities use the eight-component vector of Section 3.3, with relative Euclidean error $e_s = \|\tilde{\mathbf{s}} - \mathbf{s}\|_2 / \|\mathbf{s}\|_2$.

An approximate condensed stiffness $\hat{S}_m$ that is to replace S_m in Eq. (3) for analysis and design must satisfy four requirements. It must be symmetric and positive semidefinite with the retained rigid-body modes as its only null space, so that the supported assembled matrix is positive definite. It must be applicable without the interior factorisation of the exact operator, for every new geometry of a design iteration. Its error must be traceable to the assembled compliance and to the thickness sensitivity, since the two weight it differently. And it should admit improvement at deployment, for

a given geometry, without retraining. Section 3 derives what any admissible extension gives towards the first and third requirements; Section 4 constructs a condensed stiffness that meets all four.

Table 1: Discretisation and correction settings.

Quantity	Setting
Reference geometry	Unit box; Schwarz-P-type implicit band with trilinear corner parameters and an optional planar cut
Displacement approximation	Continuous tensor-product Q_2 solid elements on a Cartesian background mesh
Background resolution, validation geometries	$n = 32$ elements per axis; 65 Q_2 nodes per axis
Isotropic material, validation geometries	Normalised Young's modulus $E_Y = 1$, Poisson's ratio $\nu = 0.3$
Ghost-penalty coefficient, validation geometries	$\gamma = 10^{-4}$
Retained space	Active box-face DOFs and all DOFs of active elements carrying a positive-area cut-surface patch
Standard volume integration	4^3 initial subcells; one local refinement of partial subcells; clipped Kuhn tetrahedra with quadrature-rule parameter 4
Smoothing interval	$[b/30, b]$, with b equal to 1.05 times a 40-step power-iteration estimate of $\lambda_{\max}(D^{-1}A)$ (Section 4.2)
Principal interior coarse space	Trilinear vector functions on a 17^3 -vertex grid, restricted to the interior DOFs
Thickness-difference step	$h_c = 10^{-5}\tau_c$, with fixed active DOFs and ghost contribution
Correction of NICE	8 Chebyshev steps, $Q_1(17)$ coarse-grid Galerkin correction, 8 Chebyshev steps; same sequence in training and evaluation

Quantity	Setting
Arithmetic	Network in single precision; stiffness actions and energies in double precision; correction in double precision in the accuracy studies and in single precision in the timed route of Table 5 (Appendix F.3)

The mesh, material and stabilisation entries apply to all 80 validation geometries; lengths are relative to the unit box and the modulus is normalised. Coarse dimensions and smoothing counts of other correction sequences are reported with their comparisons.

3. Error of an approximate extension in analysis and design

With the retained DOFs fixed, the local approximation lies in the interior field. For any admissible linear extension, learned or not, we relate its departure from equilibrium to the condensed stiffness and examine how assembly and thickness differentiation weight it; the relations are stated for a single substructure and apply to every cell of an assembly. Appendix B.1 collects the assumptions. The Ritz identity (4), the residual form (5), the energy orthogonality behind Eq. (6), the residual-corrected functional underlying Eq. (8) and the self-adjoint compliance sensitivity are classical Fraeijns de Veubeke (1965), Toselli & Widlund (2005), Becker & Rannacher (2001), Haftka & Gürdal (1992), Bendsøe & Sigmund (2004); the share-weighted bounds of Eq. (7) and Appendix C.1, the sensitivity relations (9)–(10) with their bounds (Appendices H.1 and H.2) and the identity $\varepsilon = \delta^2 \kappa$ of Appendix B.3 (applied in Section 5.4) are derived here for approximate extensions.

3.1. Variational energy error

For the specified symmetric stiffness $K \succeq 0$, with $A = K_{II} \succ 0$ and zero interior body loads, Eq minimises the energy at prescribed q . Any admissible linear extension F , with $J_P F = J_P E = I_p$, differs from it only in the interior DOFs. Writing $H = J_I(F - E)$, interior equilibrium $J_I K E = 0$ eliminates the cross terms in the condensed stiffness $\hat{S} = F^T K F$ of F and gives the discrete Ritz identity

$$\boxed{\hat{S} - S = H^T A H \succeq 0.} \quad (4)$$

The learned extension therefore adds stiffness in proportion to the energy of its interior error; the absence of a first-order term follows from equilibrium of the reference field. For $u = Eq$, $\hat{u} = Fq$ and $d_I = Hq$, the interior residual $r_I = (K\hat{u})_I$ satisfies $r_I = Ad_I$. The relative directional energy error $\varepsilon(q) = q^T(\hat{S} - S)q/(q^T Sq)$, which measures the excess of the condensed strain energy over the exact one and is nonnegative by Eq. (4), can consequently be written as

$$\varepsilon(q) = \frac{d_I^T Ad_I}{q^T Sq} = \frac{r_I^T A^{-1} r_I}{q^T Sq}. \quad (5)$$

The squared A^{-1} -norm of the interior residual, available without the reference extension, is thus the stiffness-error measure; Appendix B gives the all-direction energy error ε_* (Eq. (B.5)); the largest error over a finite direction set bounds it from below, and an upper bound follows only from the coverage of the direction set (Appendix G.4).

3.2. Assembled compliance and energy share

Consider the exact equilibria of the two supported systems in Eq. (3), with the same local matrices, assembly maps and homogeneous supports, and the same nonzero retained load. Writing $u_m = E_m B_m U$, $\hat{u}_m = F_m B_m \hat{U}$ and $a(v, v) = \sum_m v_m^T K_m v_m$, global equilibrium and local energy orthogonality give (Appendix C)

$$C - \hat{C} = a(\hat{u} - u, \hat{u} - u) = \|\hat{U} - U\|_{\mathbb{K}}^2 + \sum_m \|H_m B_m \hat{U}\|_{A_m}^2. \quad (6)$$

Compliance underestimation is therefore the total reconstructed error energy, with orthogonal contributions from the changed retained solution and from the interior departure from equilibrium at that solution.

The influence of a local approximation depends on how much energy that substructure carries under the applied load. At the exact assembled traces $q_m = B_m U$, define the energy shares $w_m = q_m^T S_m q_m / C$ and $\beta = \sum_m w_m \varepsilon_m(q_m)$, with contributions of zero-energy rigid-body responses defined as zero. The energy shares w_m sum to one, and

$$0 \leq \frac{C - \hat{C}}{C} \leq \frac{\beta}{1 + \beta} \leq \beta. \quad (7)$$

A small energy share can thus attenuate a substructure's effect on compliance even when its local field remains inaccurate (Appendix C.1).

The energy relation (6) also separates the operator error from the error of an incomplete assembled solve. For an approximate solution $\bar{U}$, define the recomputed residual $\rho = f_g - \hat{\mathbb{K}}\bar{U}$ using the stated variational operator and the recovered fields $\bar{u}_m = F_m B_m \bar{U}$. Then

$$C - f_g^T \bar{U} = a(\bar{u} - u, \bar{u} - u) + \bar{U}^T \rho. \quad (8)$$

The signed residual work $\bar{U}^T \rho$ determines how an incomplete solution affects the compliance comparison. Appendix C.2 gives the residual-corrected functional and the additional consistency term required when the numerical action differs from the energy operator.

3.3. Field-based sensitivity and the complete design derivative

On a differentiable design interval with fixed active and retained DOFs, assembly maps, homogeneous supports and a design-independent load, the exact compliance sensitivity is $s_c = -u^T K_{,c} u$, where $K_{,c} = \partial K / \partial \tau_c$; contributions are summed over affected substructures. Interior equilibrium gives $S_{,c} = E^T K_{,c} E$, eliminating the design derivative of the exact extension from the force-controlled compliance derivative Giles & Pierce (2000), Haftka & Gürdal (1992).

The field-based sensitivity estimate $\tilde{s}_c = -\hat{u}^T K_{,c} \hat{u}$ uses the same stiffness derivative with the reconstructed field. Over the eight corners these form the thickness sensitivity vectors $\mathbf{s} = (s_c)_{c=1}^8$ and $\tilde{\mathbf{s}} = (\tilde{s}_c)_{c=1}^8$. At the same retained displacement,

$$\tilde{s}_c - s_c = -2d^T K_{,c} u - d^T K_{,c} d, \quad u = Eq, \quad d = Fq - Eq. \quad (9)$$

Interior equilibrium sets $(Ku)_I = 0$ but generally leaves $(K_{,c}u)_I \neq 0$, so the linear cross term survives, and decreasing energy error need not decrease sensitivity error monotonically. For corner thickening, nonnegative trilinear shape functions generate nested material domains, and with fixed basis, material and ghost contribution the exact integral gives $K_{,c} \succeq 0$ (Eq. H.10); a numerical derivative inherits this only while its quadrature preserves the nesting of the material domains (Appendix H.2). Even then the cross term can have either sign (Appendix H.1).

The derivative of the surrogate compliance also contains the design dependence of the extension. For a parameter affecting one substructure, differentiation at fixed retained DOFs gives

$$\hat{C}_{,c} = -\hat{u}^T K_{,c} \hat{u} - 2\hat{q}^T F_{,c}^T K \hat{u} = \tilde{s}_c - 2(F_{I,c} \hat{q})^T r_I, \quad \hat{u} = F \hat{q}. \quad (10)$$

Here $\hat{q}$ is the assembled retained solution obtained with $\hat{S}$ and $r_I = (KF\hat{q})_I$; affected-substructure contributions are summed. The second term couples the extension's design dependence to its interior imbalance and vanishes for an equilibrated extension. All sensitivities reported in this paper are the field-based estimates $\tilde{s}_c$, the first term: they estimate the exact sensitivity and are not the derivative of the surrogate compliance. Energy accuracy alone does not control the complete derivative (Appendix H.2, Section 6.2); Appendix H specifies the numerical stiffness derivatives.

These derivatives hold on intervals where the active elements, ghost faces, retained DOFs and the network's binary node indicators are fixed. Across such a switch the discrete reference compliance itself can jump, as it does for exact condensation on the same background mesh; because $0 \leq C - \hat{C} \leq \beta C$ at every design (Eq. 7), any jump that the surrogate compliance adds is bounded by its own error level (Supplementary Note S6.3).

Three requirements on an approximate extension follow. Its interior error energy must be small in the directions that the assembled solution selects, which Eq. (5) measures by a computable residual. Any correction applied at deployment must not increase that energy, so that Eq. (4) orders the corrected condensed stiffness between the uncorrected one and the exact Schur complement; Section 4.3 gives such a correction. And because the energy error does not control the sensitivity error of Eq. (9), the sensitivity must be verified independently of the compliance, which Sections 5.6 and 5.10 do. Section 4 constructs an extension to these requirements.

4. Neural-initialised condensation with equilibrium correction

The construction has three parts. A learned extension supplies the trial interior field (Section 4.4); the energy form, applied with the transpose of the complete extension, supplies the mechanical structure (Section 4.1); and the correction $\mathcal{W}$ supplies the improvability (Sections 4.2 and 4.3). Figure 2 shows how the three parts define both the condensed action and the displacement recovered after assembly. We present the mechanical object first, then the correction that improves it, and then the network that initialises it.

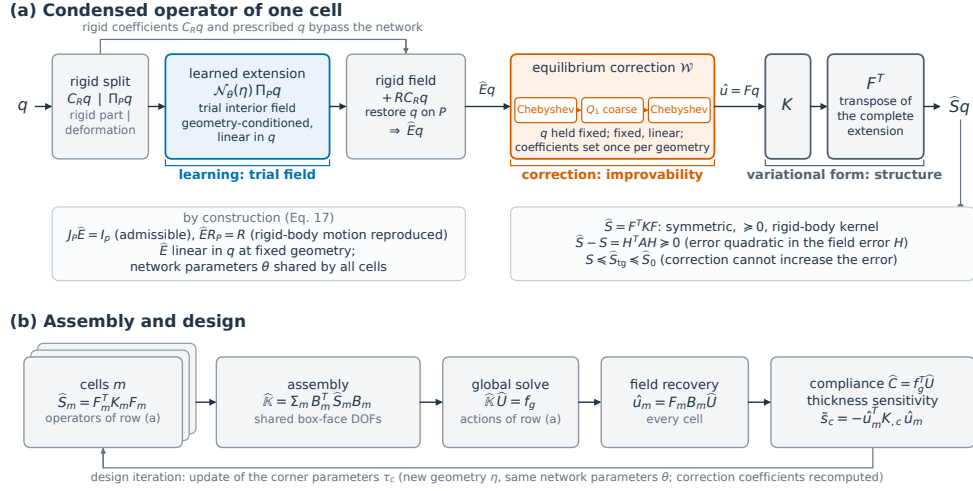


Figure 2: **Learned displacement extension, equilibrium correction and variational assembly.** (a) Rigid-body motion is separated from the retained displacement before the deformation is extended by the network. The rigid field is reconstructed and the prescribed retained values are restored; the correction $\mathcal{W}$ then reduces interior imbalance at fixed retained displacement, and applying the local stiffness and the transpose F^T of the complete extension gives the work-conjugate retained force. (b) The cell operators are assembled on the shared box-face DOFs and the assembled system is solved; the assembled solution supplies the inputs for local field recovery, the compliance and the field-based thickness sensitivity. A design iteration updates the corner thickness parameters and so regenerates every cell's geometry; the network parameters stay the same and the coefficients of $\mathcal{W}$ are recomputed. The neural extension is detailed in Figure 3 and Section 4.4; Sections 4.2 and 4.3 define the correction $\mathcal{W}$.

4.1. Variational condensed stiffness

Let $\hat{E}$ be an admissible extension, $J_P \hat{E} = I_P$, that reproduces rigid-body motion, $\hat{E} R_P = R$ for the rigid-body modes R of Section 2.2; Section 4.4 constructs it with a geometry-conditioned network (Eq. (17)). The field becomes a mechanical operator through its energy with the original stiffness. Let the interior correction $\mathcal{W}$ be one two-grid cycle (Chebyshev smoothing, coarse-grid Galerkin correction and smoothing; Sections 4.2 and 4.3) at fixed retained displacement; the method name calls it the equilibrium correction. It is linear and preserves retained values, and the identity corresponds to no correction. For each geometry, the correction schedule and its coefficients are fixed independently of the displacement query. The final extension and its condensed stiffness are

$$F = \mathcal{W} \hat{E}, \quad \hat{S} = F^T K F, \quad \hat{\mathcal{U}}(q) = \frac{1}{2} q^T \hat{S} q, \quad \hat{f}_P = \hat{S} q. \quad (11)$$

For a virtual retained displacement δq , the corresponding full virtual field is $F \delta q$, and $\delta \hat{\mathcal{U}} = (F \delta q)^T K F q = \delta q^T F^T K F q$. Applying the condensed operator therefore requires, after the stiffness action, the transpose F^T of the complete extension (rigid reconstruction, retained-value restoration and correction included). With $F_I = J_I F$, the condensed action has the block form

$$\hat{S} q = (K F q)_P + F_I^T (K F q)_I. \quad (12)$$

The second term transfers the work of the interior residual to the retained DOFs. It vanishes for an equilibrated field; otherwise it is needed for the returned force to be the derivative of the stated energy. Symmetry and positive semidefiniteness follow from $K = K^T \succeq 0$, without requiring symmetry of the learned gather, scatter or grid-transfer maps. Under the assumption of Section 2.2 on the rigid-body kernel and $F R_P = R$, the only null modes of $\hat{S}$ are the retained rigid-body modes (Appendix B); Appendix E derives the transpose F^T , and Appendix B.2 shows why it makes the operator error quadratic. By Eq. (4), $\hat{S} - S = H^T A H$ with $H = J_I (F - E)$: the operator error is the energy of the interior error of F , and Sections 4.2 and 4.3 choose $\mathcal{W}$ so that it cannot exceed that of $\hat{E}$. After an assembled solve, $F_m B_m \hat{\mathcal{U}}$ supplies the local reconstructed displacement for response and sensitivity evaluation.

4.2. Error spectrum and Chebyshev smoothing

Equations (5) and (6) suggest reducing the interior imbalance of the learned field at fixed retained motion; polynomial smoothing Adams et al. (2003) and coarse energy minimisation address different components of it.

Set $D = \text{diag}(A)$ and order the generalised eigenvectors by increasing eigenvalue, $Av_j = \lambda_j Dv_j$, with $v_j^T Dv_k = \delta_{jk}$. For $c_j = v_j^T Dd_I$,

$$d_I^T Ad_I = \sum_j \lambda_j c_j^2, \quad \chi_\ell(q) = \frac{\sum_{j=1}^\ell \lambda_j c_j^2}{d_I^T Ad_I}. \quad (13)$$

The fraction χ_ℓ locates the error energy in the first ℓ Jacobi-scaled interior modes (the exact-field fraction uses $u_I^T Au_I$).

The learned field is an initial approximation to $Au_I = -K_{IP}q$, improved by Jacobi-preconditioned Chebyshev semi-iteration Golub & Varga (1961) with q fixed; a prescribed iteration count and geometry-dependent coefficients keep the corrected extension linear. The pairing is that of smoothed-aggregation prolongation Vaněk et al. (1996) and of hybrid solvers in which relaxation removes the high-frequency error that a learned prediction leaves Zhang, E. et al. (2024).

A degree- k error polynomial maps d_I to $\Phi_k d_I$, $\Phi_k = p_k(D^{-1}A)$, multiplying each coefficient in Eq. (13) by $p_k(\lambda_j)$. The Chebyshev polynomial targets $[a, b]$, $0 < a < b$, and is nonexpansive in the A -energy norm if the actual positive spectrum of $D^{-1/2}AD^{-1/2}$ lies in $(0, b]$. Modes below a can decay slowly, which motivates a complementary coarse-grid correction. The correction sequences use $a = b/30$, with b from a power estimate increased by 5%; Section 5.2 reports its verification, and Appendix D gives the polynomial, recurrence, interval estimator and contraction bound.

4.3. Coarse-grid correction

Let $V \in \mathbb{R}^{i \times n_c}$ have full column rank, $A_c = V^T AV$, and $b_r = V^T r_I$. Minimising the energy over $\hat{u}_I + \text{range } V$, with $\hat{u}_I$ the current approximate interior field (the learned field or a smoothed iterate) and $r_I = (K\hat{u})_I$ its interior residual, gives

$$\hat{u}_I^c = \hat{u}_I - V A_c^{-1} b_r, \quad d_I^c = (I_i - V A_c^{-1} V^T A) d_I, \quad \|d_I\|_A^2 - \|d_I^c\|_A^2 = b_r^T A_c^{-1} b_r. \quad (14)$$

The coarse-grid (Galerkin) correction enforces equilibrium against the coarse variations, and the last equality measures the energy removed by that

projection Xu (1992). The principal basis consists of trilinear vector functions on a 17^3 -vertex grid restricted to interior DOFs (Appendix F); all coarse spaces preserve the full retained space.

Placing the coarse-grid correction between two k -step smoothing stages combines these corrections into one two-grid cycle, the correction $\mathcal{W}$ of Section 4.1. With $C_V = I_i - VA_c^{-1}V^TA$ and $H_{\text{net}} = J_I(\hat{E} - E)$, the two-grid error operator $\Phi_k C_V \Phi_k$ Hackbusch (1985), Trottenberg et al. (2001), Xu & Zikatanov (2002), Falgout et al. (2005) gives the resulting error

$$H_{\text{tg}} = \Phi_k C_V \Phi_k H_{\text{net}}, \quad \hat{S}_{\text{tg}} - S = H_{\text{tg}}^T A H_{\text{tg}}. \quad (15)$$

For the cycle $\mathcal{W}$ of Section 4.1, $H_{\text{tg}} = H$ and $\hat{S}_{\text{tg}} = \hat{S}$ of Eq. (11). For the exact coarse-grid correction and $\|\Phi_k\|_A \leq 1$, $S \preceq \hat{S}_{\text{tg}} \preceq \hat{S}_{\text{net}}$, $\hat{S}_{\text{net}} = \hat{E}^T K \hat{E}$: the correction cannot increase the error in the A -energy norm, and by Eq. (6) it moves exact assembled compliance toward the reference. The ordering is guaranteed; the size of the reduction is not (Appendix D), and Section 5.5 measures it. The ordering does not extend to the sensitivity error of Eq. (9) (Appendix D.1 and Supplementary Note S8), and changing the smoothing degree changes the polynomial, so the reduction need not be monotone in k .

4.4. Geometry-conditioned multilevel displacement extension

The learning target is the interior equilibrium map $q \mapsto E_I q = -A^{-1}K_{IP}q$, whose approximation provides both a displacement field and, through its energy, the condensed operator on the complete retained space. The architecture must adapt to the material and support of each cell and respect superposition at fixed geometry, which leads to a nonlinear geometry branch coupled to a linear, multilevel displacement branch (Figure 3).

Rigid-body motion is separated before learning. We take for the columns of R three translations and three rotations about a common centre. The coefficient extractor $C_R = (R_P^T R_P)^{-1} R_P^T$ and projector $\Pi_P = I_p - R_P C_R$ decompose the input into rigid coefficients $C_R q$ and retained deformation $q_d = \Pi_P q$. The network extends only q_d , so rigid reconstruction is independent of training accuracy.

In the geometry branch, element moments resolve the material distribution within each background cell, and node features identify retained and cut-band membership, weak support, local stiffness magnitude and position. Encoders form element and node feature vectors, which exchange information in two rounds; coefficient heads (small output networks) then assign weights

to prescribed element and ghost-face incidences, to transfers between latent grids and to coarse-grid convolutions. These coefficients are fixed for all displacement directions on the same geometry.

The displacement branch lifts the three components of q_d to 32 channels on retained nodes, with the interior features set to zero, defining $X^0(q)$. Local interactions propagate these values through the element and ghost-face neighbourhoods: each gathers the features on a 27-slot stencil into four weighted heads, mixes the channels within each head and scatters the result back to the incident nodes. With interior and retained masks M_I, M_P acting on nodal rows, an interaction updates the latent field by

$$X^{\ell+1} = M_I \left[X^\ell + \sum_h \mathcal{S}_{\ell h}(\eta) (\mathcal{G}_{\ell h}(\eta) X^\ell W_{\ell h}) \right] + M_P X^0(q). \quad (16)$$

Here $\mathcal{G}_{\ell h}$ and $\mathcal{S}_{\ell h}$ are geometry-weighted gather and scatter maps, $W_{\ell h}$ mixes channels, and the final term restores the retained features. A U-shaped latent hierarchy through grids with 65, 33, 17 and 9 positions per axis supplies longer-range communication, with residual convolutions on each coarser level and additive skips on the upward pass. Four local interaction pairs precede the hierarchy, four follow it, and four further pairs act on stencils containing weakly supported nodes; a linear 32-to-3 map reconstructs nodal displacement. Appendix G gives the complete order of operations and coefficient definitions.

Every displacement operation is linear, and the nonlinear encoders act only on geometry, so the raw map $\mathcal{N}_\theta(\eta)$, with network parameters θ , obeys superposition at fixed η . Adding back the rigid field and restoring the original retained values gives

$$\hat{E}q = J_P^T q + J_I^T J_I [RC_R q + \mathcal{N}_\theta(\eta) \Pi_P q]. \quad (17)$$

Consequently, $J_P \hat{E} = I_p$ and $\hat{E} R_P = R$: admissibility and rigid reproduction hold by construction.

4.5. Training directions and objective

Training probes the extension through retained displacement directions: prescribed polynomial and multiscale displacements supply broad spatial content, and responses to equilibrated nodal loads, consistent tractions, spring supports and neighbouring cells supply mechanically generated directions. Rigid components are removed and each direction is normalised

to $q_j^T S q_j = 1$ with the reference solution. For a batch of B directions, the objective is

$$\mathcal{L}(\theta) = \frac{1}{B} \sum_{j=1}^B \log(q_j^T \hat{S} q_j) + w_s \frac{1}{|\mathcal{J}_s|} \sum_{j \in \mathcal{J}_s} \frac{\|\tilde{\mathbf{s}}_j - \mathbf{s}_j\|_2^2}{\|\mathbf{s}_j\|_2^2}. \quad (18)$$

The energy term is the mean logarithm of the predicted-to-reference energy ratio. For an admissible extension, the variational identity of Section 3.1 makes its minimum correspond to the equilibrium field on each sampled direction, a Ritz principle. The sensitivity term compares the field-based thickness sensitivities (eight-component vectors) on the set $\mathcal{J}_s$ of directions with reference sensitivity labels; the reported configurations use $w_s = 1$. Gradients with respect to θ pass through the reconstructed field to the geometry encoders, coefficient heads and linear displacement maps. Directions with large energy ratios, found by a block search on the rigid complement, are added during training, and geometry augmentation uses the 48 cube symmetries (Appendix G). For NICE the objective is evaluated on the corrected extension, as described next; Table 2 identifies the variants and Table ST01 records the training and model-selection settings.

NICE is trained on the operator that is deployed: Eq. (18) is evaluated on the corrected extension $F = \mathcal{W}\hat{E}$, with the 8 / $Q_1(17)$ / 8 sequence of Section 4.3. Because $\mathcal{W}$ is linear and fixed for a given geometry, the gradient passes through the smoothing recurrence and the coarse-grid correction to the network parameters, as in solver-in-the-loop training Um et al. (2020). The smoothing interval and coarse factorisation depend only on K and are recomputed per geometry, so no trainable parameters are added.

4.6. Application of the condensed operator

Geometry-dependent quantities are prepared once per geometry and reused across retained inputs: the element moments, the network's geometry encoding, coefficients and maps, the smoothing interval and the coarse factorisation (Table ST18 gives their cost). Each application of $\hat{S}$ then requires one network pass, $2k$ smoothing steps, one coarse solve and the transposed sequence (Appendices E–F).

Algorithm 1. Corrected condensed stiffness action.

1. Extend q with $\hat{E}$ and apply the prescribed pre-smoothing, coarse-grid correction and post-smoothing sequence at fixed retained values to obtain $\hat{u} = Fq$.

2. Form $y = K\hat{u}$ and return $F^T y$: transpose the complete correction in reverse order, followed by the transpose learned extension.
3. Assemble these actions through Eq. (3); after the global solve, recover $F_m B_m \hat{U}$ for field and sensitivity evaluation.

5. Numerical examples

The examples follow the error of the learned extension from single cells, through its correction, to the assembled response and to design. Sections 5.1 and 5.2 define the geometries, variants and loads and verify the discrete reference and the deployed operator. Sections 5.3–5.5 examine the operator of a single cell: its dependence on geometry and loading, the components of its error, and what the correction removes at fixed network parameters. Sections 5.6–5.7 examine the assembled response: compliance and local sensitivity in two-cell configurations, energy share, and an ablation of the retained representation. Sections 5.8 and 5.9 assemble lattices in which every cell is learned and compare their cost with the whole-lattice direct solution and conventional exact condensation, and Section 5.10 optimises thickness fields with the operator and verifies the designs against exact condensation. All errors are measured against equilibrium of the discrete problem of Section 2.

5.1. Geometries, variants and loading conditions

The 80 validation geometries comprise 20 uncut, 20 lightly, 20 moderately and 20 heavily cut cells, with uniform, affine or mixed trilinear thickness fields and corner parameters from 0.1762 to 0.6983 (Table ST02). Each carries at most one planar cut with normal $(\cos \vartheta, \sin \vartheta, 0)$, $0 < \vartheta < \pi/4$; heavy cuts retain less than one third of the volume of the cell box, moderate cuts between one and two thirds, and light cuts more than two thirds. U, L, M and H identify the uncut, lightly, moderately and heavily cut cells used for detailed comparisons (U1, U2, L1, M1, M2 and H1–H3; Supplementary R1).

Four variants are compared (Table 2). The base network was trained without correction. Three networks continue it for 15,000 training steps: with the complete correction $\mathcal{W}$ of Sections 4.2 and 4.3 inside the training loop (NICE, Section 4.5), with eight smoothing steps and no coarse-grid correction in every training step (Smoothing-trained), or without correction (Uncorrected).

Table 2: Variants compared in the numerical examples.

Variant	Network parameters	Training set (geometries)	Correction in training	Correction at evaluation
NICE	Base network continued for 15,000 training steps	591	8 / $Q_1(17)$ / 8	8 / $Q_1(17)$ / 8
Smoothing-trained	Base network continued for 15,000 training steps	591	8 smoothing steps	8 smoothing steps
Uncorrected	Base network continued for 15,000 training steps	591	None	None
Base network	40,000 training steps, selected at 30,000	305	None	None

The base network with the correction $\mathcal{W}$ applied at deployment, without retraining, is also evaluated (Sections 5.3, 5.5 and 5.6).

All variants share the architecture of Section 4 (about 6×10^5 trainable parameters; Table ST20). The three continuations draw from the same set of 591 geometries, which contains 304 of the base network's 305 training geometries, and see the same geometries in the same order (Table ST01). On one NVIDIA GeForce RTX 5090 GPU, training the base network took 4.6 h and NICE's continuation 3.5 h, and generating the directions and exact sensitivities of the 691 training and validation geometries took about 42 GPU-hours; data generation, training and model selection together took about 60 GPU-hours (Table ST01).

Twenty of the 80 validation geometries (6 uncut, 14 cut) were used for model selection (Table ST01); statistics are therefore also given for the 60 geometries outside selection. The cells of the two-cell examples belong to

the 20 selection geometries; nine further cells outside selection are assembled in Section 5.6 as an independent check. Population statistics average the directional energy error within each geometry and loading class and weight geometries equally (Appendix A.2), so the population maximum is the largest geometry mean.

The assembly examples join a learned target to an exact, uncut neighbouring cell whose thickness is continuous across the interface (Figure 4). Each configuration has six face loads, the three Cartesian traction directions applied separately to the target and neighbour faces, and compliance and each cell's thickness sensitivity, an eight-component vector, are compared with their exact counterparts, with the nodal load fixed at its base-design value. A 3% line on compliance and on each cell's sensitivity vector is drawn as a common reference, with maxima over the six loads and, for sensitivity, over both cells. Three tractions on a target cut face are considered separately.

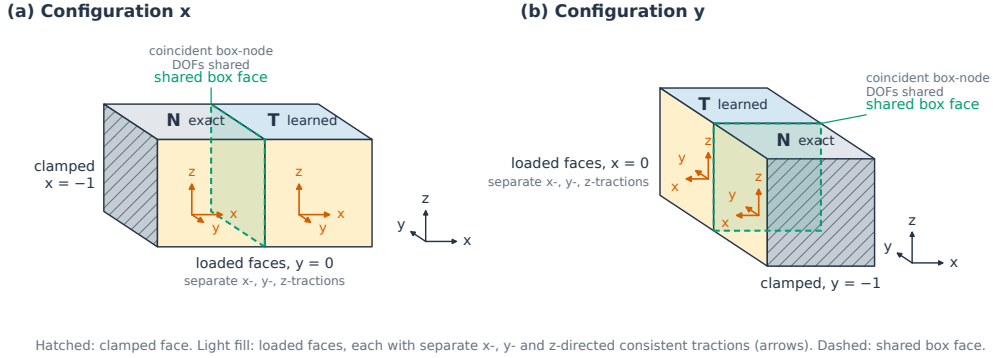


Figure 4: **Supports and loading of the two-cell examples.** (a) Configuration x: the neighbour is translated by $(-1, 0, 0)$, the face $x = -1$ is clamped, and face tractions act at $y = 0$. (b) Configuration y: the translation is $(0, -1, 0)$, the face $y = -1$ is clamped, and tractions act at $x = 0$. Each target (T) and neighbour (N) face carries separate x-, y- and z-directed consistent-traction loads. Coincident box-node DOFs are shared across the interface; non-box cut-band DOFs remain local. Cut targets also receive three cut-surface tractions, analysed separately.

5.2. Verification of the discrete reference and of the corrected operator

The $n = 32$ reference was verified by refinement on seven cells, to $n = 40$ for U1 and to $n = 48$ for M1 and M2 (Figure S01), and to $n = 64$ for H1, H2 and two further validation cells (Table ST14; Supplementary Note S2): the

compliance of uncut and moderately cut cells changes by at most 0.14% and their thickness sensitivities by at most 0.23%, those of heavily cut cells by up to 1% and 3.6%. Varying the ghost-penalty coefficient between 10^{-5} and 10^{-3} and refining the volume integration change compliance and sensitivities by at most 0.12%, and the finite-difference stiffness derivative and its step are verified in Appendix H. All comparisons that follow are made against this discrete reference (Section 2.1).

To verify the operator (arithmetic as in Table 1): on U2, M1, M2, H1 and H2, over the consistent-traction validation directions, the bilinear form $q_i^T \hat{S} q_j$ is symmetric to a relative 9×10^{-9} , the returned work $q^T \hat{S} q$ agrees with the recovered-field energy $q^T F^T K F q$ to 5×10^{-9} , the deployed operator reproduces the training-time field to 1.1×10^{-7} , and the energy of normalised rigid-body modes is at most 2×10^{-11} of a typical deformation energy; over the nodal-force directions the first, second and last of these bounds are 3×10^{-8} , 2×10^{-8} and 1×10^{-10} (Table ST04). The contraction of Section 4.2 requires $b \geq \lambda_{\max}(D^{-1}A)$: the power-iteration endpoint exceeds the converged largest eigenvalue by 2.1–5.0% on all 80 validation geometries (Appendix D), whereas the Gershgorin bound exceeds the power-iteration endpoint by factors of 4.7 to 19.4 (median 18.4). The spectrum containment thus holds on all 80 validation geometries; it was not checked for the cells of the lattices and optimisations of Sections 5.8–5.10, which use the same power-iteration estimate. Symmetry, positive semidefiniteness and $\hat{S} \succeq S$ do not depend on it.

5.3. Dependence on geometry and loading

Consistent tractions, which represent surface loads and neighbour tractions, are the principal loading class; in the five cells of Table ST04 the ghost penalty carries on average less than 0.05% of their exact field energy, and at most 0.09% in any direction. Equal nodal forces also load weakly supported nodes of small cut elements, placing on average 49–81% of the exact field energy (up to 91% in single directions) in the ghost penalty, and serve as a stress test of the stabilised problem.

NICE removes most of the geometric dependence of the uncorrected extension (Figure 5). Under consistent tractions, its mean directional energy error is 0.016%, 0.079%, 0.091% and 0.109% across the uncut, lightly, moderately and heavily cut strata (Supplementary Table ST03b; 0.074% overall, largest geometry mean 0.65%), against 1.03%, 5.33%, 7.82% and 11.2% for the Uncorrected continuation (6.33% overall, largest 42%) and 6.89% for the

base network (largest 48.6%). Over the 5,120 sampled consistent-traction directions of the 80 geometries, NICE's directional energy error has a 95th percentile of 0.33%, a 99th percentile of 0.69% and a maximum of 1.24% (0.35%, 0.73% and 1.24% on the 60 geometries outside selection).

Under the other loading classes, NICE's mean is 0.060% for neighbour-induced displacements (largest 0.38%) and 0.057% for stiffness-scaled spring supports (0.47%); for single-face nodal forces its largest geometry mean is 0.72% (Table ST03; per-geometry distributions in Figure S02). Under nodal forces, the base network's mean rises from 0.908% in uncut to 11.3% in heavily cut cells (largest 70.1%), whereas NICE's stratum means stay between 0.0135% and 0.0897% (largest 0.325%). On the 60 geometries outside selection, NICE's overall means are essentially unchanged (0.077% under consistent tractions, 0.0591% under nodal forces), but under consistent tractions its cut-stratum means are higher (0.105% moderate, 0.126% heavy) and the five largest geometry means all occur there (Supplementary Table ST03d).

The intermediate variants locate this improvement. Continuing the base network without correction changes its mean only from 6.89% to 6.33%, and Smoothing-trained reaches 1.28% (largest 7.8%). Applying the correction to the base network without retraining already gives 0.0965% (largest 0.91%), so most of the improvement comes from the correction itself; training through the correction lowers the mean by a further factor of 1.31 (95% bootstrap interval over geometries 1.20–1.40; Supplementary Table ST03).

5.4. *Components of the extension error*

The base network's error occupies different parts of the Jacobi-scaled interior spectrum in different cells (Figure 6). Under consistent tractions on M1, the lowest 200 modes contain 24.5% of the error energy but only 4.87% of the exact-field energy, and all lie below the lower endpoint $a = 0.173$ of the smoothing interval ($a = b/30$, Table 1); in H2 only 66 of the 200 modes lie below $a = 0.138$. This is consistent with the smoothing rates of Section 5.5 (Table ST05b).

Spatially (Figure 7), the exact field of M1 under one consistent-traction direction places 8% of its energy in the elements within two element widths of the cut plane, which are 13% of the elements, whereas 39–43% of the base network's error energy lies there, next to the retained cut-band DOFs whose values the extension must propagate. Relative to the base network, NICE reduces the total error energy by about two orders of magnitude (113–138

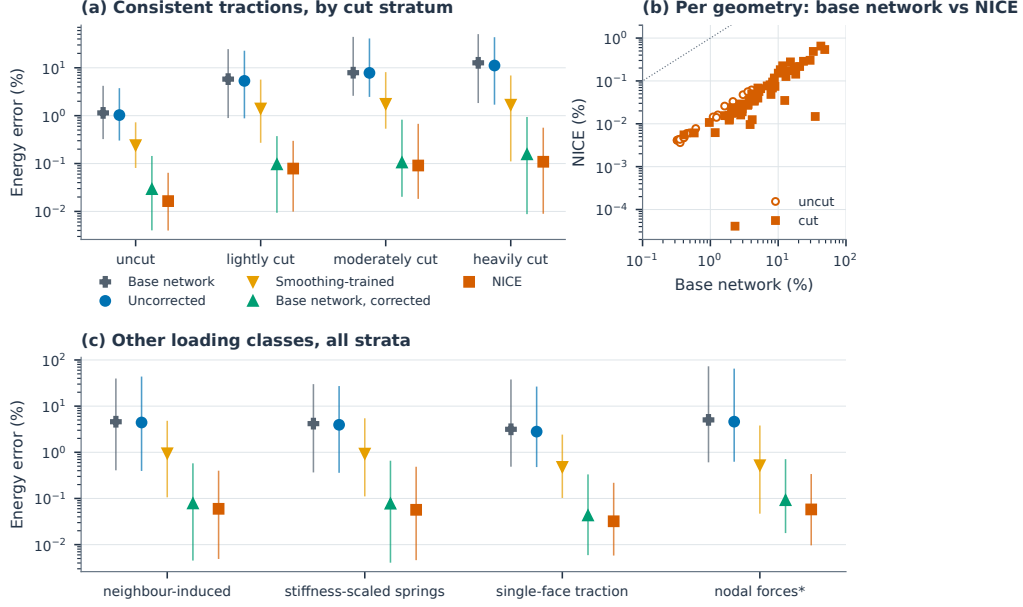


Figure 5: **Directional energy error of the learned substructures on the 80 validation geometries.** Each observation is a geometry's mean directional energy error $q^T(\hat{S} - S)q/(q^T S q)$ over the validation directions of a loading class. (a) Consistent tractions, 20 geometries per cut stratum: markers give the mean over geometries and bars the range from the 10th percentile to the maximum. (b) Geometry means of the base network and of NICE under consistent tractions, open markers for uncut and filled markers for cut geometries; the dotted line denotes equality. (c) Neighbour-induced retained displacements, stiffness-scaled spring supports, single-face consistent tractions and equal nodal forces (75 geometries for the first two classes, 80 for the others); the asterisk marks the nodal-force stress test. Twenty geometries (6 uncut, 14 cut) were used for model selection. Variants as in Table 2 and the note below it; 'Base network, corrected' is the base network with the correction applied at deployment.

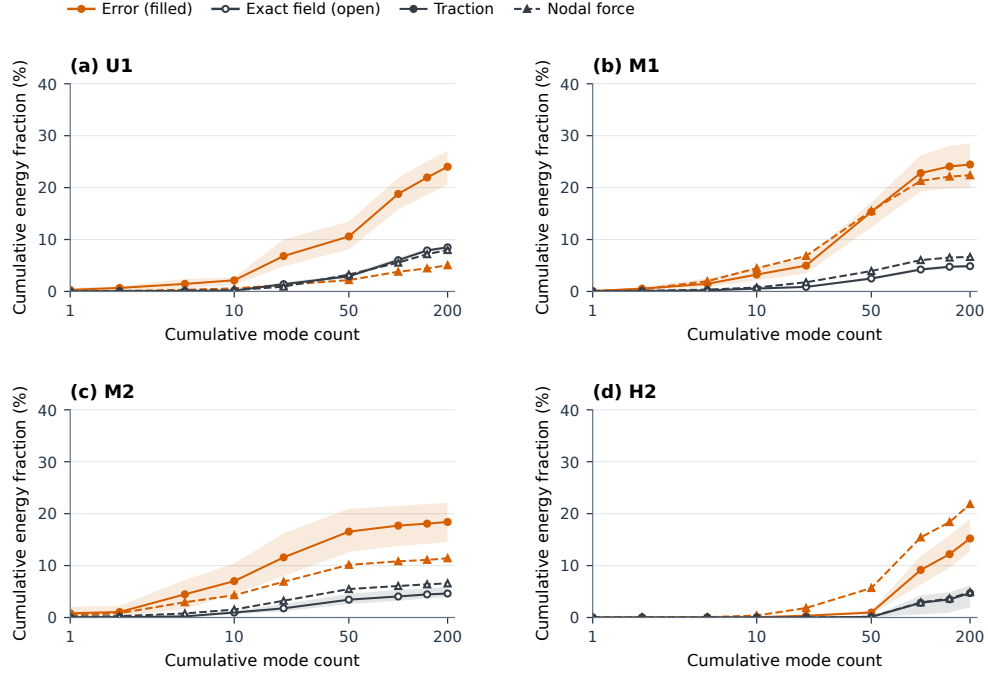


Figure 6: **Spectral distribution of the base network's extension error.** Panels show U1, M1, M2 and H2 for the base network. Modes solve $Av = \lambda Dv$, with $D = \text{diag}(A)$, ordered by increasing eigenvalue. Filled orange markers represent the extension error and open grey markers the exact interior field; solid circles correspond to consistent tractions and dashed triangles to nodal forces. Curves are directional means; bands give the 10th–90th directional percentiles under consistent tractions. Each cumulative fraction uses the total interior energy of its own field or error.

times) and the element error energies by roughly 25 to 300 times (10th–90th percentiles), and halves the share of this layer to 19–22%.

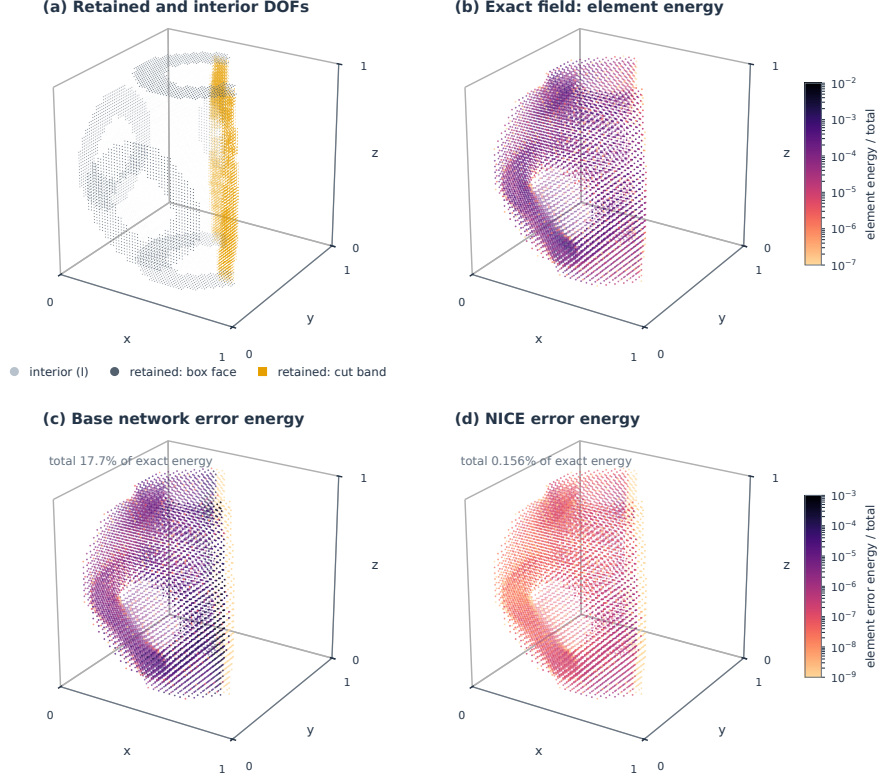


Figure 7: **Retained DOFs and spatial distribution of the extension error in M1.** (a) Retained box-face DOFs, retained cut-band DOFs and interior DOFs. (b) Bulk element energies of the exact field under one consistent-traction direction, relative to its total. (c,d) Bulk element energies of the error of the base network and of NICE for the same retained displacement, relative to the total energy of the exact field in (b), on a common colour scale. Cut-band elements carry no error because their DOFs are retained.

The size of these energy errors reflects a measurable amplification. For interior error d_I and exact interior field u_I , let $\delta^2 = d_I^T D d_I / u_I^T D u_I$ be the Jacobi-weighted relative displacement error and $\kappa = (d_I^T A d_I / d_I^T D d_I) / (u_I^T K u_I / u_I^T D u_I)$ the ratio of the Rayleigh quotients of the interior error and of the exact field (κ_W of Eq. (B.8) with $W = \text{diag}(0, D)$). Then $\varepsilon = \delta^2 \kappa$ holds exactly in every direction (Appendix B.3). Under consistent tractions, the base network's fields in M1, M2,

H1 and H2 have mean δ of 0.7–1.3% but mean κ of 124 to 7089: the displacement error is small but lies in directions far stiffer, relative to their amplitude, than the exact field, so that energy errors of 2–35% result. In the uncut U2 a similar δ of 0.74% meets $\kappa = 85$ and gives 0.40%; because δ , κ and ε are averaged over directions separately, the product of the means need not equal the mean error (0.47% against 0.40% here). NICE's fields have δ of 0.05–0.19% and κ of 33 to 582 in the cut cells (0.11% and 47 in U2).

At fixed retained displacement, M1 has mean energy and field-based sensitivity errors of 13.5% and 14.1%, and H2 of 35.0% and 75.1%; the linear term of Eq. (9) contributes only 7.2% and 0.79% of the sum of the linear- and quadratic-term norms, so the quadratic term dominates these large errors. The linear term is not negligible elsewhere: it contributes 28–43% under consistent tractions in U1, U2 and H1, whose energy errors are below 2%, and 24–72% under nodal forces in all six cells; being first order in the field error, it dominates as that error tends to zero (Table ST05, Figure S03).

5.5. Reducing interior error with fixed network parameters

Holding the base network fixed isolates the effect of the correction (Figure 8). Eight Chebyshev steps at fixed retained displacement reduce the mean energy error everywhere, but unevenly (Table ST06): under consistent tractions H2 falls from 35.0% to 0.205%, whereas M1 falls only from 13.5% to 4.68% (2.95% after 32 steps), consistent with its low-mode error below the smoothing interval; M1's sensitivity error falls from 14.1% to 2.76%, whereas U1's varies non-monotonically with the number of steps although its energy error decreases at every step (Figure 8a,b; Figure S04). A trilinear coarse-grid correction before the same eight steps brings M1 to 0.230%, and eight further pre-smoothing steps to 0.186%, with 5,601 coarse DOFs for 165,927 interior DOFs. Two, four and eight steps per stage give 1.02%, 0.422% and 0.186% on M1, and the eight-step cycle about 0.027% on M2 and U1; other coarse spaces are compared in Table ST07 and Supplementary Note S3.

Table 3 applies the same correction to four starting fields at the same retained displacements: a zero interior, a graph-harmonic extension (a volume-weighted graph Laplacian on the element connectivity, with the same exact rigid-body split), the base network's field and NICE. A zero interior leaves 28–1100% energy error and the harmonic start 0.08–17%, whereas the base network's field ends at 0.004–0.19%, lower by factors of 7 to 290 than the harmonic start and 2,400 to 12,000 than the zero interior. With 64 steps per

stage, eight times the smoothing work, the harmonic start still lies 2.6 to 50 times above the base network's eight-step result under consistent tractions in five of the six cells of Table ST08b (those of Table 3 and U1) and reaches it only in H2 (Tables ST08 and ST08b). The learned field thus supplies the part of the interior equilibrium that smoothing and the coarse space do not reach at a practical budget.

Table 3: Mean energy error (%) after the same $8 / Q_1(17) / 8$ correction applied to different starting fields (consistent tractions, 32 directions).

Cell	Base network, uncorrected	Zero interior	Harmonic	Base network, corrected	NICE
M1	13.5	1097	17.2	0.186	0.131
M2	3.64	324	6.72	0.0273	0.0358
H1	1.81	39.9	1.28	0.0131	0.0115
H2	35.0	28.3	0.0806	0.0117	0.0148
U2	0.402	47.2	1.23	0.00424	0.00465

The first column gives the base network's error without correction. Results with 32 and 64 smoothing steps per stage and under nodal forces are given in Table ST08.

5.6. Compliance, local sensitivity and energy share after assembly

Figure 9 compares the variants in the two-cell configurations of Figure 4. NICE meets both references in all fourteen configurations of the seven selection cells: its largest compliance error over the six face loads is 0.056% (M1/x), and its largest sensitivity error, taken over both cells, is 0.67% (U1/y). On nine validation cells outside model selection, chosen before evaluation as the five geometries with NICE's largest single-cell errors and one random cell in each of the uncut, lightly, moderately and heavily cut strata, the eighteen configurations give maxima of 0.27% in compliance and 1.49% in sensitivity, both on the geometry with the largest single-cell error (Supplementary Table ST09b). The five largest-error geometries account for all compliance errors above 0.02%; the four random cells stay below 0.011% and 0.22%. Assembled accuracy therefore follows the single-cell error, and remains within both references on every held-out configuration. The directions of the thickness sensitivity vectors are reproduced more closely than their

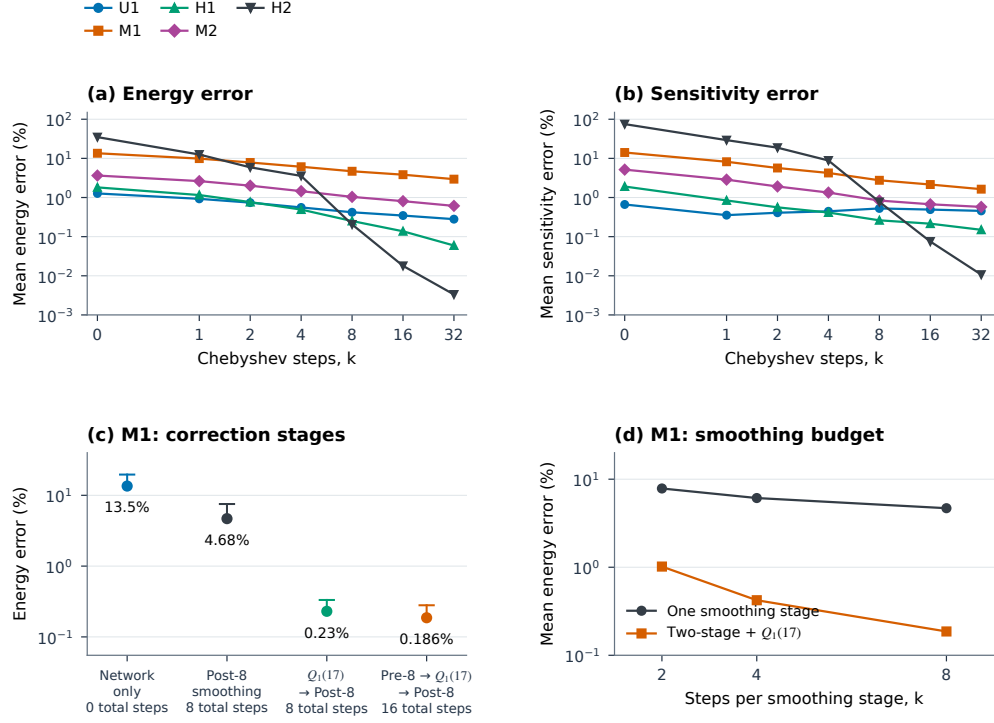


Figure 8: **Accuracy gained by correcting the base network at fixed network parameters.** (a,b) Mean directional energy error and field-based sensitivity error during Chebyshev smoothing on five cells. (c) M1 with no correction, eight smoothing steps, coarse-grid correction followed by eight steps, and eight steps on each side of the coarse-grid correction; dots show means and caps the 90th percentile. (d) M1: mean energy error against the number k of steps per smoothing stage, for one smoothing stage alone and for k pre-smoothing steps, the $Q_1(17)$ coarse-grid correction and k post-smoothing steps. All panels use consistent tractions, fixed retained displacements and $a = b/30$; in (a,b) the step axis is linear from zero to one and logarithmic thereafter.

magnitudes: in the fourteen configurations of the selection cells, NICE's vectors have cosines of at least 0.999998 with the exact ones (Supplementary Note S6.3).

The Uncorrected continuation fails the sensitivity reference in four of the eleven configurations on which it was evaluated (U1/x, U1/y, M1/x, M1/y, up to 11.4%; Table ST09) and, on M1, also the compliance reference (up to 4.1%). Smoothing-trained reduces these errors but still fails the same four configurations (3.15–4.43%): smoothing without the coarse-grid correction does not reach the slowly damped error components that dominate the sensitivity of U1 and M1. The base network with the correction applied at deployment also meets both references (largest sensitivity error 0.95%, U1/x), so the correction is what brings the assembled responses within the reference. Under the cut-face tractions, NICE reaches at most 0.10% in compliance (M1/y) and 0.16% in sensitivity, whereas the Uncorrected continuation exceeds the reference in four of its seven configurations, up to 14.1% in sensitivity and 6.9% in compliance on M1/y, among them M2/x (3.3%) and L1/y (3.7%), which meet it under the face loads (Table ST10).

Table 4 shows why both quantities are needed. Without the complete correction, an accurate compliance does not guarantee an accurate local sensitivity: under a neighbour-face load on U1/x, the Smoothing-trained compliance error is 0.00109% while the target-cell sensitivity error is 3.15%, with the target carrying 0.13% of the exact assembled energy. NICE reduces both errors: the target-cell sensitivity error of that load falls to 0.598%, and on M1/x under the target-face y load NICE gives 0.0481% and 0.145% instead of the Uncorrected 3.44% and 11.4%. The separation remains visible in NICE, whose sensitivity error exceeds its compliance error by a factor of about two on H1 and by four orders of magnitude for the U1 target, whose energy share is small, but both lie well within the reference.

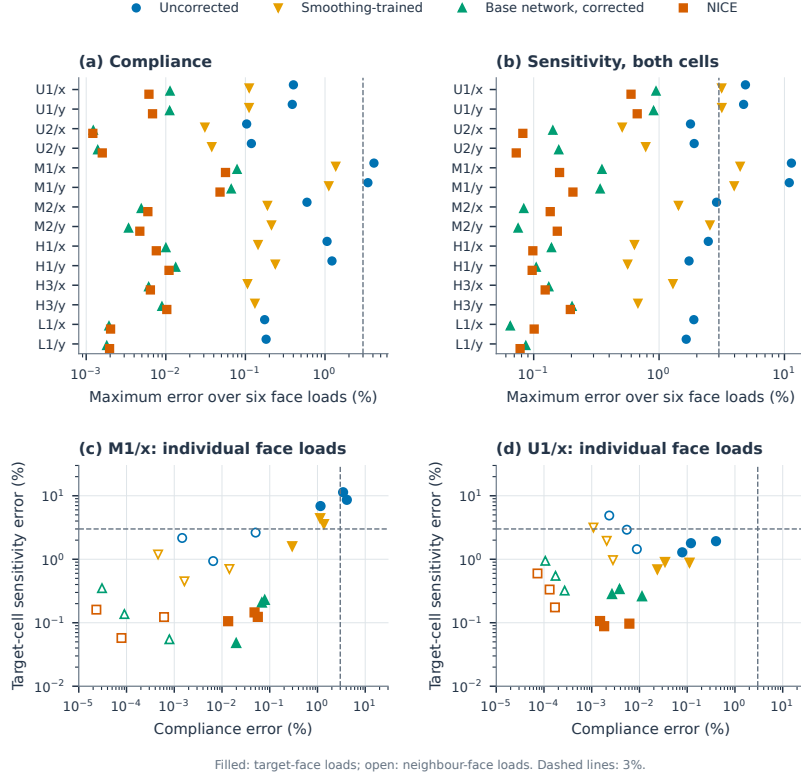


Figure 9: Compliance and thickness sensitivity in assembled cell pairs. The target cell is represented by a learned substructure and the neighbour by exact condensation. (a,b) Maximum errors over the six face loads in each configuration; the sensitivity error is also maximised over both cells. (c,d) Compliance and target-cell sensitivity errors of the individual face loads on M1/x and U1/x; filled markers denote target-face loads and open markers neighbour-face loads. Dashed lines mark the 3% reference. The base network is not shown, and not every variant was evaluated on every configuration; all results are in Table ST09. Variants as in Table 2; 'Base network, corrected' is the base network with the correction applied at deployment, without retraining.

Table 4: Compliance and target-cell sensitivity under individual face loads.

Target / configura- tion	Load	Uncorrected: compliance / sensitivity (%)	Smoothing- trained: compliance / sensitivity (%)	NICE: compliance / sensitivity (%)	Target energy share
H1/x	T-x	1.06 / 2.47	0.144 / 0.636	0.0076 / 0.0133	0.643
U1/x	N-z	0.0023 / 4.89	0.00109 / 3.15	7.3×10^{-5} / 0.598	0.0013
M1/x	T-y	3.44 / 11.4	1.14 / 4.43	0.0481 / 0.145	0.311

T and N identify the loaded face of the target or neighbour; x, y and z give the traction direction. The last column gives the target cell's share of the exact assembled energy. Table ST12 lists all six face loads of H1/x, U1/x and M1/x for Smoothing-trained.

The target's energy share (Table 4) explains how a local stiffness error can have little effect on compliance. Under the neighbour-face z load in U1/x, the base network's target cell carries 0.127% of the exact assembled energy and has a local energy error of 2.31%; their product $\beta = w\varepsilon$ gives a relative compliance bound of 0.00295%, close to the observed 0.00283%, whereas the target-cell sensitivity error is 5.83% (Supplementary Table ST13). Figure 10 shows this weighting across variants and loads. For the sensitivity at fixed retained displacement, Appendix H.1 bounds the error by the interior error energy with a linear and a quadratic term; the relative error is further divided by the local reference, which is small for a cell that carries little of the assembled energy.

5.7. Ablation: restricting the retained box-face representation

To isolate the role of the complete retained space, pairs in configuration x are solved with exact cell operators while the box-face displacements of each cell are restricted to tensor Bernstein polynomials of degree r ; the non-box cut-band DOFs remain unrestricted. The ablation restricts only the retained representation of the present cells at fixed cell size; reduced-boundary learned substructures control the resulting error by partition refinement, boundary enrichment or oversampling Huang et al. (2024), Guo et al. (2026a), Guo

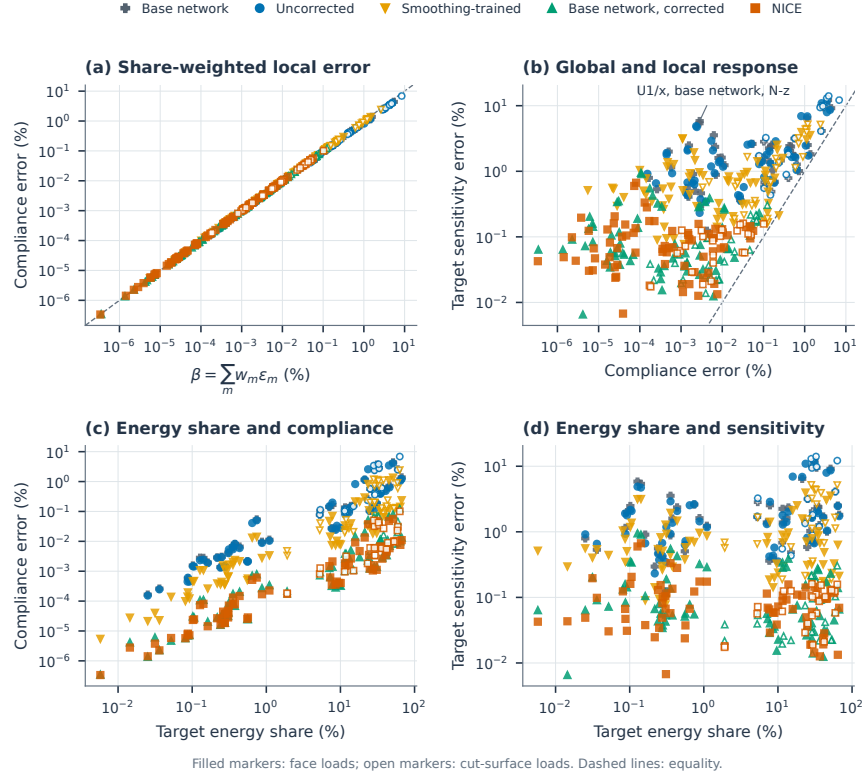


Figure 10: **Share-weighted compliance error and local sensitivity.** Each point is one load of one variant–configuration combination of Table ST09 other than L1. Filled markers denote face loads and open markers cut-surface loads. (a) Compliance error against $\beta = \sum_m w_m \varepsilon_m$, with the energy shares $w_m = q_m^T S_m q_m / C$ and energy errors $\varepsilon_m = q_m^T (\hat{S}_m - S_m) q_m / (q_m^T S_m q_m)$, evaluated at the exact assembled retained displacement; only the learned target contributes to β . (b) Compliance and target-cell sensitivity errors under the same loads; the annotation identifies the base network on U1/x under the neighbour-z load. Dashed lines denote equality. (c,d) The two response errors versus the target's exact energy share. Variants as in Table 2 and Figure 9.

et al. (2026b) and are not reproduced here (Supplementary Note S4). With every box face restricted, degree one gives compliance errors of 78–85% on U1, M1, M2 and H1, decreasing to about 4% at $r = 5$ and 0.49–0.74% at $r = 8$, where the H1 pair has 14,001 retained DOFs instead of 32,991 (Figure 11). The local sensitivity converges more slowly: at $r = 8$ the target-cell sensitivity error is 1.3–4.5% under target-face loads and 24–64% when neighbour-face loads are included. Restricting only the shared interface removes most of the compliance error (0.17–0.44% at $r = 3$ on U1, M1 and M2), but the sensitivity error under neighbour loads remains 8–21% at $r = 3$ and 2.2–3.4% at $r = 5$. A restricted interface is thus adequate for compliance at low degree but not for local sensitivity under neighbour loads; keeping the complete retained space removes this component of the error, at the cost of the larger coarse model (Table ST16).

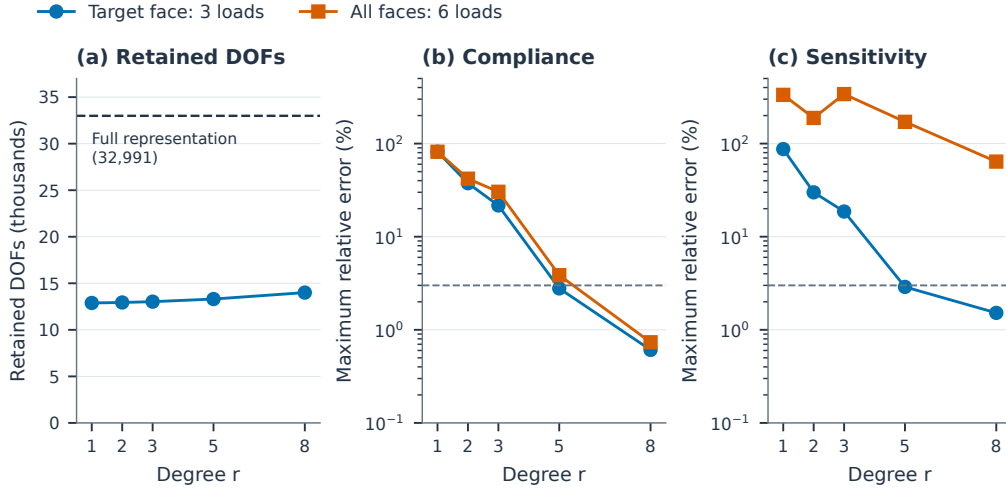


Figure 11: **Response errors caused by restricting box-face displacements.** Both cells of H1/x use exact operators and Bernstein degree r on every box face, with unrestricted non-box cut-band DOFs. (a) Number of retained DOFs; the dashed line denotes the 32,991 DOFs of the full representation. (b,c) Maximum compliance and target-cell sensitivity errors over the three target-face loads or all six target- and neighbour-face loads; cut-surface tractions are excluded. Errors are relative to the full retained-space solution; horizontal lines mark 3%.

5.8. *Heterogeneous lattices with every cell learned*

In a design every cell is learned, and the errors of neighbouring cells enter the same assembled solution. Two lattices with a continuous graded thickness field and a planar boundary cut test this: a $2 \times 2 \times 2$ block of eight distinct cells, four of them cut with retained volumes of 62% and 25%, and a $3 \times 3 \times 1$ layer of eight cells, three of them cut, with one corner position left empty. Neighbouring cells share their face corners, the corner parameters of the block range from 0.25 to 0.54, and the sixteen cells were generated for these examples and entered neither training nor model selection. The lattices are clamped on one face and loaded by unit consistent tractions on the opposite face, held fixed at the evaluated design for the sensitivities (Appendix H), and by three random load vectors. Every cell uses NICE; the reference assembles the dense exact condensed matrices and solves to a recomputed relative residual of at most 1.3×10^{-10} . The learned lattices are solved by preconditioned conjugate gradients on the free retained DOFs with a balanced two-level preconditioner: the fine action is the inverse of the assembled retained stiffness block $\mathbb{K}_{PP} = \sum_m B_m^T K_{PP,m} B_m$, factorised once per design iteration, and the coarse space consists of trilinear lattice-vertex functions multiplied by the six rigid-body modes (Supplementary Note S6.1).

In the $2 \times 2 \times 2$ block, the lattice compliance errors under the three face loads are 0.014%, 0.011% and 0.0094% (at most 0.069% under the random loads), the largest thickness-sensitivity error over all cells is 0.14% (0.45% under the random loads), and the assembled retained solution differs from the exact one by 0.13% (relative Euclidean norm over all six loads). The $3 \times 3 \times 1$ layer gives 0.015%, 0.013% and 0.010% (at most 0.056%), 0.14% (0.43%) and 0.12%. These errors are below the largest two-cell errors: the cells' energy errors combine through their shares of the assembled energy rather than accumulate. By Eq. (7), the lattice compliance error is bounded by the share-weighted sum of the cells' energy errors at the exact traces, which range from 0.006% to 0.026% in the block and 0.005% to 0.052% in the layer under the face loads (0.02% to 0.12% under the random loads), largest in the cut cells. Their weighted sums exceed the lattice compliance errors by less than 0.4% of their value under the face loads and by about 4-5% under the random loads, because the assembled solution adapts to the stiffer learned cells whereas β evaluates the errors at the exact traces.

The learned solves reach the prescribed recursive residual, while the recomputed residual $\|f_g - \mathbb{K}\bar{U}\|/\|f_g\|$ stagnates at 3.6×10^{-4} (block) and 3.2×10^{-3} (layer), consistent with rounding in the single-precision network

(repeating the block solve with the correction in single precision changes it by less than 1%). The residual work $\bar{U}^T \rho$ at the final iterate is at most 2.5×10^{-8} of the compliance, so by Eq. (8) the errors above are those of the operator (Supplementary Note S6.3, Table ST19). Aggregated over the shared corner parameters, the field-based sensitivities give the lattice gradient to 0.04–0.09% under the face loads and 0.22–0.28% under the random loads, with cosines above 0.999999.

5.9. Computational cost of a lattice design iteration

Cost is compared for one design iteration of the lattices of Section 5.8: the two eight-cell lattices and, as four-cell lattices, the two $2 \times 2 \times 1$ layers of the block, each with two uncut cells and two cut cells of 62% and 25% retained volume, with the same retained DOFs, clamp and loads. Table 5 compares three routes.

Route (a), the whole-lattice direct solution, assembles the full cut finite element model of the lattice, retained and interior degrees of freedom of every cell, and factorises it once by a sparse Cholesky factorisation (MKL PARDISO with 16 threads on one AMD EPYC 9654 CPU); its recomputed relative residuals are below 3×10^{-9} , so for the four-cell lattices it is also the exact reference.

Route (b), conventional exact condensation, condenses every cell on the same CPU by a Cholesky factorisation of its interior with PARDISO's Schur-complement option and solves the assembled condensed system by a block Cholesky factorisation that eliminates each cell's private retained DOFs before the shared interface (16 threads); its compliance agrees with (a) to 7×10^{-10} .

Route (c), NICE on the GPU, prepares the learned substructure of every cell and solves the assembled system by the preconditioned conjugate gradients of Section 5.8 to a relative residual of 10^{-6} , with the correction in single precision, including the field-based sensitivities.

All routes include cell setup and stiffness assembly, and none includes the one-off cost of data generation and training (Section 5.1).

Table 5: Cost of one design iteration of the lattices. (a) Whole-lattice direct solution by sparse Cholesky factorisation (MKL PARDISO, 16 threads on the CPU): time from cell setup to the solution of six loads (three consistent, three random); peak process memory. (b) Every cell condensed with PARDISO's Schur-complement option, then the condensed lattice system solved by block Cholesky factorisation, 16 threads: total time, peak process memory. (c) Cell preparation, preconditioner setup, conjugate gradients for the three consistent loads to 10^{-6} and field-based sensitivities; peak GPU memory, CPU memory after cell preparation; compliance error against (a) for four cells and against the exact condensation of Section 5.8 for eight cells. Memory in GiB. All phases: Supplementary Note S5 and Table ST17.

Lattice (cut / cells)	DOFs: total / free retained	(a) Whole-lattice direct, 16 threads: time (s) / memory (GiB)	(b) Con- ventional exact condensa- tion, 16 threads: time (s) / memory (GiB)	(c) NICE (one GPU): time (s) / iterations	(c) NICE memory (GiB): GPU / CPU	(c) NICE compli- ance error (%)
$2 \times 2 \times 1$, $z = 0$ (2 / 4)	957,888 / 77,310	484 / 35.1	589 / 27.9	38.6 / 114	6.3 / 2.4	0.015, 0.012, 0.010
$2 \times 2 \times 1$, $z = 1$ (2 / 4)	884,940 / 71,046	349 / 32.0	632 / 24.7	37.7 / 119	6.1 / 2.4	0.022, 0.016, 0.014
$2 \times 2 \times 2$ (4 / 8)	1,833,474 / 139,002	868 / 71.1	950 / 40.1	81.2 / 129	8.5 / 3.9	0.014, 0.011, 0.0097

Lattice (cut / cells)	DOFs: total / free retained	(a)	(b) Con- ventional exact condensa- tion, 16 threads:	(c) NICE (one GPU): time (s) / iterations	(c) NICE memory (GiB): GPU / CPU	(c) NICE compli- ance error (%)
		Whole-lattice direct, 16 threads: time (s) / memory (GiB)	time (s) / memory (GiB)			
$3 \times 3 \times 1$ (3 / 8)	2,113,611 / 143,685	1,042 / 85.9	1,218 / 42.2	110.1 / 165	9.2 / 4.7	0.015, 0.014, 0.015

For the four-cell lattices, a NICE design iteration takes 38 to 39 s, against 349 to 484 s for the whole-lattice direct solution with 16 threads and 589 to 632 s for conventional exact condensation, and the learned compliance agrees with the direct solution within 0.022%, below the reference in every case. For the eight-cell lattices, NICE takes 81 and 110 s against 868 and 1,042 s for the direct solution, about ten times longer, and 950 and 1,218 s for conventional condensation. Cell setup and stiffness assembly run on the CPU in route (a) and on the GPU in route (c) and take 41 to 53% of the direct solution's time; without them, the direct solution takes 165 to 610 s and NICE, without its whole cell preparation, which also contains the network encoding and the correction setup, 33 to 99 s, still five to eight times less. For the eight-cell lattices, the timed errors include the algebraic error of the 10^{-6} solve (Eq. 8) and exceed the operator errors of Section 5.8 by up to 0.005 percentage points. The Cholesky factor grows slightly faster than the number of degrees of freedom, from 29 to 32 GiB for four cells to 66 and 81 GiB for eight, and the process peak of the direct solution reaches 71 and 86 GiB; conventional condensation needs 40 and 42 GiB, most of it for the condensed cell matrices held until their elimination (Supplementary Table ST17). NICE needs at most 9.2 GiB of GPU and 4.7 GiB of CPU memory. Per cell, NICE's stored state takes 0.25 to 1.37 GiB, against 0.60 to 9.4 GiB for the interior Cholesky factor of conventional condensation (Supplementary Note S5, Table ST18).

A design iteration changes the geometry of every cell, so every cell's preparation, interior factorisation or network encoding and correction setup, is repeated; the assembled solve can start from the previous solution, and the conventional routes can reuse their symbolic factorisation while the active-element pattern is unchanged.

5.10. Thickness design optimisation

The corner thickness parameters at the lattice vertices, shared through $\boldsymbol{\tau}_m = \boldsymbol{\tau}_m(\boldsymbol{\tau}_g)$, are optimised for minimum compliance under one unit consistent face traction subject to $V \leq V^* = 0.8 V(\boldsymbol{\tau}^0)$, with V the material volume of the discrete model. The parameters at the vertices in the plane of the loaded face (9 of 27 in case A, 10 of 74 in the plates) are fixed, which keeps the nodal load independent of the design and removes the load-derivative term $2f_{g,c}^T \hat{U}$ of Appendix H. The optimiser receives the field-based estimates of Eq. (9) summed over the cells at each vertex, $\sum_m (\partial \boldsymbol{\tau}_m / \partial \boldsymbol{\tau}_g)^T \tilde{\mathbf{s}}_m$. The bounds $0.18 \leq \tau \leq 0.69$ and limits of 0.45 on each cell's corner span and gradient norm keep every cell inside the trained domain $[0.175, 0.699]$, at most

0.47; Section 6.4). Each design iteration regenerates every cell's geometry and learned substructure, solves the lattice by the preconditioned conjugate gradients of Section 5.8 from the previous solution and takes one step of the method of moving asymptotes Svanberg (1987), with a move limit of 5% of the parameter range and no line search (Table ST21).

Case A, the $2 \times 2 \times 2$ block of Section 5.8 loaded normal to the face opposite the clamp, is optimised from its graded design with NICE and, as a twin, with exact condensation (Table 6, Figure 12a). NICE stops after 24 design iterations and exact condensation after 23, both when the relative change of the objective has stayed below 10^{-4} in three consecutive iterations (Table ST21); no KKT residual is used for stopping, since the discrete model, and with it the objective, changes between design iterations (Section 3.3, Supplementary Note S9.1). Both lower the compliance by 2.0% with 20% less material; their final corner parameters differ by at most 0.0051, and the exact compliances of their final designs by 1.25×10^{-5} of their value. At iterations 0, 12 and 23, the NICE compliance lies 0.011%, 0.018% and 0.028% below the exact one, the sign given by Eq. (7), and the lattice gradient has errors of 0.069%, 0.17% and 0.33%, cosines of at least 0.999996, 95th-percentile per-variable errors of at most 0.28% of the largest component and the exact sign in every component (Table ST22). Both errors grow along the path as the design approaches the bounds, yet of the two final designs the NICE design has the lower exact compliance.

Plates B1 and B2 are a single layer of 8×4 cells from which a planar cut removes eight, leaving 16 uncut and eight cut cells (Table ST23). With the in-plane axes interchanged, the cut normal ($\vartheta = 33.7^\circ$) lies in the orientation in which NICE was mainly evaluated ($0 < \vartheta < \pi/4$, Sections 5.1 and 6.4). Clamped along one long edge and started from the uniform $\tau = 0.40$, the plate is loaded on the opposite side face, which the cut leaves two cells wide, in plane along the long edge (B1) or out of plane (B2, bending). NICE lowers the compliance by 7.7% in 23 design iterations (B1) and 8.3% in 30 (B2), both final designs reaching the bounds and the span and gradient limits. Checked with exact condensation (Table ST27), the NICE compliance lies 0.013% and 0.021% below the exact one at the uniform start of B1 and B2 and 0.023% and 0.037% below it at their final designs; there the lattice gradient has errors of 0.22% and 0.32%, cosines of at least 0.999995, 95th-percentile per-variable errors of 0.26% of the largest component and the exact sign in all 64 components.

A homogenised model, as in homogenisation-based graded design Li et al.

(2018), with the tensor of the uniform-thickness cell from periodic homogenisation on the same discrete model (Table ST23), is optimised with the same variables, bounds and limits, volume fraction 0.8 and optimiser; its volume, from the homogenised material volume fraction $V^H(\tau)$, equals the discrete material volume at the uniform start to 1.2×10^{-7} . It underestimates the exact compliance of the initial design by 26.7% under the in-plane load and 36.6% in bending, consistent with the single layer offering no separation of scales through its thickness.

Evaluated with NICE, the homogenisation designs Hom- y and Hom- z have 1.1% higher compliance than B1 and 0.34% lower than B2, at volumes within 0.05% of V^* . Continuing from the homogenisation designs with NICE lowers their compliance by a further 1.0% (X- y) and 0.53% (X- z) in 12 design iterations, so the optimisations are local; X- z ends 0.87% below B2 and X- y 0.060% above B1; since the NICE compliance does not exceed the exact one (Eq. (7)), the exact compliance of X- y is at least 87.14, above the exact 87.109 of B1, which therefore remains the better in-plane design (Figure 12b, Figure 13a,b; Table ST23).

The discrete model switched (Section 3.3) between every pair of consecutive design iterations (Table ST24), and the residual work of Eq. (8) stayed below 3.1×10^{-7} of the compliance. In five design iterations over all runs the geometry generator required a perturbation of at most 5.8×10^{-4} in a corner parameter (Supplementary Note S9.1).

On one GPU, the design iteration was timed on plates with the proportions, cut and in-plane load of B1 and 24, 51, 88 and 110 cells (8 to 18 cut cells; 6.5 to 32.7 million degrees of freedom of the cut finite-element model, condensed to 0.39 to 1.62 million free retained DOFs), each over at most its first four design iterations, with the learned substructures beyond 4 GiB of GPU memory streamed from CPU memory (Figure 13c, Table ST26). A design iteration took 9, 20, 35 and 43 min, about in proportion to the number of cells (22 to 24 s per cell), with 138 to 174 conjugate-gradient iterations and at most 78 GiB of CPU memory. At 135 cells the factorisation of $\mathbb{K}_{PP}$ on the GPU failed for lack of device memory. At 43 min per design iteration, the 23 to 30 iterations of B1 and B2 would take about 17 to 22 h for the 110-cell plate.

Table 6: Thickness optimisation cases. Compliance under a unit consistent face traction. Start: initial design, whose volume exceeds V^* by 25% in A, B1 and B2; final: last design iteration. Compliances are NICE values except where marked. Exact verification: exact discrete compliance of the NICE designs, with the amount by which the NICE compliance lies below it in parentheses. Time: mean over the design iterations, including geometry generation for every cell. Times are not comparable with Table 5: the optimisation runs regenerate every cell's geometry, integrate the moment derivatives without the fused kernels of the timed route and solve for one load (Supplementary Note S9, Table ST22c). Hom: homogenised macroscale model (4,464 trilinear elements). Details: Supplementary Note S9 and Tables ST21–ST27.

Case	Cells (cut)	Load	Design iterations	Start compliance	Final compliance	Exact verification	Time per design iteration (s)
A, NICE	8 (4)	Normal traction on the face opposite the clamp	24	24.5908	24.1038	24.5936 (0.011%), 26.0063 (0.018%), 24.1105 (0.028%) at iterations 0, 12, 23	141
A, exact condensation	8 (4)	As A	23	24.5936 ^a	24.1108 ^a	Exact throughout; final design 0.0013% above the NICE design	469
B1	24 (8)	Face opposite the clamp, in plane along the long edge	23	94.34	87.09	94.353 (0.013%), 87.109 (0.023%) at iterations 0, 22	574
B2	24 (8)	Face opposite the clamp, out of plane (bending)	30	1,467.7	1,346.4	1,468.05 (0.021%), 1,346.85 (0.037%) at iterations 0, 29	639
Hom- y	Macroscale	As B1	23	69.18 ^b (NICE 94.34)	66.57 ^b (NICE 88.02)	Not checked	—
Hom- z	Macroscale	As B2	27	930.7 ^b (NICE 1,467.7)	859.9 ^b (NICE 1,341.8)	Final design: 1,342.30 (0.038%)	—
X- y : NICE from Hom- y	24 (8)	As B1	12	88.02	87.14	Not checked	545
X- z : NICE from Hom- z	24 (8)	As B2	12	1,341.8	1,334.7	1,335.20 (0.038%) at iteration 11	643
Scale: plates of 24, 51, 88, 110 cells	24–110 (8–18)	As B1	Up to 4 timed	94.34, 89.12, 86.75, 85.38	—	Not checked	534; 1,215; 2,126; 2,606

^a Exact compliance. ^b Homogenised macroscale model; in parentheses the NICE compliance of the same design on the cut geometry.

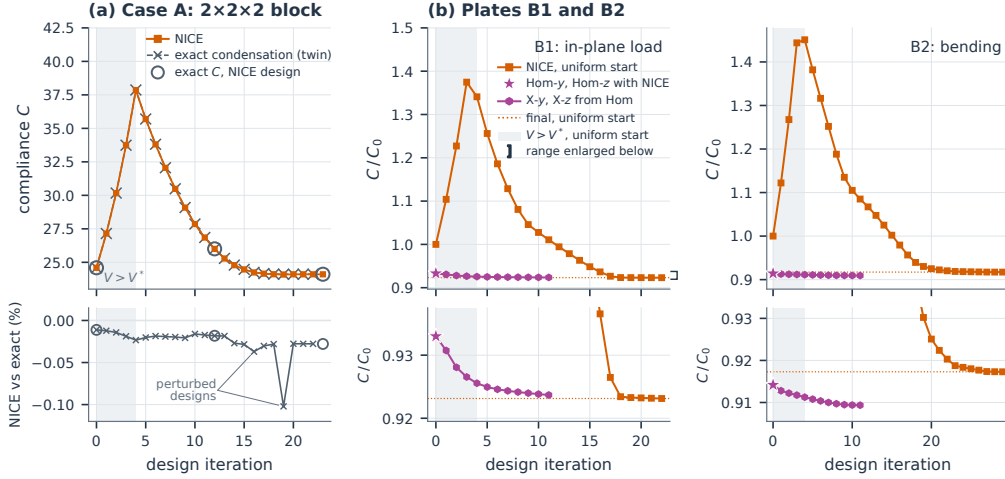


Figure 12: **Thickness optimisation with NICE.** (a) Case A: compliance histories of the NICE optimisation and of the twin optimisation with exact condensation, with the exact compliance of the NICE designs at iterations 0, 12 and 23 (open circles); the compliance first rises while the volume is reduced from $1.25V^*$ to V^* (shading, iterations 0–3). Lower strip: NICE compliance relative to the twin run at the same design iteration (line; the designs of the two runs coincide at iterations 0–4 and differ afterwards) and to the exact compliance of the same design (circles: -0.011% , -0.018% , -0.028%); at iterations 16 and 19 (labelled 'perturbed designs') the geometry-generation fallback had perturbed the NICE design (Supplementary Note S9.4, Table ST24a). (b) Plates B1 (in-plane load, left) and B2 (bending, right): compliance divided by C_0 , the initial NICE compliance of B1 or B2 at the uniform start (94.34 and 1,467.7). The homogenisation designs Hom- y and Hom- z evaluated with NICE (stars) and the NICE continuations X- y and X- z started from them are divided by the same C_0 and drawn over their own design iterations. Shading: design iterations with $V > V^*$ of the runs from the uniform start; the continuations start within 0.05% of V^* . Lower strips: the range marked by the bracket, enlarged.

6. Discussion

6.1. What is retained, what is learned and what is corrected

Fixing the retained DOFs separates two decisions that a reduced substructure model otherwise combines: which displacement patterns neighbouring cells may exchange, and how accurately the interior responds to each of them. Learning and correction alter only the second. The ablation of Section 5.7

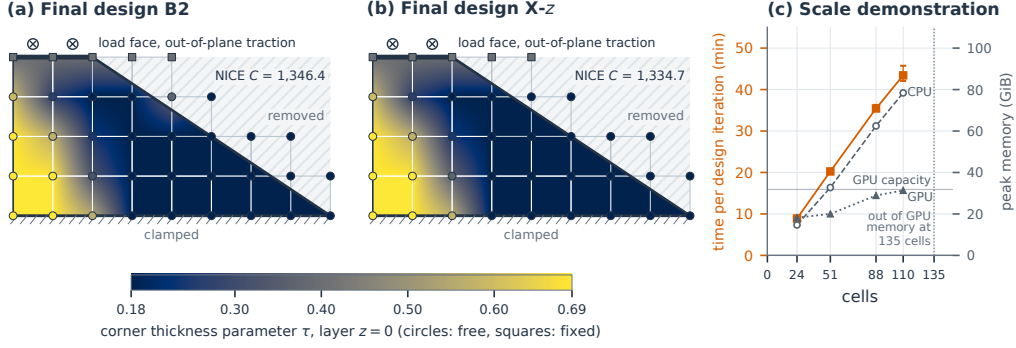


Figure 13: **Final plate designs and scale demonstration.** (a,b) Corner thickness parameters of the final designs B2 and X-z, drawn with the long side horizontal, layer $z = 0$; circles: design variables; squares: vertices in the plane of the loaded face, held fixed; vertices drawn in the removed region are corners of cut cells. $\otimes$: load face, traction normal to the plate. (c) Scale demonstration on plates with the geometry and load of B1: time per design iteration (mean; bars: range over the timed iterations), peak CPU memory of the main process and peak GPU memory in use, against the number of cells. Plate compliances are NICE values; exact checks of the plate designs in Table 6 and Table ST27.

shows the role of the first: even with exact interiors, restricting box-face displacements changes the assembled response, and a degree that gives accurate compliance can leave appreciable sensitivity error. For the local sensitivity of the cut cells examined, the boundary restriction produces larger errors than the interior approximation of NICE (Sections 5.6 and 5.7, for single cells at fixed size). The retained cut band keeps cut-surface loads and supports available without retraining; in the lattices, whose cut surfaces are traction-free, its DOFs off the box faces account for 37 to 50% of the free retained DOFs (Table ST17a).

Within the interior, prediction and correction are complementary, and the division of labour between them is what the variational structure makes visible. An interior correction that does not increase the A -energy error for any retained input yields a condensed stiffness between the learned stiffness and the exact Schur complement, independently of the network (Section 4.3). Smoothing removes the error components with large Rayleigh quotients, which amplify a displacement error of about one percent into an energy error of 2 to 35% (Section 5.4); the coarse-grid correction removes the smooth components that smoothing reaches slowly. The learned field supplies what

neither reaches at a practical budget: under the same correction, a harmonic start leaves 7 to 290 times its error on the five cells of Table 3, and eight times the smoothing work closes the gap in only one of six cells. Conversely, the correction relieves the network of what it represents poorly: a base-network error of 13.5% on M1 becomes 0.19% without any change of network parameters, and the base network with the correction applied at deployment already meets the assembled accuracy reference, so an existing network need not be retrained to benefit from the correction, and training through the correction adds a further factor of 1.31 (Section 5.3).

6.2. What the guarantees cover

The variational properties hold for every input, and the ordering for every geometry whose spectrum the smoothing interval contains (Section 5.2). Design differentiation adds a requirement that the energy error does not control. At a fixed retained displacement, with $H = J_I(F - E)$ as in Eq. (4), $E_{I,c} = J_I E_c$, $H_c = \partial H / \partial \tau_c$ and $K_{II,c} = J_I K_{,c} J_I^T$, Eqs. (9) and (10) give $\tilde{s}_c - s_c = 2(Hq)^T A E_{I,c} q - (Hq)^T K_{II,c} Hq$ and $\hat{C}_{,c} - s_c = -2(H_{,c}q)^T A Hq - (Hq)^T K_{II,c} Hq$: the residual term of Eq. (10) removes the linear field-error term and replaces it by one weighted by the design derivative of the extension error, so which of the two is the more accurate gradient depends on whether $\|H_{,c}q\|_A$ is small compared with $\|E_{I,c}q\|_A$. A central-difference check of $\hat{C}_{,c}$ could not resolve it: at least one of the discrete choices listed in Section 3.3 switches in 21 to 48 of the 49 rebuilds of each lattice cell and the difference quotient grows as the step decreases (Supplementary Note S6.3, Table ST19). The field-based estimate needs no difference quotient and gives the shared-variable gradient of the two eight-cell lattices of Section 5.8 and the two four-cell layers of the block (Section 5.9) to 0.02–0.13% under the face loads and 0.22–0.35% under the random loads (Table ST19). For optimisers with a line search the inconsistency between the surrogate objective and the field-based gradient matters (Section 3.3).

6.3. Computational value

A correction budget adequate for compliance can leave local sensitivity inaccurate, so the relevant cost is the total work needed to reach the prescribed accuracy of both. For the lattices examined, a NICE design iteration takes less time and memory than the whole-lattice direct solution and conventional exact condensation (Section 5.9). Unlike inexact BDDC and FETI-DP methods Li & Widlund (2007), Klawonn & Rheinbach (2007), Hirschler et

al. (2024), which use approximate subdomain and coarse solves only inside the preconditioner and therefore converge to the exact discrete solution, its lattice solve iterates on the assembled condensed stiffnesses $\hat{S}$ of the learned substructures, so that the approximation lies in the operator itself. Its output is a condensed operator per cell with a quantified and reducible error; whole-lattice iterative solvers are a different route (Section 6.4). Against the offline cost of about 60 GPU-hours (Section 5.1), each eight-cell design iteration saves 787 to 932 s compared with the parallel whole-lattice direct solution, so the offline cost is recovered after a few hundred design iterations, roughly ten optimisations of the size of case A in Section 5.10, if GPU and CPU time are counted alike. Fusing the operations of an application or reducing the smoothing budget would lower its cost, at an accuracy cost that Section 5.5 quantifies.

6.4. Limitations

The construction and the error relations are not specific to the cells studied: they apply to any discrete model with a symmetric positive semidefinite stiffness and a positive definite interior block. The trained network is. It applies to the setting in which it was trained and tested: Schwarz-P-type cells of Eq. (1) with eight corner parameters in $[0.175, 0.699]$, corner span and gradient norm at most 0.47 (Table ST02), and at most one planar cut whose normal is a cube-symmetry image of $(\cos \vartheta, \sin \vartheta, 0)$. It is tied to one discretisation ($n = 32$, $E_Y = 1$, $\nu = 0.3$, $\gamma = 10^{-4}$), to which the 65/33/17/9 latent hierarchy is matched. Other TPMS families were not tested; they, cells with several cuts or curved boundaries, other resolutions and other materials require new training data and network parameters, at the cost of Section 5.1. The network is not equivariant under the cube symmetries, as symmetry-decoupled shape functions are for voxel substructures Jiang, C. et al. (2026), and NICE was evaluated mainly in the identity orientation: in a second cube-symmetry orientation of the 80 validation geometries, which also entered model selection, its mean directional energy error over the load classes is 4% higher, and 12% higher under consistent tractions (Supplementary Table ST03c). The variational properties of Section 4.1 hold for any input, but the accuracy does not: no certified error bound is provided (Appendix G.4): the interior residual of Eq. (5) is available at run time, but the energy error it determines needs one interior solve per direction, and the label-free lower bound of Appendix I can detect a large error in a direction but cannot confirm a small one. Accuracy is measured against the discrete reference, whose

error for heavily cut cells at $n = 32$ is of order one percent in compliance and several percent in sensitivity (Section 5.2). The assembled evidence comprises fourteen two-cell configurations of selection cells, eighteen of held-out cells, two eight-cell lattices and the two four-cell layers of the block (Sections 5.8 and 5.9), three exact-checked designs of the block (case A) and six of two cut 24-cell plates (Section 5.10); all training comparisons use one seed, and the effect of the sensitivity term in Eq. (18) was not isolated. The reported sensitivities are field-based estimates that omit the design dependence of the extension in Eq. (10). Fine-scale iterative solvers of the whole lattice, such as algebraic multigrid Vaněk et al. (1996), Henson & Yang (2002), Falgout & Yang (2002) or BDDC and FETI-DP preconditioners Dohrmann (2003), Farhat et al. (2001), were not compared. An optimiser must keep designs inside the parameter domain above, as the bounds and limits of Section 5.10 do; the optimisations there are local, and on plates of more than 24 cells, whose size on one GPU is limited by the direct factorisation of $\mathbb{K}_{PP}$, the accuracy of NICE was not verified.

7. Conclusions

A learned reduced model of a structural component acquires the structure of a mechanical model from where its approximation is placed and how its condensed stiffness is formed. Approximating only the interior extension on the complete retained space, evaluating it in the energy form with its transpose F^T and adding a fixed, geometry-specific two-grid correction gives a condensed operator that is symmetric and positive semidefinite with the rigid-body kernel, bounded below by the exact Schur complement with an error quadratic in the interior error, assemblable and differentiable, and improvable at deployment: a correction whose smoothing interval contains the spectrum cannot increase the error, and in every example a larger correction budget reduced the energy error further, whether the correction started from the base network's field or from a deterministic field. Learning supplies the trial field, the variational form the structure and the correction the improvability; the three are complementary, since under the same correction a deterministic start leaves up to four orders of magnitude more error than the learned field, and the correction alone, applied to the base network without retraining, lowers its mean energy error under consistent tractions on the 80 validation geometries from 6.89% to 0.0965% and keeps its compliance and sensitivity errors below 1% in the fourteen two-cell configurations evaluated.

The error relations fix what such a model must be checked against. The interior error enters the assembled compliance weighted by the energy the cell carries, so that a cell with a small energy share can leave the compliance accurate; it enters the thickness sensitivity through the stiffness derivative with a linear term that the energy error does not control; and in the cut cells examined a displacement error of about one percent lies in stiff directions and becomes an energy error of 2 to 35%. Accurate compliance therefore does not imply accurate local sensitivity, and joint accuracy of both is the criterion for design use.

Realised for cut thin-walled Schwarz-P cells, with tens of thousands of retained degrees of freedom and a discrete space that changes with the cut, NICE keeps the mean energy error under consistent tractions at 0.074% on 80 validation geometries and the compliance and sensitivity errors below 0.3% and 1.5% in every two-cell configuration and below 0.02% and 0.15% in lattices of eight learned cells under face loads. An eight-cell design iteration takes 81–110 s on one GPU against 868–1,042 s for the whole-lattice Cholesky solution on 16 CPU threads. Driven by the field-based sensitivities, the method of moving asymptotes reproduces on an eight-cell lattice the design obtained with exact condensation to within 0.0051 in every corner parameter; on cut 24-cell plates the exact checks give compliance errors below 0.04% and gradient errors below 0.33%, whereas a homogenised model underestimates the compliance by 27–37%, although its designs, evaluated with NICE, have compliances within 1.1% of those of the NICE designs; a design iteration of a 110-cell plate with 32.7 million degrees of freedom takes 43 min. The construction transfers to other component families with new training data; the structure and the error relations do not depend on the family.

Code and data availability

The implementation of the neural extension, the equilibrium correction and its transpose, the cut-cell preprocessing and reference condensation, and the assembly, benchmark and figure scripts will be archived as a cleaned, frozen snapshot, together with the trained network parameters of the variants of Table 2, the parameter files of all validation and lattice geometries and the result records from which every table and figure is generated, in a public Zenodo repository upon acceptance. Training data are available from the corresponding author on reasonable request.

CRedit authorship contribution statement

[Author contributions (CRedit taxonomy).]

Declaration of competing interest

[Competing-interest statement.]

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

[To be written by the author according to the journal’s current policy (tool or service, purpose, and the authors’ responsibility for the content).]

Appendix A. Discrete construction and metric definitions

Appendix A.1. Active degrees of freedom and element integration

The active background elements use tensor-product Q_2 displacements, with 27 nodes and 81 displacement degrees of freedom per element. Active DOFs are stored in a fixed node-major x, y, z order. The retained box set is the union of the nine face nodes of every certified positive-area material patch. The cut set is the union of all 27 nodes of each active element carrying a positive-area cut-surface patch. Their union is deduplicated in active-node order. The off-plane cut-band DOFs are retained because their basis functions determine displacement and virtual work on the cut plane.

The discrete stiffness is assembled as

$$K(\eta) = \sum_e L_e^T K_e(\eta) L_e + \gamma \sum_{f \in \mathcal{F}_g} L_f^T G_f^T G_f L_f, \quad K_e(\eta) = \sum_{\alpha \in \{0, \dots, 4\}^3} M_{e\alpha}(\eta) T_\alpha. \quad (\text{A.1})$$

Here L_e, L_f extract element and ghost-face DOFs. The fixed elasticity templates $T_\alpha \in \mathbb{R}^{81 \times 81}$ use the physical basis derivatives and isotropic Lamé constants $\lambda_L = E_Y \nu / [(1 + \nu)(1 - 2\nu)]$ and $\mu_L = E_Y / [2(1 + \nu)]$. The 125 moments $M_{e\alpha}$ integrate local monomials with coordinatewise exponents from

zero to four over the material part of an element. The ghost-face set, its factors and coefficient specify the stabilisation contribution. The sum defines the stabilised discrete energy, including its ghost-penalty contribution.

The ghost-face set $\mathcal{F}_g$ consists of interior background faces whose two neighbouring elements are active and are not both certified as completely filled with material. These faces lie within each substructure. For background-element width $h = 1/n$, the unit-coefficient stabilisation bilinear form is

$$g_h(v, u) = (\lambda_{\text{ref}} + 2\mu_{\text{ref}}) \sum_{f \in \mathcal{F}_g} \sum_{j=1}^2 h^{2j-1} \int_f [\partial_{n_f}^j v] \cdot [\partial_{n_f}^j u] \, dA.$$

The derivative uses a common positive coordinate normal on both neighbouring elements, and the jump is the value on the first element minus that on the second. Integration covers the complete background face. The stabilisation form g_h , and hence G_f , uses fixed reference values $E_{\text{ref}} = 1$, $\nu_{\text{ref}} = 0.3$ and their Lamé constants $\lambda_{\text{ref}}, \mu_{\text{ref}}$; these equal the material values E_Y, ν of the 80 validation geometries. Each face integral uses a tensor-product three-point Gauss rule in its two tangential coordinates. Stacking the square-root-weighted jump evaluations for both derivative orders and all three displacement components gives $G_f \in \mathbb{R}^{54 \times 135}$, acting on the 45 distinct nodes of the two-element patch. Thus the second term in Eq. (A.1) is the matrix representation of γg_h .

All 80 validation geometries use $n = 32$, normalised Young's modulus $E_Y = 1$, Poisson's ratio $\nu = 0.3$, and ghost-penalty coefficient $\gamma = 10^{-4}$, with length expressed relative to the unit reference box. This gives 65 Q2 node positions per axis; the network's three transfer levels use 33, 17 and 9 positions per axis. The moment evaluator begins with 4^3 subcells per element and refines partial subcells once, then clips Kuhn tetrahedra using the tetrahedral rule parameter four. Appendix F.4 gives the full integration sequence. Geometry generation accepts a cell only if its active background elements form a single face-connected set carrying retained box-face DOFs; 10 of the 1,109 cells generated as training candidates failed this check, and every training, validation, neighbour and lattice cell used in this work passes it. Training populations are identified in Table ST01.

Appendix A.2. Response metrics and aggregation

With $K_{,c} = \partial K / \partial \tau_c$ on the stated design interval with fixed DOF sets, the metrics are

$$\begin{aligned}
\varepsilon(q) &= \frac{\hat{u}^T K \hat{u}}{q^T S q} - 1, & C &= f_g^T U, & \hat{C} &= f_g^T \hat{U}, \\
e_C &= \left| \frac{\hat{C}}{C} - 1 \right|, & \tilde{s}_c(\hat{u}) &= -\hat{u}^T K_{,c} \hat{u}, \\
e_s &= \frac{\|\tilde{\mathbf{s}} - \mathbf{s}\|_2}{\|\mathbf{s}\|_2}, & \mathbf{s} &= \left(-u^T K_{,c} u \right)_{c=1}^8.
\end{aligned} \tag{A.2}$$

The local same-retained-displacement comparison uses $u = Eq$. The assembled comparison uses the recovered fields at the respective global equilibria, whose retained displacements generally differ. All relative denominators are nonzero. A geometry-level energy score averages over the evaluated directions in one class; population means then give equal weight to each available geometry. A population maximum of geometry means is distinct from a worst-direction operator error.

The pair-assembly comparison joins a learned target to an exact neighbour with matched thickness parameters on the common face. Its six face-load cases define the common 3% reference for compliance and sensitivity, with sensitivity maxima taken over both cells. Cut-traction cases are reported separately (Table ST10). Table 2 and Tables ST03, ST01 and ST09 identify the populations and loading sets, and Supplementary Note S6 specifies the two-cell configurations, the pair solver and the preconditioner of the lattice solves.

Appendix B. Variational identity, rigid-body kernel, and directional norms

Appendix B.1. Assumptions

In Appendices B and C, $f \equiv f_g$ denotes the assembled retained load.

For each cell, $K = K^T \succeq 0$, the retained and interior DOF sets P, I are fixed, and $A = K_{II} \succ 0$. The exact extension is defined by $E_P = I_p$ and $E_I = -A^{-1}K_{IP}$. A linear extension F has $F_P = I_p$. Set $H = F_I - E_I$, so $F = E + J_I^T H$, $S = E^T K E$, and $\hat{S} = F^T K F$. All transposes include the entire extension and correction. Interior body loads are zero; equivalently, the load functional factors through the retained DOFs. Consistent boundary loads have this property when all DOFs with nonzero boundary virtual work are retained.

The cell interiors are disjoint, all intercell couplings act through the retained DOFs, and the assembly maps B_m include fixed homogeneous supports. The supported exact assembled stiffness $\mathbb{K} = \sum_m B_m^T S_m B_m$ is positive definite. The same local stabilised matrices define both the reference and the learned assemblies. The applied global load $f \neq 0$ is fixed. These assumptions distinguish the modular reference from a different monolithic discretisation that introduces additional intercell stabilisation.

Design statements additionally use a differentiable interval on which the active and retained DOF sets, assembly maps, supports, and load are fixed. Stiffness derivatives include every varying term in the chosen discrete specification. In the sensitivities reported here, moment differences vary the body stiffness while the selected ghost matrix and its coefficient are fixed. The algebraic field-error expansions also hold for the symmetric finite-difference derivative matrices; interpreting them as exact design derivatives requires convergence to the derivative of that discrete system.

The full retained nodal representation need not be a minimal independent basis of surface traces. The cut band keeps the active DOFs needed for the cut-surface load functional. This sufficiency, and containment of all intercell couplings in P , is the property used by condensation and assembly.

Appendix B.2. Variational identity, transpose of the complete extension and rigid-body kernel

Let the selectors satisfy $J_P^T J_P + J_I^T J_I = I_{n_a}$, and assume the symmetric stiffness in Eq. (2) has $A \succ 0$. For any field v with $J_P v = q$, there is a unique $w \in \mathbb{R}^i$ such that $v = E q + J_I^T w$. Since $J_I K E = 0$,

$$v^T K v = q^T S q + w^T A w. \quad (\text{B.1})$$

This proves the energy-minimising property and uniqueness of $E q$. Applying the same expansion to a linear admissible extension $F = E + J_I^T H$ yields Eq. (4). At fixed q , the stabilised discrete strain energy of $F q$ therefore exceeds that of $E q$ by one half of $q^T H^T A H q$; the reported energy error uses the quadratic energy without the factor one half. Moreover, $r_I = J_I K F q = A H q$, proving both expressions in Eq. (5). These identities require trace admissibility and the specified symmetric stiffness; a contraction property of the approximation is a separate condition.

Suppose additionally that $\ker K = \text{range } R$, R_P has rank six, and $F R_P = R$. The exact field associated with $R_P a$ is $R a$, because it has that retained

trace and satisfies interior equilibrium. Thus $ER_P = R$ and $HR_P = 0$. Rigid reproduction gives $\hat{S}R_P = F^TKR = 0$. Conversely, if $q^T\hat{S}q = 0$, positive semidefiniteness implies $Fq \in \ker K$. Write $Fq = Ra$; selecting the retained entries gives $q = R_Pa$. Hence

$$\ker S = \ker \hat{S} = \text{range } R_P. \quad (\text{B.2})$$

The retained rigid coefficient map and deformation projector are

$$C_R = (R_P^TR_P)^{-1}R_P^T, \quad \Pi_P = I_p - R_PC_R, \quad q_d = \Pi_Pq. \quad (\text{B.3})$$

Rigid reproduction and symmetry give

$$\hat{S}R_P = 0, \quad R_P^T\hat{S} = 0. \quad (\text{B.4})$$

For the construction in Eq. (17), $C_RR_P = I_6$ and $\Pi_P R_P = 0$. A linear raw displacement map sends zero to zero, so $\hat{E}R_P = R$. Corrections driven by the interior residual leave this field unchanged. These observations establish the two-sided rigid annihilation in Eq. (B.4) without altering any stiffness eigenvalue. Positive semidefiniteness alone does not imply rigid reproduction, and rigid reproduction alone does not exclude extra modes in a reference K with a larger kernel.

Let $Z \in \mathbb{R}^{p \times (p-6)}$ have orthonormal columns spanning the retained rigid complement, and define $S_* = Z^TSZ \succ 0$. The all-direction energy error on this space is

$$\varepsilon_* = \sup_{q \perp R_P, q \neq 0} \varepsilon(q) = \|A^{1/2}HZS_*^{-1/2}\|_2^2, \quad \mu_* = 1 + \varepsilon_*. \quad (\text{B.5})$$

Let $q = Za$, with $Z^TZ = I$ spanning the rigid complement. Its relative energy error is

$$\frac{a^TZ^TH^T A H Z a}{a^TS_*a} = \frac{\|A^{1/2}HZS_*^{-1/2}b\|_2^2}{\|b\|_2^2}, \quad b = S_*^{1/2}a. \quad (\text{B.6})$$

Taking the supremum proves Eq. (B.5). The basis Z is used for analysis; the operator application does not construct these whitened coordinates.

The retained block of the unbalanced discrete force, used alone, would be

$$(KF)_P = S + K_{PI}H.$$

The additional variational reaction is $F_I^T(KF)_I = (E_I + H)^T AH$. Since $E_I^T A = -K_{PI}$, its term $E_I^T AH$ cancels $K_{PI}H$. Thus

$$(KF)_P + F_I^T(KF)_I = S + H^T AH. \quad (\text{B.7})$$

The transpose F^T supplies both reciprocity and cancellation of the first-order stiffness error. A retained-force extraction without this term generally has an $O(H)$ error and need not be symmetric.

For any global vector, the quadratic error is the sum of nonnegative local errors in Eq. (C.1). Consequently $\hat{\mathbb{K}} \succeq \mathbb{K} \succ 0$. This also proves supported solvability even if some local cells are only semidefinite. A high condition number can still increase the cost and sensitivity of the numerical solve.

Appendix B.3. Displacement magnitude and directional stiffness

The energy metric assigns different significance to displacement errors of equal magnitude. To make that dependence explicit, fix the retained rigid gauge by taking $q \perp R_P$, let $d = Fq - Eq$, and introduce a displacement norm induced by $\mathbf{W} \succ 0$:

$$\delta_{\mathbf{W}}(q) = \frac{\|d\|_{\mathbf{W}}}{\|Eq\|_{\mathbf{W}}}, \quad \mathcal{R}_{K,\mathbf{W}}(v) = \frac{v^T K v}{v^T \mathbf{W} v}, \quad \kappa_{\mathbf{W}}(q, d) = \frac{\mathcal{R}_{K,\mathbf{W}}(d)}{\mathcal{R}_{K,\mathbf{W}}(Eq)}. \quad (\text{B.8})$$

For nonzero field error and nonzero reference energy, cancellation of $d^T \mathbf{W} d$ and $u^T \mathbf{W} u$, with $u = Eq$, gives

$$\boxed{\varepsilon(q) = \delta_{\mathbf{W}}(q)^2 \kappa_{\mathbf{W}}(q, d)}. \quad (\text{B.9})$$

Section 5.4 and Table ST04 use $\mathbf{W} = \text{diag}(0, D)$ with $D = \text{diag}(K_{II})$, i.e. a Jacobi-weighted interior norm. This weighting is only positive semidefinite, but the identity still holds because d vanishes on the retained DOFs, and the rigid gauge is then not needed. The factor $\kappa_{\mathbf{W}}$ compares the stiffness content of the error and the equilibrium response; it is not a matrix condition number. An error concentrated in stiffer deformation than the loaded response has a large $\kappa_{\mathbf{W}}$, making even a small relative displacement error mechanically significant. Thus an energy target ε_{tar} requires $\delta_{\mathbf{W}} \leq \sqrt{\varepsilon_{\text{tar}}/\kappa_{\mathbf{W}}}$. Correction must address this energy content as well as displacement magnitude.

Changing the displacement normalisation changes the corresponding amplification factor. An alternative based on the retained displacement is

$$\delta_P = \frac{\|d_I\|_2}{\|q\|_2}, \quad \kappa_P = \frac{d_I^T A d_I / \|d_I\|_2^2}{q^T S q / \|q\|_2^2}, \quad \varepsilon(q) = \delta_P^2 \kappa_P. \quad (\text{B.10})$$

In Eq. (B.9), the same displacement weighting enters both Rayleigh quotients, and the denominator of κ_W measures the physical response in the chosen trace direction. The smoothing instead acts on the spectrum of $Av = \lambda Dv$ with the trace fixed, so the two notions of softness are distinct. For example, $A = \text{diag}(\epsilon, 1)$ has a very soft unscaled direction, while $D^{-1/2} A D^{-1/2} = I$. An interior eigenmode also has zero retained trace and is not itself a free-cell Schur response. Consequently, cumulative Jacobi-scaled low-mode error energy identifies components that the smoothing reduces slowly; it does not determine the measured κ , a continuum bending-mode label, or a model-capacity lower bound.

Appendix C. Assembly and compliance ordering

For a global retained vector U , minimising the sum of substructure energies over their disjoint interior DOFs separates into the individual minimisations in Eq. (B.1). The minimised quadratic form is $\sum_m U^T B_m^T S_m B_m U$. This proves the equivalence of local condensation followed by assembly and elimination from the modular assembled system under the coupling assumptions in Section 2.3.

Let $\mathbb{D} = \hat{\mathbb{K}} - \mathbb{K}$. From Eq. (4),

$$U^T \mathbb{D} U = \sum_m \|H_m B_m U\|_{A_m}^2 \geq 0. \quad (\text{C.1})$$

If homogeneous supports make $\mathbb{K} \succ 0$, then both global systems are invertible and matrix inversion reverses their order. For any fixed nonzero load,

$$\hat{C} = f_g^T \hat{\mathbb{K}}^{-1} f_g \leq C = f_g^T \mathbb{K}^{-1} f_g. \quad (\text{C.2})$$

For completeness, the order reversal follows by setting $Q = \mathbb{K}^{-1/2} \hat{\mathbb{K}} \mathbb{K}^{-1/2} \succeq I$. Its eigenvalues are at least one, so $Q^{-1} \preceq I$; a congruence gives the inverse inequality. If every substructure also satisfies

$0 \preceq \hat{S}_m - S_m \preceq \varepsilon_* S_m$, applying the same argument to the upper and lower bounds gives

$$\mathbb{K} \preceq \hat{\mathbb{K}} \preceq (1 + \varepsilon_*)\mathbb{K}, \quad \frac{C}{1 + \varepsilon_*} \leq \hat{C} \leq C, \quad \frac{C - \hat{C}}{C} \leq \frac{\varepsilon_*}{1 + \varepsilon_*}. \quad (\text{C.3})$$

The required local bound is uniform over directions. A mean over a finite set of sampled directions has a different statistical meaning.

The assembled state error obeys the exact identity

$$\hat{U} - U = -\hat{\mathbb{K}}^{-1}\mathbb{D}U. \quad (\text{C.4})$$

Consequently, for a fixed finite assembly with uniformly stable support and bounded inverse, a family $H_m = O(t)$ gives $\mathbb{D} = O(t^2)$ and $\hat{U} - U = O(t^2)$ as $t \rightarrow 0$ (Eq. (C.5)).

Appendix C.1. Compliance error as the reconstructed error energy

Let $U = \mathbb{K}^{-1}f$, $\hat{U} = \hat{\mathbb{K}}^{-1}f$, $q_m = B_m U$, $\hat{q}_m = B_m \hat{U}$, $u_m = E_m q_m$, and $\hat{u}_m = F_m \hat{q}_m$. The matrix $\mathbb{D} = \hat{\mathbb{K}} - \mathbb{K}$ of Eq. (C.1) equals $\sum_m B_m^T H_m^T A_m H_m B_m$.

For conforming cell fields, use the energy form $a(v, v) = \sum_m v_m^T K_m v_m$. Shared retained DOFs are counted through their separate cell energy contributions. Exact local equilibrium and the global equation give $a(u, v) = f^T V_P$, where V_P denotes the free assembled trace. At the equilibrium $\hat{\mathbb{K}}\hat{U} = f$, $a(\hat{u}, \hat{u}) = f^T \hat{U} = \hat{C}$. Hence

$$C - \hat{C} = a(\hat{u} - u, \hat{u} - u) = \|\hat{U} - U\|_{\mathbb{K}}^2 + \sum_m \|H_m \hat{q}_m\|_{A_m}^2.$$

The second equality follows from $\hat{u}_m - u_m = E_m(\hat{q}_m - q_m) + J_{I,m}^T H_m \hat{q}_m$. The two terms are K_m -orthogonal because $J_{I,m} K_m E_m = 0$. Thus compliance underestimation measures the total energy error of the reconstructed field, whereas the retained-solution error accounts for only one part of it. In particular,

$$\frac{\|\hat{U} - U\|_{\mathbb{K}}}{\|U\|_{\mathbb{K}}} \leq \sqrt{\frac{C - \hat{C}}{C}}.$$

This nonasymptotic bound is valid without a small-error assumption. The sharper asymptotic retained-solution order is derived below.

Define $T = \mathbb{K}^{-1/2} \mathbb{D} \mathbb{K}^{-1/2} \succeq 0$, $z = \mathbb{K}^{-1/2} f$, and

$$\beta = \frac{U^T \mathbb{D} U}{C} = \sum_{m: q_m^T S_m q_m > 0} w_m \varepsilon_m(q_m), \quad w_m = \frac{q_m^T S_m q_m}{C}.$$

For zero-energy cells with exact rigid reproduction, the corresponding numerator is also zero and its contribution is defined directly as zero. Spectral calculus gives

$$\frac{C - \hat{C}}{C} = \frac{z^T T (I + T)^{-1} z}{z^T z}, \quad \boxed{\frac{\beta}{1 + \|T\|_2} \leq \frac{C - \hat{C}}{C} \leq \frac{\beta}{1 + \beta} \leq \beta.}$$

The lower bound uses $\lambda/(1 + \lambda) \geq \lambda/(1 + \|T\|_2)$. For the sharper upper bound, insert $V = \alpha U$ in the maximum principle

$$\hat{C} = \max_V \{2f^T V - V^T \hat{\mathbb{K}} V\}.$$

Optimising the scalar gives $\alpha = (1 + \beta)^{-1}$ and $\hat{C} \geq C/(1 + \beta)$. The share-weighted bound $e_C \leq \beta$ of Eq. (7) follows. This is a load-specific statement: w_m and ε_m are evaluated at the same exact assembled trace.

A uniform local inequality $0 \preceq \hat{S}_m - S_m \preceq \varepsilon_* S_m$ implies $T \preceq \varepsilon_* I$, which recovers the load-independent bound $e_C \leq \varepsilon_*/(1 + \varepsilon_*)$ of Eq. (C.3). Rigid reproduction extends a quotient-space bound to arbitrary local traces.

Now take $H_m(t) = tH_{0,m}$ and hold the reference matrices and maps fixed. Set $\mathbb{D}_0 = \sum_m B_m^T H_{0,m}^T A_m H_{0,m} B_m$. Expanding the inverse at zero yields

$$\begin{aligned} \hat{\mathbb{K}}_t &= \mathbb{K} + t^2 \mathbb{D}_0, \\ \hat{U}_t - U &= -t^2 \mathbb{K}^{-1} \mathbb{D}_0 U + O(t^4), \\ C - \hat{C}_t &= t^2 U^T \mathbb{D}_0 U + O(t^4), \\ \hat{u}_{m,t} - u_m &= t J_{I,m}^T H_{0,m} q_m - t^2 E_m B_m \mathbb{K}^{-1} \mathbb{D}_0 U + O(t^3). \end{aligned} \tag{C.5}$$

Thus the retained displacement and compliance errors are $O(t^2)$, the local full-field error is generically $O(t)$, and the retained-error contribution in Eq. (6) is $O(t^4)$. If $\mathbb{D}_0 U = 0$, then $H_{0,m} q_m = 0$ for every affected cell, and this load is reproduced exactly for all t . Uniform estimates across a family of geometries require uniform coercivity and bounded maps; fixed-geometry big-O constants do not supply those estimates automatically, nor do they

determine which contribution dominates the sensitivity error at a finite error level.

Appendix C.2. Inexact assembled solves

Let $\bar{U}$ be an approximate solution of the same fixed variational operator, $\rho = f - \hat{\mathbb{K}}\bar{U}$, $\bar{C} = f^T\bar{U}$, and $\bar{u}_m = F_m B_m \bar{U}$. The exact identity becomes

$$C - \bar{C} = a(\bar{u} - u, \bar{u} - u) + \bar{U}^T \rho, \quad J(\bar{U}) = \bar{C} + \bar{U}^T \rho \leq \hat{C} \leq C.$$

The error of the corrected functional is $\hat{C} - J(\bar{U}) = \rho^T \hat{\mathbb{K}}^{-1} \rho$, and $\hat{C} - \bar{C} = \bar{U}^T \rho$. Thus the work $f^T \bar{U}$ can violate the variational compliance ordering if the algebraic residual is appreciable, while the concave work functional retains a quadratic residual error. A Euclidean residual bound converts through

$$\frac{|\bar{U}^T \rho|}{C} \leq \frac{\|\bar{U}\|_2 \|f\|_2}{C} \frac{\|\rho\|_2}{\|f\|_2},$$

with a load- and stiffness-dependent constant.

If an applied action $y(\bar{U})$ is not numerically identical to the variational energy, define $\omega = \bar{U}^T y(\bar{U}) - \sum_m \bar{u}_m^T K_m \bar{u}_m$ and $\rho = f - y(\bar{U})$. Then the first identity acquires an additional $+\omega$. This separates solve residual from action/energy inconsistency. A recursive Krylov residual need not equal this applied-action residual. Evaluating the terms in Eq. (8) additionally requires the signed residual work and ω . For the pair configurations and lattices, both are recorded together with the dual-norm bound $|\bar{U}^T \rho| \leq \sqrt{\bar{U}^T y(\bar{U})} \sqrt{\rho^T \mathbb{K}^{-1} \rho}$, which holds because $\hat{\mathbb{K}} \succeq \mathbb{K}$ (Supplementary Note S6.3): at the final iterates, $|\bar{U}^T \rho|$ and $|\omega|$ stay below 6×10^{-8} of the compliance and the bound below 2×10^{-5} , so the reported errors are those of the operator.

Appendix D. Polynomial smoothing and coarse projections

For $0 < a < b$, the degree- k Chebyshev error polynomial and its energy bound are

$$p_k(t) = \frac{T_k((a+b-2t)/(b-a))}{T_k((a+b)/(b-a))}. \quad (\text{D.1})$$

$$d_I^{\text{sm}} = p_k(D^{-1}A)d_I, \quad \|d_I^{\text{sm}}\|_A^2 \leq \psi_k^2 \|d_I\|_A^2, \quad \psi_k = \max_{\lambda \in \sigma(D^{-1/2}AD^{-1/2})} |p_k(\lambda)|. \quad (\text{D.2})$$

Here $D = \text{diag}(A)$.

Write $\bar{\lambda} = (a + b)/2$, $\eta_s = (b - a)/2$, and $\sigma = \bar{\lambda}/\eta_s > 1$ for the centre and half-width of the target interval and their ratio. Starting from u_0 , keep its retained entries fixed and define the interior residual with the sign convention $r_j = (Ku_j)_I$. A recurrence producing the polynomial in Eq. (D.1) is

$$\begin{aligned} \varrho_0 &= 1/\sigma, & z_0 &= -D^{-1}r_0/\bar{\lambda}, \\ u_{j+1} &= u_j + J_I^T z_j, & \varrho_{j+1} &= (2\sigma - \varrho_j)^{-1}, \\ z_{j+1} &= \varrho_{j+1}\varrho_j z_j - \frac{2\varrho_{j+1}}{\eta_s} D^{-1}r_{j+1}. \end{aligned} \quad (\text{D.3})$$

The last line is evaluated only when another step is needed. Zero steps return the initial field. To verify the polynomial, observe that $\varrho_j = T_j(\sigma)/T_{j+1}(\sigma)$, and apply $T_{j+2}(x) = 2xT_{j+1}(x) - T_j(x)$. After the first step the error is $d_1 = (I - D^{-1}A/\bar{\lambda})d_0$, and the same recurrence gives $d_k = p_k(D^{-1}A)d_0$.

The matrices $D^{-1}A$ and $D^{-1/2}AD^{-1/2}$ are similar, $p_k(D^{-1}A)$ is self-adjoint in the A -inner product, and

$$\|p_k(D^{-1}A)d\|_A^2 = \sum_j \lambda_j p_k(\lambda_j)^2 c_j^2, \quad c_j = v_j^T D d. \quad (\text{D.4})$$

This proves Eq. (D.2). If $a \leq \lambda \leq b$, the argument of T_k lies in $[-1, 1]$, where $|T_k| \leq 1$. For $0 < \lambda < a$, it lies between 1 and σ , where T_k increases from 1 to $T_k(\sigma)$. Hence $|p_k(\lambda)| \leq 1$ throughout $(0, b]$, so the polynomial is nonexpansive in the A -energy norm when the actual positive spectrum is contained in $(0, b]$. Modes below a can contract slowly, and eigenvalues above b can be amplified.

The operational upper endpoint is estimated using a random power vector, 40 iterations of $D^{-1}A$, Euclidean normalisation, and a factor of 1.05. The correction implementation fixes the random generator seed and caches the interval for the geometry. The separate fixed-weight smoothing study of the base network uses its own random start. The estimates are used to set the polynomial, while the result in Eq. (D.2) is conditional on its actual spectral values: a finite power estimate multiplied by a safety factor is an estimate

of the endpoint, not the containment used above. For the 80 validation geometries we therefore compared the operational endpoint with $\lambda_{\max}(D^{-1}A)$ computed by Lanczos iteration with full reorthogonalisation on $D^{-1/2}AD^{-1/2}$ (78–150 steps; relative Ritz residual at most 9.6×10^{-4} , median 2.2×10^{-6}). The operational endpoint exceeds the converged value by 2.1–5.0% (median 3.9%) on every geometry, so the containment required by Eq. (D.2) holds for all evaluated cells. The strict Gershgorin bound $\max_i \sum_j |A_{ij}|/A_{ii}$ is 4.7–19.4 times the operational endpoint (median 18.4); used as b , it would stretch the targeted interval by that factor and slow the contraction of the upper spectrum accordingly.

For a full-column-rank coarse basis V , with $A_c = V^T A V \succ 0$, define $Q_V = V A_c^{-1} V^T A$. Direct multiplication gives $Q_V^2 = Q_V$ and $Q_V^T A = A Q_V$. Thus Q_V is an A -orthogonal projection and $C_V = I - Q_V$ is its complementary projection. With $b_r = V^T A d$,

$$\|d\|_A^2 = \|C_V d\|_A^2 + \|Q_V d\|_A^2, \quad \|Q_V d\|_A^2 = b_r^T A_c^{-1} b_r. \quad (\text{D.5})$$

This proves Eq. (14). Applying Eq. (D.2) before and after this projection yields $\|\Phi_k C_V \Phi_k d\|_A \leq \psi_k^2 \|d\|_A$. Because $\psi_k < 1$ whenever the positive spectrum lies in $(0, b]$, the cycle is an energy contraction, which gives the operator ordering of Eq. (15) (Appendix D.1). For the spectra of U1, M1 and M2 ($\lambda_1 \approx 4\text{--}9 \times 10^{-4}$ against $a \approx 0.17$; Supplementary Table ST05b), however, $\psi_8^4 \geq 0.98$, so the estimate guarantees little more than non-expansion; it uses no approximation property of the coarse space, and the reductions by one to two orders of magnitude in Section 5.5 are empirical. A two-grid convergence estimate would require an approximation property of the Q_1 space for walls about one element thick Xu & Zikatanov (2002). This is the standard energy-projection mechanism of subspace correction Xu (1992).

Appendix D.1. Corrections, orderings and approximate coarse inverses

A geometry-fixed linear correction acts on H through $H_{\text{new}} = \Theta H$. If $\Theta^T A \Theta \preceq A$, then

$$S \preceq \hat{S}_{\text{new}} \preceq \hat{S}_{\text{old}}.$$

Thus exact assembled compliance increases monotonically toward the reference, and Eq. (6)'s total reconstructed energy error decreases; in particular, repeating a fixed cycle cannot increase the energy error. Changing the

smoothing degree, or stopping the recurrence of Eq. (D.3) at an intermediate step, changes the polynomial, so the energy error need not decrease monotonically in k . Neither a signed component of displacement nor the field-based sensitivity norm is ordered by this matrix inequality.

For a symmetric approximate inverse G in place of A_c^{-1} in the coarse-grid correction $C_V = I - VA_c^{-1}V^TA$ of Eq. (D.5), the precise condition is

$$\|d\|_A^2 - \|d - VGV^TAd\|_A^2 = b_r^T(2G - GA_cG)b_r, \quad b_r = V^TAd. \quad (\text{D.6})$$

Consequently $2G - GA_cG \succeq 0$ suffices for nonexpansiveness. For an accurate A_c , a nonnegative shift $G = (A_c + \Lambda)^{-1}$, $\Lambda \succeq 0$, satisfies this condition, although it is not the exact projection. This statement concerns the coarse inverse while the prescribed fine-grid stiffness K remains fixed. If a probed matrix differs from A_c , contraction depends on the approximate inverse relative to the true A_c .

At the operator-application level the entire affine correction of the field, including the retained forcing, is transposed in reverse order, as in Eqs. (E.3)–(E.4). Geometry-fixed coefficients and a fixed iteration count ensure linearity. A fixed number of ordinary CG iterations generally does not: its coefficients depend on the right-hand side. Arbitrary direction-dependent stopping also need not define one linear extension.

The benefit of learning under a fixed correction budget is measured by the error surviving the correction: $\|\Theta H_{\text{net}}q\|_A^2$ versus $\|\Theta H_{\text{start}}q\|_A^2$. Initial errors with the same energy can leave different corrected errors. This connects direction-sensitive training to the numerical correction actually used.

Boundary-space restriction and interior correction affect different trial spaces. Exact interior condensation followed by a restriction $U = G_r y$ solves over a subspace of the retained space; nested boundary spaces give nondecreasing Ritz compliance. Interior correction preserves all retained DOFs and changes the interior graph $u = FB U$; its matrix ordering comes from contraction of H , even though two such graph spaces need not be nested. In neither case is a local sensitivity norm ordered.

Appendix E. Transpose of the complete extension

All transposes below use the Euclidean pairing of the stored displacement and nodal-force vectors. Write $M_I = J_I^T J_I$. Transposing Eq. (17) gives

$$\hat{E}^T y = J_P y + C_R^T R^T M_I y + \Pi_P^T \mathcal{N}_\theta(\eta)^T M_I y. \quad (\text{E.1})$$

The overwrite therefore selects the direct retained contribution and masks the field sent through the network transpose. The geometry coefficients are held fixed during this displacement transpose.

Set $\varphi_k(t) = [1 - p_k(t)]/t$, with the polynomial continuation at zero. The full-field smoothing map $\mathcal{T}_k$ has the block representation

$$\mathcal{T}_k = \begin{bmatrix} I_p & 0 \\ -\varphi_k(D^{-1}A)D^{-1}K_{IP} & p_k(D^{-1}A) \end{bmatrix}, \quad \mathcal{T}_k^T y = \begin{bmatrix} y_P - K_{PI}\varphi_k(D^{-1}A)D^{-1}y_I \\ Dp_k(D^{-1}A)D^{-1}y_I \end{bmatrix}. \quad (\text{E.2})$$

Indeed, the final interior field is its equilibrium value plus $p_k(D^{-1}A)$ times the initial error. Replacing $(I - p_k(D^{-1}A))A^{-1}$ by $\varphi_k(D^{-1}A)D^{-1}$ yields the first expression. The second follows from $(D^{-1}A)^T D = A = D(D^{-1}A)$, which implies $p_k(D^{-1}A)^T = Dp_k(D^{-1}A)D^{-1}$ and the corresponding identity for φ_k . Although the forward map preserves retained displacement, its transpose contributes a retained force through the upper block in Eq. (E.2).

For the full-field coarse-grid correction,

$$\mathcal{W}_c = I_{n_a} - J_I^T V A_c^{-1} V^T J_I K, \quad \mathcal{W}_c^T = I_{n_a} - K J_I^T V A_c^{-1} V^T J_I. \quad (\text{E.3})$$

A cycle $\mathcal{W} = \mathcal{T}_{\text{post}} \mathcal{W}_c \mathcal{T}_{\text{pre}}$ therefore uses the reverse sequence $\mathcal{W}^T = \mathcal{T}_{\text{pre}}^T \mathcal{W}_c^T \mathcal{T}_{\text{post}}^T$. The complete reaction is

$$\hat{S}q = \hat{E}^T \mathcal{W}^T K \mathcal{W} \hat{E}q. \quad (\text{E.4})$$

The same formulas hold for the symmetric shifted inverse defined below when it replaces the coarse inverse consistently in both actions. Iteration counts, coarse bases, and polynomial coefficients depend on geometry and remain fixed across input directions. An input-dependent iterative stopping rule would require a separate analysis of linearity and differentiation.

Appendix F. Coarse spaces, factorisation, and arithmetic

Appendix F.1. Explicit coarse-space study

Coarse tensor-product shape functions are evaluated at the active background-node coordinates and restricted to I . Standard Q_1 and Q_2

vector spaces carry three displacement DOFs per coarse node. The enriched partition-of-unity space carries 12 DOFs per vertex through fields of the form $\sum_v N_v(x)[a_v + B_v(x - x_v)]$, with $a_v \in \mathbb{R}^3$ and $B_v \in \mathbb{R}^{3 \times 3}$. The spaces examined use Q_1 grids of 9, 17, or 33 vertices per axis, a Q_2 grid of 17 nodes per axis, and enriched Q_1 grids of 9 or 17 vertices per axis.

Columns with an absolute interior-support sum at most 10^{-14} are removed. In the fixed-weight CPU study, the symmetric interior matrix is formed from its stored upper triangle. The coarse matrix is explicitly computed as $V^T A V$ and symmetrised. Columns whose diagonal energy is at most 10^{-12} times the largest coarse diagonal are then removed. The resulting sparse matrix is factorised with PARDISO. This construction adds no explicit diagonal shift. The projection formulas in Eqs. (14)–(15) apply to linearly independent surviving coarse columns and the exact Galerkin action. Interpreting a recorded numerical solve through those formulas additionally requires verification of its solve accuracy.

The enriched generating functions are linearly dependent, and the surviving columns are not certified independent; the enriched entries of Table ST07 are therefore reported as numerical observations in Supplementary Note S3, and the $Q_1(17)$ result is the principal coarse-correction result.

For each input direction, the study evaluates the exact field and the uncorrected network field once. It compares the network alone, a smoothing tail, a coarse-grid correction, a coarse-grid correction followed by smoothing, smoothing on both sides of the coarse-grid correction, and a zero-interior initialisation followed by the same complete cycle. Its energy ratios use a recomputed exact energy for the supplied direction. The fields are converted to float64 before correction.

Appendix F.2. Differentiable correction implementation

The differentiable implementation constructs coarse support metadata and recovers a coarse matrix using coloured stiffness probes. Its configured stencil reach is four fine-grid node spacings. With h_g fine-node spacings per coarse-vertex spacing, the probe radius is $R_g = 2 + \lceil 4/h_g \rceil$; colours use vertex coordinates modulo $2R_g + 1$ and the component or enrichment slot. Recovery assumes the resulting colour separation resolves every interacting coarse pair. This setup supports the Q_1 and enriched Q_1 spaces. The explicit Q_2 study uses the CPU construction in Appendix F.1.

The recovered matrix is Jacobi scaled with the corresponding column scaling of V . Cholesky factorisation reads its lower triangle, trying diagonal shifts

in the order $0, 10^{-12}, 10^{-10}, 10^{-8}, 10^{-6}, 10^{-4}$. The selected value is stored with the geometry factorisation (the coarse-factor shift); in the recorded setups it was 0 on six cells and 10^{-12} on H1 (data archive record of the setup timings). On three lattice cells, the probed factor agrees with the element-assembled Galerkin factor to a relative 1.3×10^{-11} or better (data archive record of the probing test). To characterise such a solve algebraically, let V_s denote the scaled basis, $A_s = V_s^T A V_s \succeq 0$, $\xi \geq 0$, $A_s + \xi I \succ 0$, $G_s = (A_s + \xi I)^{-1}$, and $b_s = V_s^T r_I$. Then

$$\|d_I\|_A^2 - \|d_I - V_s G_s b_s\|_A^2 = b_s^T G_s (A_s + 2\xi I) G_s b_s \geq 0. \quad (\text{F.1})$$

Expanding the squared norms, as in Eq. (D.6) with basis V_s , $G = G_s$, and A_c replaced by A_s , gives $b_s^T (2G_s - G_s A_s G_s) b_s$, and multiplication by $A_s + \xi I$ establishes the equality. A positive shift changes the exact projection but preserves this energy decrease when the solved matrix is the stated scaled Galerkin matrix. The identity assumes that matrix equality, symmetry, and consistent forward and transpose solves.

Appendix F.3. Precision and normalisation

The learned displacement extension, rigid-body reconstruction, and prescribed retained entries use float32. Quadratic energies and sparse stiffness products use float64. The differentiable correction implementation computes its smoothing and coarse algebra in float64 in training and in the accuracy studies of Sections 5.2–5.8, and returns to the input displacement dtype; the timed lattice route of Table 5 runs the smoothing and the coarse solve in float32 (Supplementary Note S5). Convolutions use true float32 arithmetic; TF32 and lower precisions are not used. The explicit inference transpose also crosses the float64/float32 boundary around its network action. Sensitivity contractions use float32 element products with float64 accumulation. These arithmetic choices approximate the real linear maps in the preceding derivations.

Evaluation path	Initial field and correction	Condensed action
Fixed-weight coarse study (base network)	Float32 network field converted to float64; explicit CPU Galerkin matrix and float64 correction	The saved comparison evaluates corrected field energies; it is separate from the deployment timing study.
Differentiable correction (training; accuracy evaluation of NICE, the corrected base network and Smoothing-trained)	Float32 learned field; smoothing and coarse algebra in float64; correction routines return to their input dtype	The correction and its transpose enter the stated variational action. Smoothing-trained uses eight smoothing steps and no coarse-grid correction.
Timed lattice route (Table 5)	Float32 network and rigid reconstruction; smoothing and coarse solve in float32; float64 stiffness product of the condensed action	Same operations and transpose as above; the fp32 correction changes the $2 \times 2 \times 2$ compliance errors by less than 3×10^{-9} (Supplementary Note S5).
Explicit inference implementation	Float32 network and rigid reconstruction; float64 stiffness product; dtype conversions before the transpose action	The transpose F^T follows the actual configured correction sequence, with finite-precision consistency assessed separately.

In the mixed-precision two-grid implementation, each smoothing routine returns its input dtype. Consequently a float32 pre-smoothed field can be rounded before entering the float64 coarse solve; the final corrected field returns to the original dtype. These casts belong to the numerical implementation rather than the real-linear maps used in the identities.

The role of actual retained values can be isolated algebraically. Suppose a computed field v has retained entries $q' = J_P v = q + b_P$, and set $d = v - E q$. Without assuming $b_P = 0$,

$$v^T K v - q^T S q = d^T K d + 2b_P^T S q. \quad (\text{F.2})$$

Using the actual trace instead gives the nonnegative variational difference $v^T K v - q'^T S q'$. In exact arithmetic applied to those actual vectors, with $r_I = (K v)_I$,

$$v^T K v - 1 = r_I^T A^{-1} r_I + (q'^T S q' - 1). \quad (\text{F.3})$$

Thus a score formed by subtracting one also depends on the normalisation of the stored direction. Recomputed exact denominators, stored unit-energy assumptions, and finite-precision quadratic forms are separate parts of a numerically evaluated energy error.

Appendix F.4. Moment integration

The reference integration initialises 4^3 subcells per active background element and refine partially occupied subcells once into 2^3 children. Thus the finest subcell width near the material boundary is $h/8$. Fully occupied subcells contribute analytic tensor-product monomial moments. At the finest partial level, each subcell is divided into six Kuhn tetrahedra and clipped successively by the three interpolated inequalities defining the band and the cut plane. The tetrahedra are integrated with a rule of order four. The accumulated moments are multiplied by the physical Jacobian $(2n)^{-3}$. Refinement is local to partial subcells; it does not uniformly subdivide every active element to the finest level.

Appendix G. Network coefficients and directional training

Appendix G.1. Geometry features and displacement maps

The architecture in Figure 3 separates geometry-dependent coefficient generation from displacement propagation. Let N denote the number of active nodes, N_P the number of retained nodes and B the number of simultaneous displacement directions. Then $n_a = 3N$, $p = 3N_P$, and the deformational input $q_d = \Pi_P q$ has shape $p \times B$. The fine displacement features have shape $N \times B \times 32$; a coarse latent level k with $N^{(k)}$ participating vertices has shape $N^{(k)} \times B \times 32$, where $k = 1, 2, 3$ index the 33, 17 and 9 grids and $k = 0$ the fine 65 grid. Geometry embeddings, the 64-component element and node feature vectors of Section 4.4, carry no displacement-direction dimension. Element, face, retained-node and grid-transfer incidences are constructed before the network is evaluated.

Geometry encoding. Each element has 126 input features: material volume fraction, its logarithm, and the remaining 124 moments normalised

by material volume. The 11 node features contain retained, box, cut-band and weak-support indicators, a normalised logarithmic stiffness diagonal, and six sine/cosine coordinate entries. The stiffness summary is the Frobenius norm of the node's diagonal 3×3 block. A node is designated weakly supported when this norm is less than 0.01 times its median over active nodes. These features supply information about both material occupancy and its mechanical support.

The element encoder maps 126 inputs to a 64-component embedding. Mean aggregation of incident element embeddings, concatenated with the 11 node features, supplies the 75-input node encoder. Two residual message-passing rounds (the two rounds of exchange in Section 4.4) then update elements from the mean embeddings of their 27 nodes and update nodes from their incident elements. Each update uses the current embedding and the aggregated neighbouring embedding as a 128-component input. A ghost-face embedding is formed from the two adjacent element embeddings and a three-component indicator of the face axis. All encoders and coefficient heads use two affine layers separated by GELU, with hidden width 64. Every variant uses only the features described here.

Local linear interactions. The three retained displacement components are lifted by a shared matrix $W_{\text{in}} \in \mathbb{R}^{3 \times 32}$, while interior features are initially zero. Denote this initial feature field by $X^0(q)$, with its dependence on q occurring through q_d . For each element or face stencil t , local node slot s and head h , the interaction first gathers and mixes channels:

$$Z_{th} = \left(\sum_{s=1}^{27} a_{tsh}(\eta) X_{i(t,s)} \right) W_{\ell h}, \quad \Delta X_i = \sum_{(t,s): i(t,s)=i} \sum_{h=1}^4 \omega_{ts} b_{tsh}(\eta) Z_{th}.$$

Here $W_{\ell h} \in \mathbb{R}^{32 \times 32}$ is a shared learned channel map for layer ℓ and head h . The scalar gather and scatter coefficients a and b are generated separately from the element or face embedding, the incident node embedding and an eight-dimensional slot embedding, a learned 27×8 table indexed by the local slot (one table for element stencils and one for face stencils). Their 136-component concatenation identifies both the local material context and the position within the stencil. The four heads provide distinct gather--mix--scatter contributions on the same fixed incidence. Equation (16) adds their update to the incoming state and restores the prescribed retained features after every local interaction.

The element stencil contains all 27 Q_2 nodes. The learned ghost-face stencil contains 18 owner-side nodes and nine neighbour-side nodes, ordered consistently by face direction. It defines a learned communication map on a face neighbourhood; the full mechanical ghost-penalty contribution remains in K .

The node scatter accounts for the number and relative stiffness of incident contributions. If k_{es} is the Frobenius norm of the element's diagonal 3×3 block at slot s , the element weights are

$$\pi_{es} = \frac{k_{es}}{\sum_{(e',s') \rightarrow i} k_{e's'}}, \quad \omega_{es} = \frac{1 - \lambda_{\text{mix}}}{\deg(i)} + \lambda_{\text{mix}} \pi_{es}, \quad (e, s) \mapsto i. \quad (\text{G.1})$$

The face weights use the corresponding ghost-contribution diagonal blocks and face incidences. When all incident stiffness norms at a node vanish, the stiffness share is defined by inverse incidence count. Each layer has its own learned λ_{mix} , initialised at zero (plain incidence averaging), without a convex-interval constraint. This weighting changes coefficients on the prescribed scatter map. For the additional weak-region layers, a stencil is selected when it contains a weak node; its weights retain the normalisation of the full element or face incidence set.

Multilevel propagation. The latent hierarchy enlarges the spatial range of communication while retaining the fine field through additive skips. Let t_{ia} be the fixed trilinear incidence weight between a fine node i and a coarse vertex a . Positive geometry coefficients r_i and p_i define restriction and prolongation componentwise:

$$(\mathcal{R}X)_a = \frac{\sum_i t_{ia} r_i X_i}{\sum_i t_{ia} r_i}, \quad (\mathcal{P}X_c)_i = p_i \frac{\sum_a t_{ia} X_{c,a}}{\sum_a t_{ia}}.$$

Only coarse vertices reached by the stored trilinear incidences participate. The denominators depend on geometry alone, so both maps are linear in displacement. Restriction and prolongation are separately parameterised. Their relationship does not impose symmetry on the raw extension; the transpose in Eq. (11) is obtained from the complete composition of operations.

On each coarse level, a residual convolution takes the form

$$X \leftarrow X + g_j^{(k)}(\eta) \odot \text{Conv}_j^{(k)}(X).$$

Each convolution has a $3 \times 3 \times 3$ kernel, 32 input and output channels, unit padding and no bias or displacement activation. Participating latent

nodes are inserted into the corresponding background grid, with unused positions set to zero; the convolution output is sampled back at those nodes. The geometry branch supplies one multiplicative coefficient per participating node, channel and convolution. Its coarse embeddings are obtained by fixed trilinear weighted averaging.

The full displacement sequence is specified below. Grid sizes count background positions per axis, whereas the participating node counts depend on geometry. Every row carries 32 displacement channels until the final projection.

Stage	Operations in execution order
Fine input, 65 grid	Lift q_d on retained nodes, set interior features to zero; apply four element-then-face interaction pairs; save $Y^{(0)}$.
Downward, 33 grid	Restrict 65→33; apply two residual convolutions; save $Y^{(1)}$.
Downward, 17 grid	Restrict 33→17; apply two residual convolutions; save $Y^{(2)}$.
Downward, 9 grid	Restrict 17→9; apply two residual convolutions.
Upward, 9 grid	Apply two residual convolutions; prolong 9→17; add $Y^{(2)}$.
Upward, 17 grid	Apply two residual convolutions; prolong 17→33; add $Y^{(1)}$.
Upward, 33 grid	Apply two residual convolutions; prolong 33→65; add $Y^{(0)}$.
Fine output, 65 grid	Restore retained features; apply four element-then-face interaction pairs, then four weak-region element-then-face pairs; project 32→3.

At the upward step that returns to level $k \in \{0, 1, 2\}$, the combination is $X = Y^{(k)} + \sigma^{(k)} \mathcal{P}^{(k)} X_c$, where X_c is the field on level $k + 1$ and $\sigma^{(k)}$ is a learned scalar. Thus convolution precedes prolongation, the skips are additive, and the 9-position grid receives two downward and two upward convolutions. There are twelve coarse convolution updates in total. The fine retained values are restored after the hierarchy returns to the active nodes. These latent operations communicate within the extension; the mechani-

cal operator still accepts and returns all p retained DOFs. The coarse-grid (Galerkin) correction of Section 4.3 instead uses the interior stiffness and the interior equilibrium residual.

Coefficient bounds and output. The local heads produce separate gather/scatter values for eight ordinary element layers, twelve face layers and four additional weak-region element layers, each with four heads and 27 slots. Raw head outputs are multiplied by 0.2. For a finite positive group bound $a_{\max}$, the coefficient map is

$$\mathcal{C}(z) = \begin{cases} z, & |z| \leq k_b a_{\max}, \\ \text{sgn}(z) \left[k_b a_{\max} + (1 - k_b) a_{\max} \tanh\left(\frac{|z| - k_b a_{\max}}{(1 - k_b) a_{\max}}\right) \right], & |z| > k_b a_{\max}. \end{cases}$$

All variants of Table 2 set $k_b = 0.5$; each $a_{\max}$ is twice the largest magnitude of its coefficient group recorded in a calibration pass over training geometries, so the map is the identity up to that recorded maximum. Table ST20 (Supplementary Note S7) lists the groups, the initialisation and the parameter count of every block. An infinite bound denotes the identity map. Bounds are fixed model-state arrays indexed by coefficient type, layer and gather/scatter role. Positive transfer coefficients apply the corresponding bound to $\text{softplus}(z)$ and add 10^{-3} ; convolution coefficients use $2 \text{sigmoid}(z)$. All these nonlinearities act on geometry-derived coefficients. The channel matrices, convolution kernels and skip scalars are shared learned parameters, whereas the coefficient-head outputs vary with geometry.

A bias-free matrix $W_{\text{out}} \in \mathbb{R}^{32 \times 3}$ returns the fine features to nodal displacement in the prescribed coordinate order. Rigid reconstruction and retained-value restoration then produce Eq. (17). The energy action Eq. (11) uses this complete map, together with the correction $\mathcal{W}$, and its transpose; it therefore determines the force from the same displacement representation.

Frame consistency. For an orthogonal cube transformation, the deterministic signed permutation maps satisfy

$$K' = T_a K T_a^T, \quad E' = T_a E T_P^T, \quad S' = T_P S T_P^T. \quad (\text{G.2})$$

Here $T_a \in \mathbb{R}^{n_a \times n_a}$ and $T_P \in \mathbb{R}^{p \times p}$ are the signed permutation matrices that the cube transformation induces on the active and the retained DOFs of the cell, and primes denote quantities of the transformed cell. The augmented geometry and vector field are passed through the network and the predicted

displacement is returned to the original frame for mechanical evaluation. These relations describe the transformed target and the frame-consistency condition. Sampling the cube views trains the geometry-conditioned extension towards that condition; displacement linearity alone does not establish rotational equivariance.

Appendix G.2. Direction sets

The normalised retained directions are

$$q_j = \frac{\Pi_P q_j^{\text{raw}}}{\sqrt{(\Pi_P q_j^{\text{raw}})^T S (\Pi_P q_j^{\text{raw}})}}, \quad (\text{G.3})$$

where q_j^{raw} is the direction produced by its class before rigid-body removal and energy normalisation.

The direction classes used by the trainer have the following mechanical definitions.

Class	Construction before rigid removal and energy normalisation
force	Equilibrated nodal loads from smooth plane waves or localised Gaussian patches, with a subset loading the cut boundary.
force_c	Consistently integrated self-equilibrated tractions on material box faces; a subset also loads the cut face. Equilibrium is imposed in traction-quadrature space.
face / face_c	Single-face equilibrated nodal loads or consistently integrated tractions.
support	Responses with soft spring support on one box face and equilibrated loads on other faces.
support_k	Consistent loads with springs scaled by the local stiffness diagonal and a logarithmically sampled factor in $[0.3, 3]$; the cut face remains unloaded.

Class	Construction before rigid removal and energy normalisation
<code>macro</code>	Equal sampling of uniform strain, quadratic, and cubic imposed displacement fields.
<code>grf</code>	Multiscale displacement fields spanning 0.5–24 spatial cycles across the reference box.
<code>glued</code>	Traces induced by a neighbouring cell with shared interface degrees of freedom and far-face clamping or springs; one quarter of samples load only the neighbour.
<code>adv</code>	Directions selected by the generalised-energy search described below.

The exact reference computation for an isolated cell removes the rigid-body component and normalises the resulting retained direction by Eq. (G.3). Glued responses therefore supply a contextual direction while the energy and sensitivity labels refer to the target cell's operator. The direction sets store the normalised vectors in single precision.

Appendix G.3. Optimisation and difficult-direction search

The variants of Table 2 use batches of 16 directions and change the geometry at every step; three geometries of the training set are held on the GPU, and one of them is replaced every 100 steps. Their nominal class weights, in the order `force`, `force_c`, `support`, `support_k`, `face`, `face_c`, `macro`, `grf`, `glued`, `adv`, are 0.10, 0.10, 0.075, 0.075, 0.10, 0.10, 0.10, 0.125, 0.075, 0.15. The class weights are renormalised over the available classes and assigned by systematic quotas. A run of N_{st} training steps therefore visits at most $N_{\text{st}}/100 + 3$ distinct geometries of its training set, which contains 305 geometries for the base network and 591 for the continuations (Table ST01).

Optimisation uses Adam, a one-cycle learning-rate schedule with peak 3×10^{-4} , 5% warm-up, cosine decay, and final division factor 100. The gradient norm is clipped at one. Model selection and evaluation use an exponential moving average of the network parameters, with decay 0.9997 and zero-initialisation bias correction. Table 2 identifies the variants, and

Table ST01 gives the training and model-selection settings of each. The energy term in Eq. (18) uses a 10^{-12} lower clamp inside the logarithm, and the sensitivity term, the squared relative error of the thickness sensitivity vector, has unit weight.

NICE and the Smoothing-trained variant continue the base network from its selected parameters for 15,000 training steps with the same schedule, class weights, training set and seed as the Uncorrected continuation, so that the three continuations differ only in the correction used in the forward map of Eq. (18): none for Uncorrected, eight Chebyshev smoothing steps for Smoothing-trained, and the eight-step, $Q_1(17)$, eight-step sequence for NICE. The correction is recomputed for every geometry held on the GPU (smoothing interval by power iteration, coarse matrix by coloured probing, Appendix F.2) and is treated as a fixed linear map of the network output. Gradients of Eq. (18) are propagated through the smoothing recurrence and the coarse-grid correction by automatic differentiation in double precision; the transposes used at evaluation (Appendix E) are the explicit adjoints of the same operations. Model selection (Section 5.1, Table ST01) compared the network parameters after 7,500 and 15,000 training steps of each continuation and after 10,000, 20,000, 30,000 and 40,000 steps of the base network; the continuations use those after 15,000 steps and the base network those after 30,000.

The adversarial search initialises eight candidate directions and performs four block iterations whenever a geometry enters the set held on the GPU. Applying $S_{\perp}^{-1}$ uses an equilibrated Neumann solve: six deterministically selected displacement pins remove the rigid-body motion, and the result is projected into the retained rigid complement. A search within a stored direction set instead forms $G = Q_b^T S Q_b$ and $\hat{G} = (F Q_b)^T K (F Q_b)$, solves a generalised eigenproblem after flooring the search metric at $10^{-3} \lambda_{\max}(G)$, and renormalises selected candidates with the original G . The floor applies only within the search. Candidate energy ratios remain directional observations; their maximisation does not supply an all-direction upper bound.

Appendix G.4. Direction coverage, symmetry and spectra

With Z and $S_* = Z^T S Z \succ 0$ as in Eq. (B.5), let $T_* = S_*^{-1/2} Z^T H^T A H Z S_*^{-1/2}$. For an energy-normalised direction, $x_j = S_*^{1/2} Z^T q_j$ has unit Euclidean norm and, by Eq. (B.6), $\varepsilon_j = x_j^T T_* x_j$. For N_q sampled directions, define the coverage matrix $M_{N_q} = N_q^{-1} \sum_{j=1}^{N_q} x_j x_j^T$. Then

$$\bar{\varepsilon} = \text{tr}(T_* M_{N_q}), \quad M_{N_q} \succeq \alpha I, \alpha > 0 \implies \varepsilon_* \leq \text{tr}(T_*) \leq \bar{\varepsilon}/\alpha. \quad (\text{G.4})$$

If the directions do not span the complement, choose a unit v orthogonal to them and $T_* = \Theta vv^T$ with $\Theta > 0$. Every sampled error is zero and the worst error is Θ , arbitrarily large. Such a PSD perturbation can be realised as $H^T A H$ if there is at least one interior degree of freedom. Thus retaining all DOFs in $\mathbf{P}$ defines the approximation target, while adequate direction coverage controls its learned accuracy.

The logarithmic loss is nonnegative for the exact variational construction and behaves as $\log(1+\varepsilon) = \varepsilon + O(\varepsilon^2)$ near zero. It weights large directional errors differently from their arithmetic mean. A finite average log loss L_{N_q} over these directions only supplies the weak sample bound $\sum_j \varepsilon_j \leq e^{N_q L_{N_q}} - 1$; it has no unsampled-direction implication without coverage. With exact reference normalisation, Ritz searches give attained Rayleigh quotients and hence lower bounds on the true largest quotient. Geometry-averaged direction means, their population maxima, and the full operator supremum are different statistics.

For an orthogonal signed-permutation action, the transformations in Eq. (G.2) together with $q' = T_P q$ and $B'_m = T_P B_m T_g^T$, where T_g is the signed permutation that the same transformation of the assembly induces on the free global retained displacement U , preserve energies, nullspaces and assembly. A correspondingly transformed extension has the same property. Training on transformed examples (Appendix G.1) encourages this relation but does not prove that an unconstrained learned map satisfies it on every group element.

Appendix H. Sensitivity identities and design intervals

The exact condensed derivative and the fixed-displacement energy derivative are

$$S_{,c} = E^T K_{,c} E, \quad \frac{\partial}{\partial \tau_c} \left(\frac{1}{2} q^T S q \right) = \frac{1}{2} (E q)^T K_{,c} (E q). \quad (\text{H.1})$$

For an assembly, the full local field difference is

$$F \hat{q} - E q = (F - E) q + E(\hat{q} - q) + (F - E)(\hat{q} - q). \quad (\text{H.2})$$

On a differentiable interval with fixed active topology and fixed selectors J_P and J_I , differentiating $J_P E = I_p$ gives $J_P E_{,c} = 0$. Since $J_I K E = 0$, both

terms $E_{,c}^T K E$ and $E^T K E_{,c}$ vanish. Differentiating $S = E^T K E$ therefore proves Eq. (H.1).

For a supported equilibrium, differentiating $\mathbb{K}U = f_g$ at a fixed load gives $U_{,c} = -\mathbb{K}^{-1}\mathbb{K}_{,c}U$ and hence $C_{,c} = -U^T \mathbb{K}_{,c}U$. With fixed gathers and Eq. (H.1), this becomes the sum of $-u_m^T K_{m,c} u_m$ over affected cells. For an isolated energy-normalised direction, this label holds the base-design force fixed. The assembled sensitivity labels likewise hold the base-design nodal load fixed, including loads generated from consistent tractions. Reassembling a prescribed surface traction over a changing material boundary introduces the load-derivative term stated below; differentiating the direction normalisation also defines a different quantity.

Expanding the field-based quadratic estimate at $u + d$ proves Eq. (9). In the Euclidean norm, it implies

$$|\tilde{s}_c - s_c| \leq 2\|K_{,c}u\|_2\|d\|_2 + \|K_{,c}\|_2\|d\|_2^2. \quad (\text{H.3})$$

For the corner-thickness band in Eq. (1), exact integration with fixed basis functions and ghost stabilisation gives $K_{,c} \succeq 0$ and nonpositive equilibrium compliance derivatives; Appendix H.2 proves this (Eq. H.10) and shows that the discrete moments of this study preserve it only approximately. The linear cross term in Eq. (9) can have either sign, while under exact integration the quadratic term is nonpositive; their relative importance depends on derivative-weighted alignment and error amplitude. The numerical moment-difference implementation is specified below. In an assembly, substituting Eq. (H.2) into the same quadratic expansion includes both the change of trace and its interaction with the extension error (Appendix H.1).

For the condensed stiffness $\hat{S} = F^T K F$, direct differentiation gives

$$\hat{S}_{,c} = F_{,c}^T K F + F^T K_{,c} F + F^T K F_{,c}. \quad (\text{H.4})$$

Apply the equilibrium compliance derivative to the assembly of learned substructures and use $J_P F_{,c} = 0$. The two extension terms become $2(F_{I,c}\hat{q})^T r_I$, proving Eq. (10). Its magnitude is bounded by

$$|\hat{C}_{,c} - \tilde{s}_c| \leq 2\|F_{I,c}\hat{q}\|_A\|r_I\|_{A^{-1}} \quad (\text{H.5})$$

for a single affected cell, with a sum of such bounds for several cells. This bound involves the extension's design derivative as well as its equilibrium residual. It provides no fixed error order in the field amplitude without a

corresponding assumption on that derivative. If the load depends on design, the full compliance derivative also contains $2f_{g,c}^T \widehat{U}$, and correspondingly $2f_{g,c}^T U$ for the exact assembly; changes of the DOF selectors, assembly maps or support maps contribute their own chain-rule terms, including the derivatives of $B_m^T S_m B_m$ when B_m varies. Fixed active topology does not, on its own, imply differentiability of quadrature branches or a moving-load map.

In the reference computation, moment derivatives are approximated by

$$M_{e\alpha,c} \approx \frac{M_{e\alpha}(\boldsymbol{\tau} + h_c e_c) - M_{e\alpha}(\boldsymbol{\tau} - h_c e_c)}{2h_c}, \quad h_c = 10^{-5} \tau_c. \quad (\text{H.6})$$

The active set is fixed for this calculation, and the elasticity templates and ghost contribution are held constant. The moment quadrature is recomputed at each perturbed thickness, so its clipping and integration branches may change. The resulting matrices represent a numerical design derivative on that prescribed topology. Training differentiates the field-based sensitivity loss with respect to the predicted field and network parameters. Computing the complete design derivative in Eq. (10) additionally differentiates the geometry-dependent extension and correction maps.

Appendix H.1. Sensitivity error bounds

For a fixed retained q , set $u = Eq$, $d = J_I^T Hq$, and $\mathcal{E} = d^T K d = (Hq)^T A(Hq)$. The derivatives $K_{,c}$, $c = 1, \dots, 8$, are symmetric. Define the reference vector $\mathbf{s}$ with components $s_c = -u^T K_{,c} u$, and assume $\|\mathbf{s}\|_2 > 0$. Eq. (9) then reads

$$\tilde{s}_c - s_c = -2d^T K_{,c} u - d^T K_{,c} d.$$

Let $b_c = J_I K_{,c} u$, $K_{II,c} = J_I K_{,c} J_I^T$, and let the rows of $\mathbf{B}_s \in \mathbb{R}^{8 \times i}$ be $b_c^T A^{-1/2}$. Define

$$L = \|\mathbf{B}_s\|_2, \quad Q = \left(\sum_{c=1}^8 \|A^{-1/2} K_{II,c} A^{-1/2}\|_2^2 \right)^{1/2}.$$

Cauchy–Schwarz and the quadratic-form operator bound give

$$\|\tilde{\mathbf{s}} - \mathbf{s}\|_2 \leq 2L\sqrt{\mathcal{E}} + Q\mathcal{E}, \quad e_s \leq \frac{2L\sqrt{q^T S q}}{\|\mathbf{s}\|_2} \sqrt{\varepsilon} + \frac{Qq^T S q}{\|\mathbf{s}\|_2} \varepsilon. \quad (\text{H.7})$$

All eight components are included, and the denominator is the norm of the exact sensitivity vector for that particular direction. The constants distinguish derivative coupling, interior coercivity, and reference sensitivity scale. For fixed q and $H = tH_0$, the linear coefficient vanishes precisely when $(H_0q)^T b_c = 0$ for every corner. It vanishes for all interior errors if $J_I K_{,c} E q = 0$ for every corner. A useful special case is $K_{,c} = \alpha_c K$: interior equilibrium removes the linear term and the relative sensitivity error equals the energy error when the reference vector is nonzero.

For a re-equilibrated assembly, the quadratic expansion remains exact with $d_m = \hat{u}_m - u_m$, but d_m now includes a changed retained trace. Eq. (C.5) gives, componentwise,

$$\begin{aligned} \tilde{s}_{m,c}(t) - s_{m,c} = & -2t(J_I^T H_{0,m} q_m)^T K_{m,c} u_m \\ & + t^2 \{ 2(E_m B_m \mathbb{K}^{-1} \mathbb{D}_0 U)^T K_{m,c} u_m \\ & - (J_I^T H_{0,m} q_m)^T K_{m,c} (J_I^T H_{0,m} q_m) \} + O(t^3). \end{aligned}$$

The leading term is the same as the fixed-trace term. The solution-only replacement is $O(t^2)$; its coefficient can be large on a soft assembly or relative to a small local reference. If only one cell has a learned extension, an exact neighbouring cell has no $O(t)$ reconstruction term, although its solved trace can change at order t^2 .

There is also a global-to-local bound. Suppose K_m has its fixed rigid-body kernel, each $K_{m,c}$ annihilates that kernel, and

$$\Gamma_m = \left(\sum_c \|K_{m,*}^{-1/2} K_{m,c,*} K_{m,*}^{-1/2}\|_2^2 \right)^{1/2} < \infty$$

on its rigid complement. Since $u_m^T K_m u_m = q_m^T S_m q_m = w_m C > 0$, set $\chi_m = \|\mathbf{s}_m\|_2 / (w_m C) > 0$ and $e_C = (C - \hat{C})/C$. The field expansion and Eq. (6) yield

$$\boxed{e_{s,m} \leq \frac{\Gamma_m}{\chi_m} \left(2\sqrt{\frac{e_C}{w_m}} + \frac{e_C}{w_m} \right)}. \quad (\text{H.8})$$

Indeed, the local error energy is at most $e_C C$. Substitution into $\|\delta \mathbf{s}_m\|_2 \leq \Gamma_m \left(2\sqrt{w_m C a_m(d_m, d_m)} + a_m(d_m, d_m) \right)$ proves the result. This bound makes the roles of a small energy share and a small reference sensitivity explicit. The recorded results do not measure Γ_m , χ_m , or the local error energy, so

Eq. (H.8) is a quantitative theoretical explanation, not a fitted attribution of a reported percentage.

The field-only and solution-only replacements of Table ST11 do not form an additive sensitivity decomposition. Denote the three terms of Eq. (H.2) by $d^{(1)} = (F - E)q$, $d^{(2)} = E(\hat{q} - q)$ and $d^{(3)} = (F - E)(\hat{q} - q)$, so that the full field error is $d^{(1)} + d^{(2)} + d^{(3)}$. In addition to the field-only and solution-only vector differences, its sensitivity discrepancy contains $-2d^{(3)T}K_{,c}u - d^{(3)T}K_{,c}d^{(3)} - 2d^{(1)T}K_{,c}d^{(2)} - 2d^{(1)T}K_{,c}d^{(3)} - 2d^{(2)T}K_{,c}d^{(3)}$. Knowing only the three output norms does not determine these terms or their angles.

Appendix H.2. The complete design derivative and the sign of the thickness derivative

Differentiating $J_I K E = 0$ with fixed selectors and $J_P E_{,c} = 0$ gives

$$A E_{I,c} = -J_I K_{,c} E,$$

while the condensed derivative $S_{,c} = E^T K_{,c} E$ is Eq. (H.1). The exact assembly compliance derivative $C_{,c} = -U^T \mathbb{K}_{,c} U$ and, from the equilibrium of the assembly with $\hat{S}$, Eq. (10) were derived above: $\hat{C}_{,c}$ is the sum over affected cells of $\hat{s}_{m,c} - 2(F_{I,m,c} \hat{q}_m)^T r_{I,m}$. The design derivative of the solved trace has already been eliminated using equilibrium; the residual term differentiates F at fixed trace. It includes geometry-conditioned coefficients and every design-dependent correction operation.

Equivalently, with $K_{II,c}$ as in Appendix H.1, differentiate the variational stiffness error:

$$\boxed{(\hat{S} - S)_{,c} = H_{,c}^T A H + H^T K_{II,c} H + H^T A H_{,c}.} \quad (\text{H.9})$$

Its norm is bounded by $2\|A\|\|H\|\|H_{,c}\| + \|K_{II,c}\|\|H\|^2$. Thus a value error $H = O(t)$ by itself does not imply a quadratic derivative error. A uniformly C^1 -small extension error, $H = tH_0(\tau)$ with bounded $H_0, H_{0,c}$, gives $(\hat{S} - S)_{,c} = O(t^2)$. With uniformly bounded exact solutions and inverse stiffnesses, this also yields $\hat{C}_{,c} - C_{,c} = O(t^2)$. One useful exact splitting is

$$\hat{C}_{,c} - C_{,c} = -\hat{U}^T \mathbb{D}_{,c} \hat{U} - (\hat{U} - U)^T \mathbb{K}_{,c} (\hat{U} + U),$$

where $\mathbb{D} = \hat{\mathbb{K}} - \mathbb{K}$ as in Eq. (C.1).

The $O(t)$ field-estimate term can cancel against the residual term. At the same trace, $J_I K_{,c} E q = -A E_{I,c} q$. For $H = tH_0$, the linear field-estimate

error is $+2t(H_0q)^T AE_{I,c}q$, while the linear residual-chain contribution is $-2t(E_{I,c}q)^T AH_0q$. This cancellation explains how a first-order field estimate and a second-order complete derivative can coexist.

Under a fixed basis and exact integration, increasing one band parameter τ_c with nonnegative Q_1 shape functions enlarges the material domain. With fixed ghost stabilisation, this gives $K_{,c} \succeq 0$, $S_{,c} \succeq 0$, and $C_{,c} \leq 0$. The field estimate is then also nonpositive for any field. Pointwise variational stiffness dominance does not itself enforce monotonicity of the surrogate compliance with respect to design: the residual-chain term can change the sign of its complete derivative. This is a separate mechanical consistency question from symmetry and positive definiteness at a fixed design. Numerical moment derivatives inherit the monotonicity statement only when they represent the same nested-domain integration rule.

More explicitly, for $h > 0$, $N_c^{Q_1}(x) \geq 0$ implies $\Omega(\tau) \subseteq \Omega(\tau + he_c)$. For any discrete coefficient vector v , with interpolated displacement field v_h ,

$$v^T[K(\tau + he_c) - K(\tau)]v = \int_{\Omega(\tau + he_c) \setminus \Omega(\tau)} \nabla^s v_h : \mathbb{C} : \nabla^s v_h \, dx \geq 0. \quad (\text{H.10})$$

Here ∇^s is the symmetric gradient, the elasticity tensor $\mathbb{C}$ and the basis are fixed, and the ghost contribution cancels. Taking the differentiable limit proves positive semidefiniteness of the thickness derivative. A centred difference of stiffness matrices assembled from exactly nested domains is also PSD. Changes of adaptive integration subdivision, approximate moments, or independently selected stabilisation can interrupt that discrete nesting relation; fixed active topology alone does not verify its numerical preservation. Figure S01(c) reports a step-refinement study of the numerical derivative on four cells: relative to the production step $10^{-5}\tau_c$, the sensitivity changes by at most 2.6×10^{-7} at $10^{-3}\tau_c$, 2.5×10^{-9} at $10^{-4}\tau_c$ and 1.6×10^{-10} at $10^{-6}\tau_c$; the hundredfold reduction per decade is the second-order truncation of the central difference, so no integration branch changes within $\pm 10^{-3}\tau_c$ at the fixed active set on these cells. On the same cells, the central-difference stiffness derivative agrees with differences of the re-solved compliance to 4×10^{-8} . The element-level check of Supplementary Table ST15 shows that the discrete moments do not preserve the nesting everywhere. The element stiffnesses are positive semidefinite to rounding, but the exact derivative of the discrete moments at fixed clipping topology gives element derivative matrices with negative eigenvalues: in the uncut cells the smallest ratio $\lambda_{\min}/\max|\lambda|$ is -2.3×10^{-3} , and in four of the six cut cells between 2 and 11 partially

filled elements (of 762 to 6,033) have a uniform-thickening derivative whose most negative eigenvalue exceeds 10^{-6} of its largest in magnitude, some of them negative semidefinite. The production central difference agrees with this exact discrete derivative to 2×10^{-8} . Eq. (H.10), and the bound (H.11) that rests on it, therefore describe exact integration; the discrete model satisfies them only approximately. The reference sensitivities of this study are derivatives of the discrete model and are compared as such.

This positive structure strengthens the sensitivity interpretation without requiring an indefinite derivative. Put $a_c = u^T K_{,c} u = -s_c \geq 0$, $\zeta_c = d^T K_{,c} d \geq 0$. Positive-semidefinite Cauchy–Schwarz gives

$$|\tilde{s}_c - s_c| \leq 2\sqrt{a_c \zeta_c} + \zeta_c, \quad \|\tilde{\mathbf{s}} - \mathbf{s}\|_2 \leq 2\sqrt{\sum_c a_c \zeta_c} + \|\boldsymbol{\zeta}\|_2. \quad (\text{H.11})$$

If $\gamma_c = \|A^{-1/2} K_{II,c} A^{-1/2}\|_2$ (so that $\|\boldsymbol{\gamma}\|_2 = Q$ of Eq. (H.7)), then $\zeta_c \leq \gamma_c \mathcal{E}$, giving $2\sqrt{\sum_c a_c \gamma_c} \sqrt{\mathcal{E}} + \|\boldsymbol{\gamma}\|_2 \mathcal{E}$. The quadratic term in the signed sensitivity discrepancy is nonpositive; the cross term can have either sign. The reference sensitivity vector has nonpositive components. A small reference norm reflects small derivative-weighted local strain energy. For a general variable that moves a cut or redistributes material, an indefinite derivative is possible; that generality is unnecessary to explain the present corner-thickening mechanism.

Design-dependent loads and assembly maps add the chain-rule terms stated after Eq. (H.5). These changes are different derivative problems rather than modifications of Eq. (H.9).

Appendix I. A computable lower bound on the energy error

For any nonzero interior test vector z , Cauchy–Schwarz applied to $A^{1/2}z$ and $A^{-1/2}r_I$ gives

$$\Delta(q) := q^T(\hat{S} - S)q = r_I^T A^{-1} r_I \geq L_z := \frac{(z^T r_I)^2}{z^T A z}. \quad (\text{I.1})$$

A preconditioned residual, a coarse approximation, or a Krylov iterate can supply z . Evaluating the final one-vector quotient with the original A makes the inequality independent of how that vector was generated. For $e_h = \hat{u}^T K \hat{u}$ and $e_h - L_z > 0$, the map $x \mapsto x/(e_h - x)$ is increasing on the relevant interval. Therefore

$$\varepsilon(q) \geq \frac{L_z}{e_h - L_z}, \quad \mu_* \geq 1 + \frac{L_z}{e_h - L_z}. \quad (\text{I.2})$$

If the exact reference energy is known, $L_z/(q^T S q)$ gives a sharper lower estimate. A large value detects an inaccurate direction; a small value alone does not bound the error from above. These are real-arithmetic inequalities, with numerical evaluation requiring the stated original quadratic forms.

References

- Adams, M., Brezina, M., Hu, J., Tuminaro, R. (2003). Parallel multigrid smoothing: polynomial versus Gauss–Seidel. *Journal of Computational Physics*, 188(2), 593–610.
- Alexandersen, J., Lazarov, B. S. (2015). Topology optimisation of manufacturable microstructural details without length scale separation using a spectral coarse basis preconditioner. *Computer Methods in Applied Mechanics and Engineering*, 290, 156–182.
- Amir, O., Bendsøe, M. P., Sigmund, O. (2009). Approximate reanalysis in topology optimization. *International Journal for Numerical Methods in Engineering*, 78(12), 1474–1491.
- Amir, O., Stolpe, M., Sigmund, O. (2010). Efficient use of iterative solvers in nested topology optimization. *Structural and Multidisciplinary Optimization*, 42(1), 55–72.
- Ballani, J., Huynh, D. B. P., Knezevic, D. J., Nguyen, L., Patera, A. T. (2018). A component-based hybrid reduced basis/finite element method for solid mechanics with local nonlinearities. *Computer Methods in Applied Mechanics and Engineering*, 329, 498–531.
- Becker, R., Rannacher, R. (2001). An optimal control approach to a posteriori error estimation in finite element methods. *Acta Numerica*, 10, 1–102.
- Bendsøe, M. P., Sigmund, O. (2004). *Topology Optimization: Theory, Methods, and Applications*, 2nd edn. Springer, Berlin, Heidelberg.
- Bonilla Moreno, G., Guarino, G., Antolin, P. (2027). A ROM-based BDDC solver for unfitted p-FEM level-set-based two-dimensional lattice structures. *Computer Methods in Applied Mechanics and Engineering*, 463, 119304.
- Boullé, N., Townsend, A. (2023). Learning elliptic partial differential equations with randomized linear algebra. *Foundations of Computational Mathematics*, 23(2), 709–739.

- Burman, E. (2010). Ghost penalty. *Comptes Rendus Mathématique*, 348(21–22), 1217–1220.
- Burman, E., Claus, S., Hansbo, P., Larson, M. G., Massing, A. (2015). CutFEM: Discretizing geometry and partial differential equations. *International Journal for Numerical Methods in Engineering*, 104(7), 472–501.
- Capuano, G., Rimoli, J. J. (2019). Smart finite elements: A novel machine learning application. *Computer Methods in Applied Mechanics and Engineering*, 345, 363–381.
- Chen, W., Li, M. (2026). Conditioned numerical shape functions on unfitted reduced coarse elements for robust analysis of complex solid structures. *Computer-Aided Design*, 193, 104038.
- Dohrmann, C. R. (2003). A preconditioner for substructuring based on constrained energy minimization. *SIAM Journal on Scientific Computing*, 25(1), 246–258.
- Dohrmann, C. R. (2007). An approximate BDDC preconditioner. *Numerical Linear Algebra with Applications*, 14(2), 149–168.
- E, W., Yu, B. (2018). The Deep Ritz method: a deep learning-based numerical algorithm for solving variational problems. *Communications in Mathematics and Statistics*, 6(1), 1–12.
- Efendiev, Y., Galvis, J., Hou, T. Y. (2013). Generalized multiscale finite element methods (GMsFEM). *Journal of Computational Physics*, 251, 116–135.
- Eftang, J. L., Patera, A. T. (2013). Port reduction in parametrized component static condensation: approximation and a posteriori error estimation. *International Journal for Numerical Methods in Engineering*, 96(5), 269–302.
- Falgout, R. D., Vassilevski, P. S., Zikatanov, L. T. (2005). On two-grid convergence estimates. *Numerical Linear Algebra with Applications*, 12(5–6), 471–494.
- Falgout, R. D., Yang, U. M. (2002). hypre: A library of high performance preconditioners. In *Computational Science — ICCS 2002*, Lecture Notes in Computer Science, 2331, 632–641. Springer, Berlin, Heidelberg.
- Farhat, C., Lesoinne, M., LeTallec, P., Pierson, K., Rixen, D. (2001). FETI-DP: a dual–primal unified FETI method—part I: A faster alternative to the two-level FETI method. *International Journal for Numerical Methods in Engineering*, 50(7), 1523–1544.
- Fraeijs de Veubeke, B. (1965). Displacement and equilibrium models in the finite element method. In Zienkiewicz, O. C., Holister, G. S. (eds.), *Stress Analysis*, Chapter 9, 145–197. John Wiley & Sons. Reprinted in *International Journal for Numerical Methods in Engineering*, 52(3) (2001), 287–342, doi:10.1002/nme.339.

- Giles, M. B., Pierce, N. A. (2000). An Introduction to the Adjoint Approach to Design. *Flow, Turbulence and Combustion*, 65(3–4), 393–415.
- Gogu, C. (2015). Improving the efficiency of large scale topology optimization through on-the-fly reduced order model construction. *International Journal for Numerical Methods in Engineering*, 101(4), 281–304.
- Golub, G. H., Varga, R. S. (1961). Chebyshev semi-iterative methods, successive overrelaxation iterative methods, and second order Richardson iterative methods. Part I. *Numerische Mathematik*, 3(1), 147–156.
- Greenfeld, D., Galun, M., Basri, R., Yavneh, I., Kimmel, R. (2019). Learning to optimize multigrid PDE solvers. In *Proceedings of the 36th International Conference on Machine Learning*, PMLR 97, 2415–2423.
- Guo, Y., Liu, C., Du, Z., Jia, Y., Jiang, C., Guo, X., Shen, C. (2026a). High-Generalization AI-Enhanced Mechanical Analysis and Topology Optimization via Cubic Bézier Interpolation of Substructure Boundary Displacements. *Computer Methods in Applied Mechanics and Engineering*, 456, 118955.
- Guo, Y., Liu, C., Du, Z., Liu, J., Feng, J., Zhang, X., Li, Y., Yang, T., Shen, C., Guo, X. (2026b). PIML-OFEM: A New Large-Scale Structural Analysis Method Based on Problem-Independent Machine Learning and Overlapping Finite Element Technique. arXiv:2607.22019v1 [Preprint].
- Guyan, R. J. (1965). Reduction of stiffness and mass matrices. *AIAA Journal*, 3(2), 380.
- Hackbusch, W. (1985). *Multi-Grid Methods and Applications*. Springer, Berlin.
- Haftka, R. T., Gürdal, Z. (1992). *Elements of Structural Optimization*, 3rd edn. Kluwer, Dordrecht.
- He, J., Liu, X., Xu, J. (2024). MgNO: Efficient parameterization of linear operators via multigrid. In *International Conference on Learning Representations (ICLR 2024)*, 53409–53428. OpenReview: 8OxL034uEr.
- Heinlein, A., Klawonn, A., Lanser, M., Weber, J. (2021). Combining machine learning and adaptive coarse spaces—a hybrid approach for robust FETI-DP methods in three dimensions. *SIAM Journal on Scientific Computing*, 43(5), S816–S838.
- Henson, V. E., Yang, U. M. (2002). BoomerAMG: A parallel algebraic multigrid solver and preconditioner. *Applied Numerical Mathematics*, 41(1), 155–177.
- Hirschler, T., Bouclier, R., Antolin, P., Buffa, A. (2024). Reduced order modeling based inexact FETI-DP solver for lattice structures. *International Journal for Numerical Methods in Engineering*, 125(8), e7419.
- Hou, T. Y., Wu, X.-H. (1997). A Multiscale Finite Element Method for Elliptic

- Problems in Composite Materials and Porous Media. *Journal of Computational Physics*, 134(1), 169–189.
- Huang, M., Du, Z., Liu, C., Zheng, Y., Cui, T., Mei, Y., Li, X., Zhang, X., Guo, X. (2022). Problem-independent machine learning (PIML)-based topology optimization—A universal approach. *Extreme Mechanics Letters*, 56, 101887.
- Huang, M., Cui, T., Liu, C., Du, Z., Zhang, J., He, C., Guo, X. (2023). A Problem-Independent Machine Learning (PIML) enhanced substructure-based approach for large-scale structural analysis and topology optimization of linear elastic structures. *Extreme Mechanics Letters*, 63, 102041.
- Huang, M., Liu, C., Guo, Y., Zhang, L., Du, Z., Guo, X. (2024). A mechanics-based data-free Problem Independent Machine Learning (PIML) model for large-scale structural analysis and design optimization. *Journal of the Mechanics and Physics of Solids*, 193, 105893.
- Huynh, D. B. P., Knezevic, D. J., Patera, A. T. (2013). A static condensation reduced basis element method: approximation and a posteriori error estimation. *ESAIM: Mathematical Modelling and Numerical Analysis*, 47(1), 213–251.
- Irons, B. (1965). Structural eigenvalue problems—elimination of unwanted variables. *AIAA Journal*, 3(5), 961–962.
- Jiang, C., Liu, C., Guo, Y., Du, Z., Zhang, W., Guo, X. (2026). Mechanically consistent equivariant machine learning for substructural analysis. *Composite Structures*, 397, 120865.
- Jiang, H., Zhan, J., Zhang, C., Wang, F. (2026). Convex Neural Energy Elements: Monolithic Finite-Element Assembly of Geometry-Parameterized Neural Operators with Stability and Error Guarantees. arXiv:2608.02036v1 [Preprint].
- Klawonn, A., Rheinbach, O. (2007). Inexact FETI-DP methods. *International Journal for Numerical Methods in Engineering*, 69(2), 284–307.
- Kopaničáková, A., Karniadakis, G. E. (2025). DeepONet Based Preconditioning Strategies for Solving Parametric Linear Systems of Equations. *SIAM Journal on Scientific Computing*, 47(1), C151–C181.
- Li, D., Liao, W., Dai, N., Dong, G., Tang, Y., Xie, Y. M. (2018). Optimal design and modeling of gyroid-based functionally graded cellular structures for additive manufacturing. *Computer-Aided Design*, 104, 87–99.
- Li, J., Widlund, O. B. (2007). On the use of inexact subdomain solvers for BDDC algorithms. *Computer Methods in Applied Mechanics and Engineering*, 196(8), 1415–1428.
- Li, Z., Huang, D. Z., Liu, B., Anandkumar, A. (2023a). Fourier neural operator with learned deformations for PDEs on general geometries. *Journal of Machine*

- Learning Research*, 24(388), 1–26.
- Li, Z., Kovachki, N., Choy, C., Li, B., Kossaifi, J., Otta, S., Nabian, M. A., Stadler, M., Hundt, C., Azizzadenesheli, K., Anandkumar, A. (2023b). Geometry-informed neural operator for large-scale 3D PDEs. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, 35836–35854.
- Luz, I., Galun, M., Maron, H., Basri, R., Yavneh, I. (2020). Learning algebraic multigrid using graph neural networks. In *Proceedings of the 37th International Conference on Machine Learning*, PMLR 119, 6489–6499.
- Oden, J. T., Prudhomme, S. (2001). Goal-oriented error estimation and adaptivity for the finite element method. *Computers & Mathematics with Applications*, 41(5–6), 735–756.
- Parish, E., Lindsay, P., Shelton, T., Mersch, J. (2024). Embedded symmetric positive semi-definite machine-learned elements for reduced-order modeling in finite-element simulations with application to threaded fasteners. *Computational Mechanics*, 74(6), 1357–1381.
- Pfaff, T., Fortunato, M., Sanchez-Gonzalez, A., Battaglia, P. W. (2021). Learning mesh-based simulation with graph networks. In *International Conference on Learning Representations (ICLR 2021)*.
- Smetana, K., Patera, A. T. (2016). Optimal Local Approximation Spaces for Component-Based Static Condensation Procedures. *SIAM Journal on Scientific Computing*, 38(5), A3318–A3356.
- Svanberg, K. (1987). The method of moving asymptotes—a new method for structural optimization. *International Journal for Numerical Methods in Engineering*, 24(2), 359–373.
- Toselli, A., Widlund, O. B. (2005). *Domain Decomposition Methods — Algorithms and Theory*. Springer, Berlin.
- Trottenberg, U., Oosterlee, C. W., Schüller, A. (2001). *Multigrid*. Academic Press, San Diego. ISBN 0-12-701070-X.
- Um, K., Brand, R., Fei, Y., Holl, P., Thuerey, N. (2020). Solver-in-the-loop: learning from differentiable physics to interact with iterative PDE-solvers. In *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*, 6111–6122.
- Vaněk, P., Mandel, J., Brezina, M. (1996). Algebraic multigrid by smoothed aggregation for second and fourth order elliptic problems. *Computing*, 56(3), 179–196.
- Wu, Z., Xia, L., Wang, S., Shi, T. (2019). Topology optimization of hierarchical lattice structures with substructuring. *Computer Methods in Applied Mechan-*

- ics and Engineering*, 345, 602–617.
- Xing, Y., Liu, Y., Xue, T., Lu, L. (2026). GMT: A Geometric Multigrid Transformer Solver for Microstructure Homogenization. *ACM Transactions on Graphics*, 45(4), 1–15.
- Xu, J. (1992). Iterative Methods by Space Decomposition and Subspace Correction. *SIAM Review*, 34(4), 581–613.
- Xu, J., Zikatanov, L. (2002). The method of alternating projections and the method of subspace corrections in Hilbert space. *Journal of the American Mathematical Society*, 15(3), 573–597.
- Xu, W., Liu, C., Guo, Y., Huang, M., Guo, X. (2025). Problem-Independent Machine Learning (PIML) enhanced 3D lattice composite structures optimization via moving morphable components approach. *Composite Structures*, 369, 119330.
- Yu, S., Sun, J., Bai, J. (2019). Investigation of functionally graded TPMS structures fabricated by additive manufacturing. *Materials & Design*, 182, 108021.
- Zhang, E., Kahana, A., Kopaničáková, A., Turkel, E., Ranade, R., Pathak, J., Karniadakis, G. E. (2024). Blending neural operators and relaxation methods in PDE numerical solvers. *Nature Machine Intelligence*, 6(11), 1303–1313.
- Zhang, L., Huang, M., Liu, C., Du, Z., Cui, T., Guo, X. (2024). Problem-independent machine learning-enhanced structural topology optimization of complex design domains based on isoparametric elements. *Extreme Mechanics Letters*, 72, 102237.
- Zhang, L., Liu, C., Guo, Y., Jiang, C., Guo, X. (2026). Problem-independent transfer learning for complex-domain 3D topology optimization. *International Journal of Mechanical Sciences*, 328, 112007.